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(54) **CARBON CAPTURE IN FERMENTATION FOR COMMODITY CHEMICALS**

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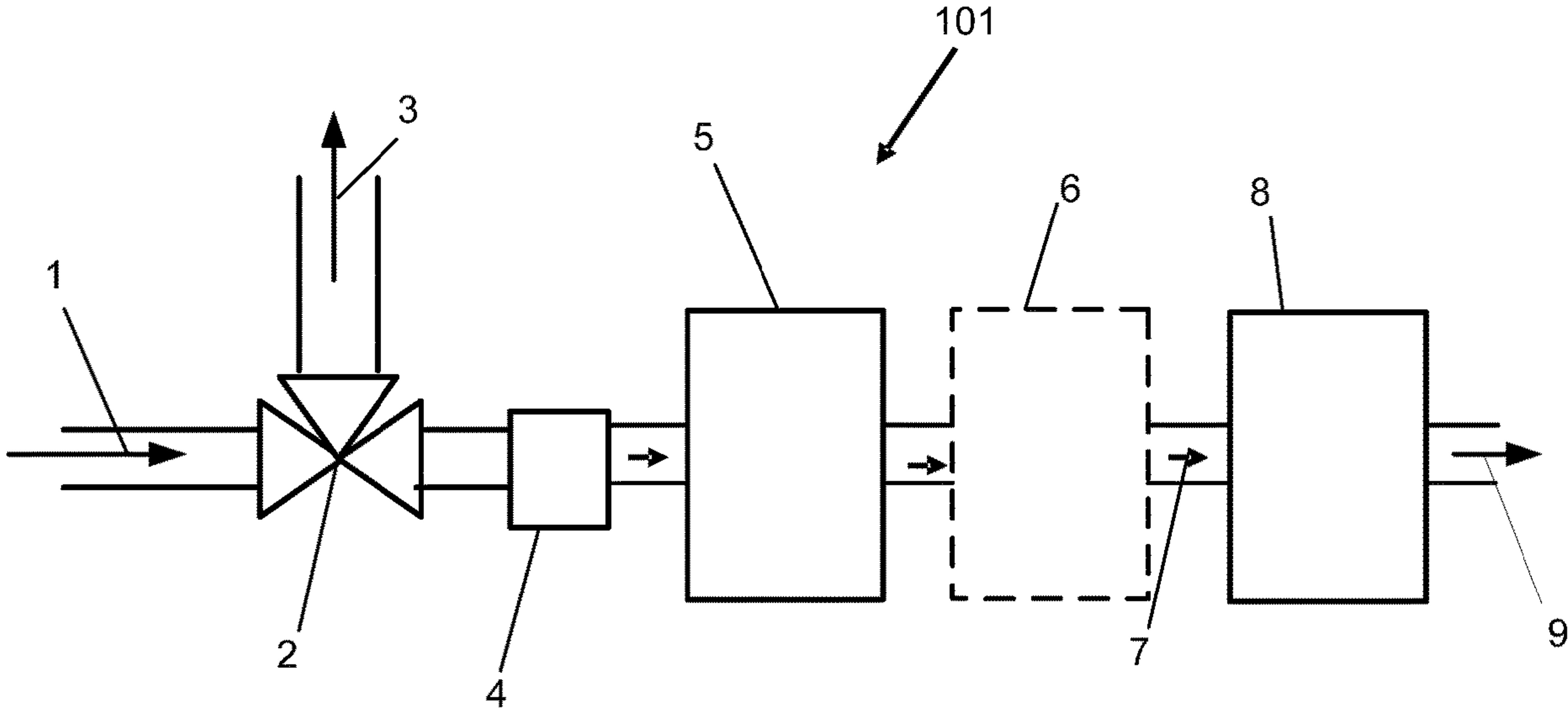
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(57) **ABSTRACT**

The disclosure relates to methods of capturing carbon by microbial fermentation of a gaseous substrate comprising CO into one or more first products which, in turn, may be incorporated into an article of manufacture or one or more second products. Further, the disclosure relates to improving carbon capture and/or efficiency.



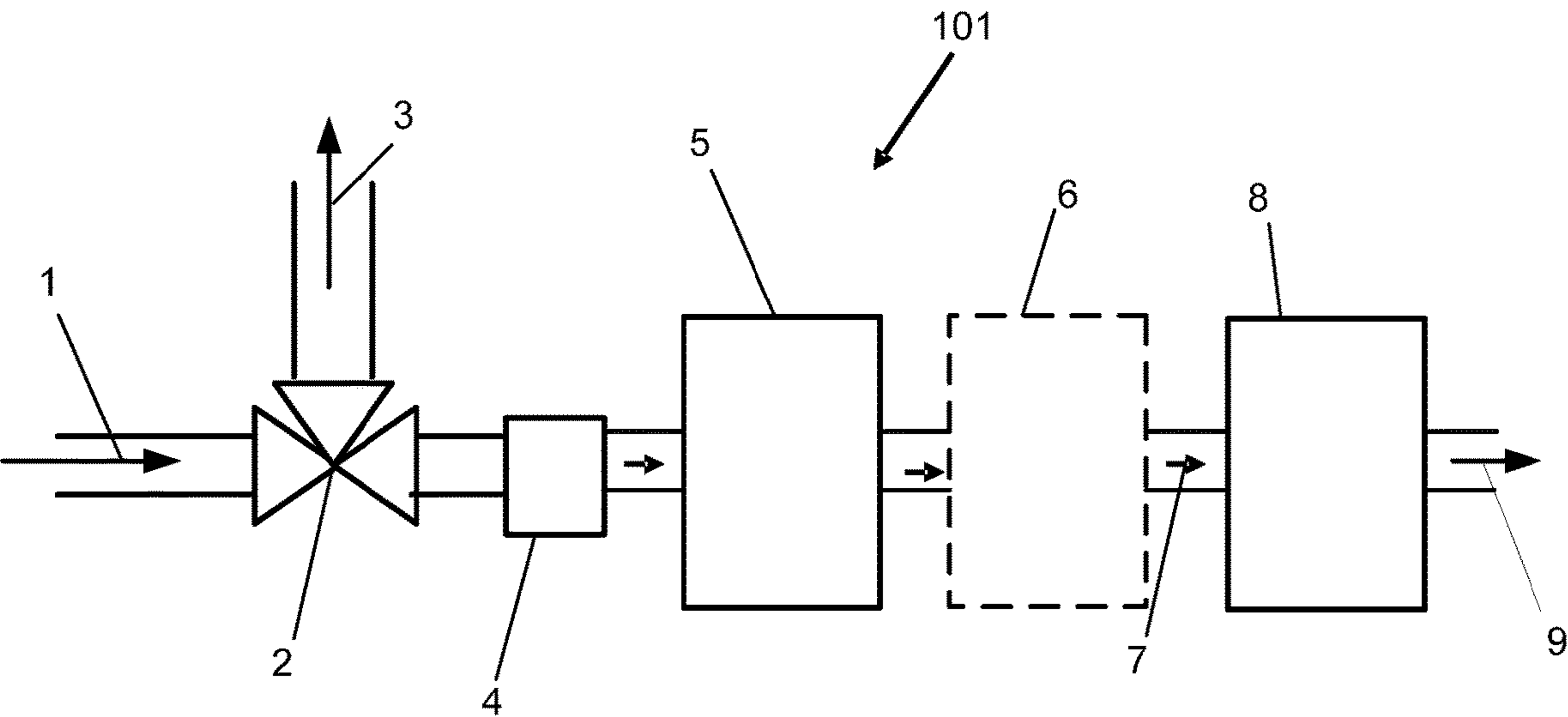


FIGURE 1

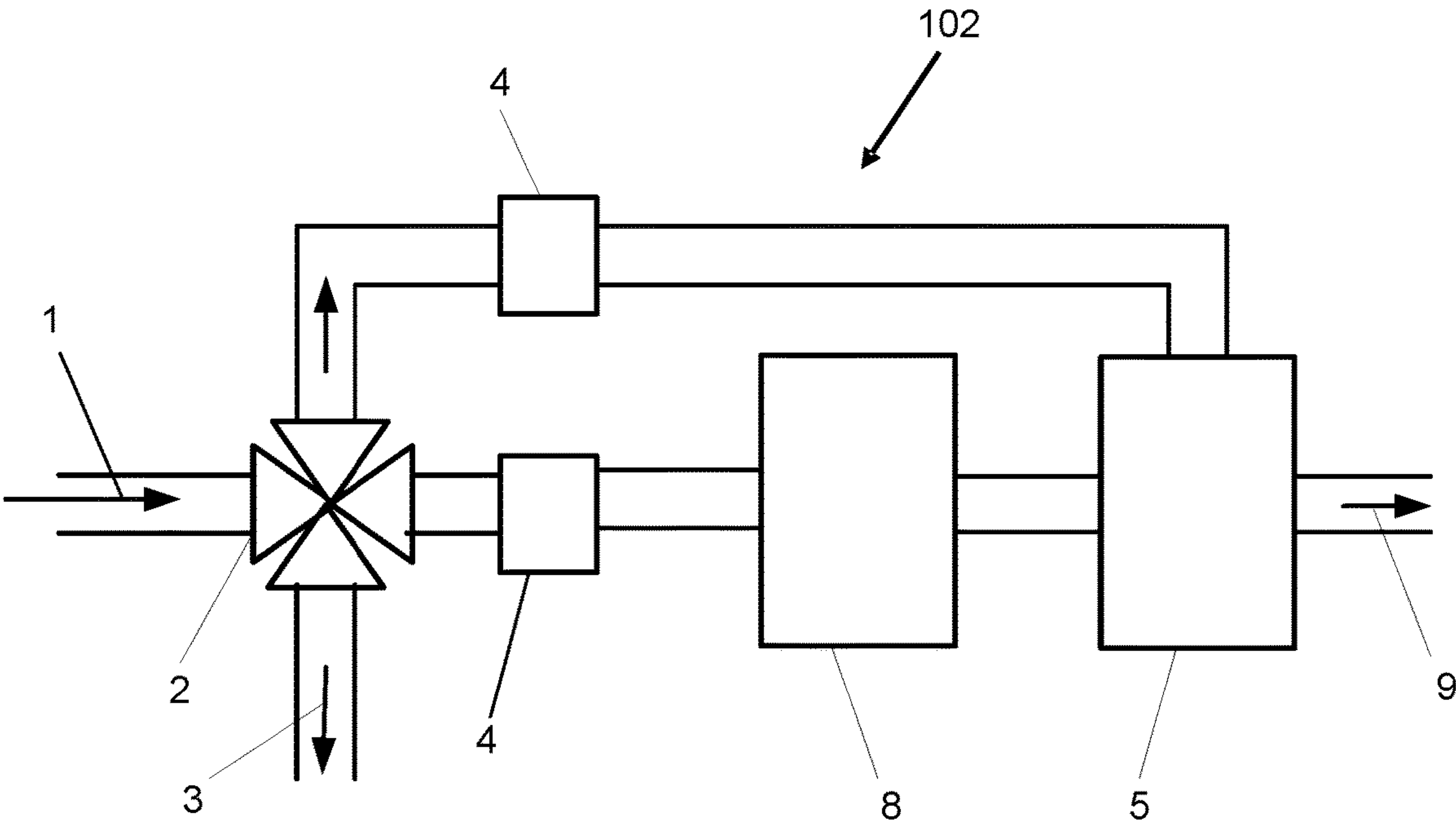


FIGURE 2

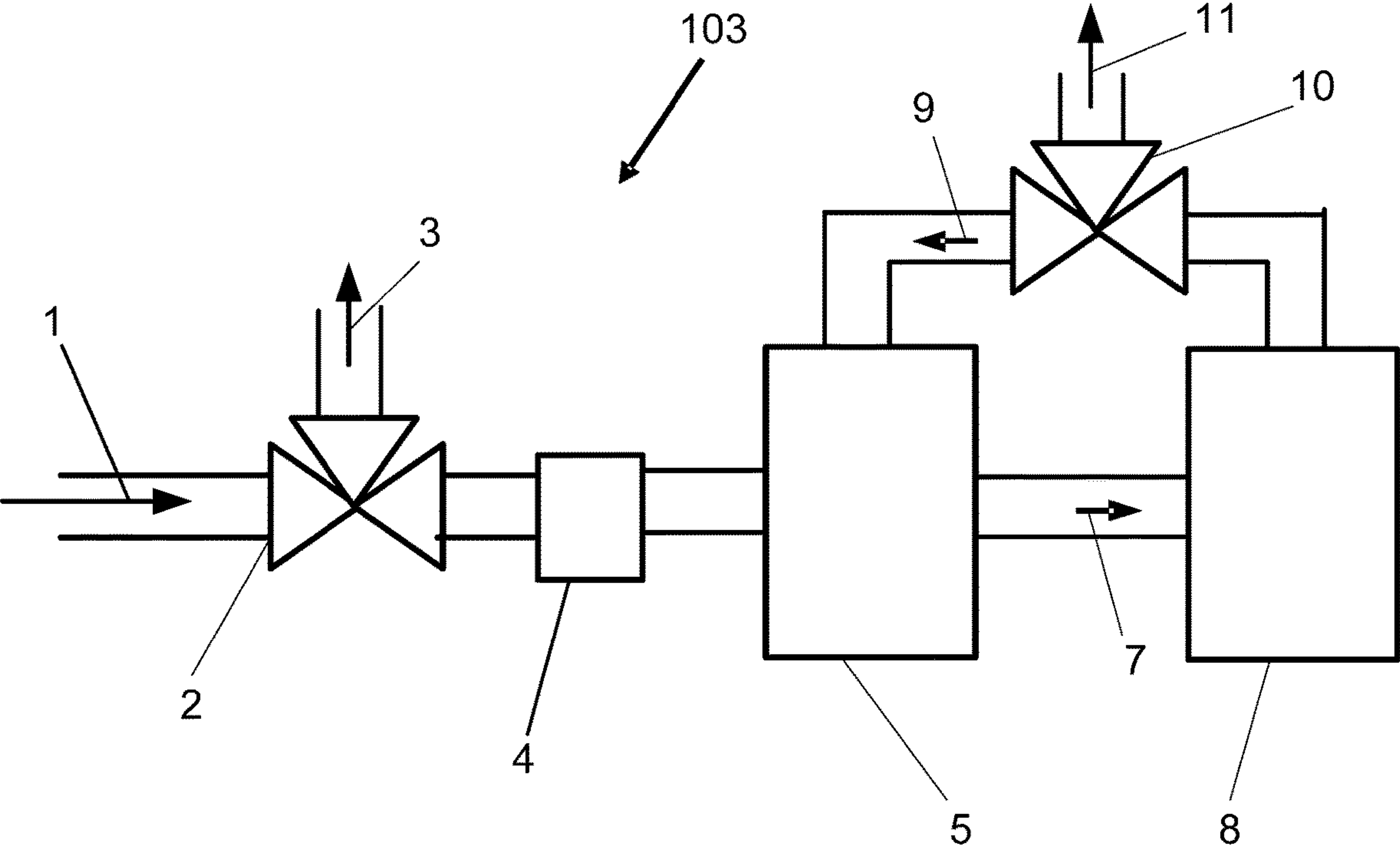


FIGURE 3

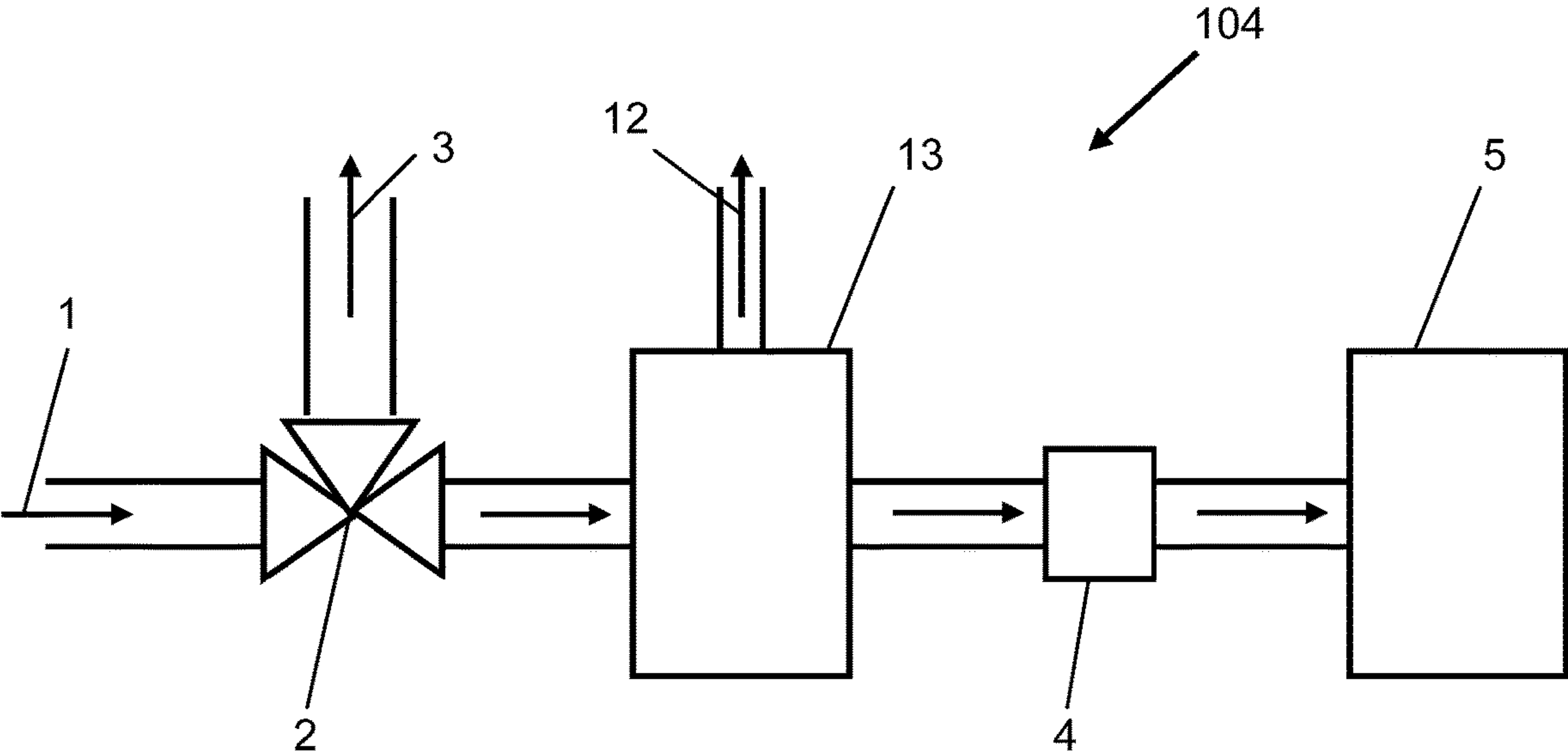


FIGURE 4

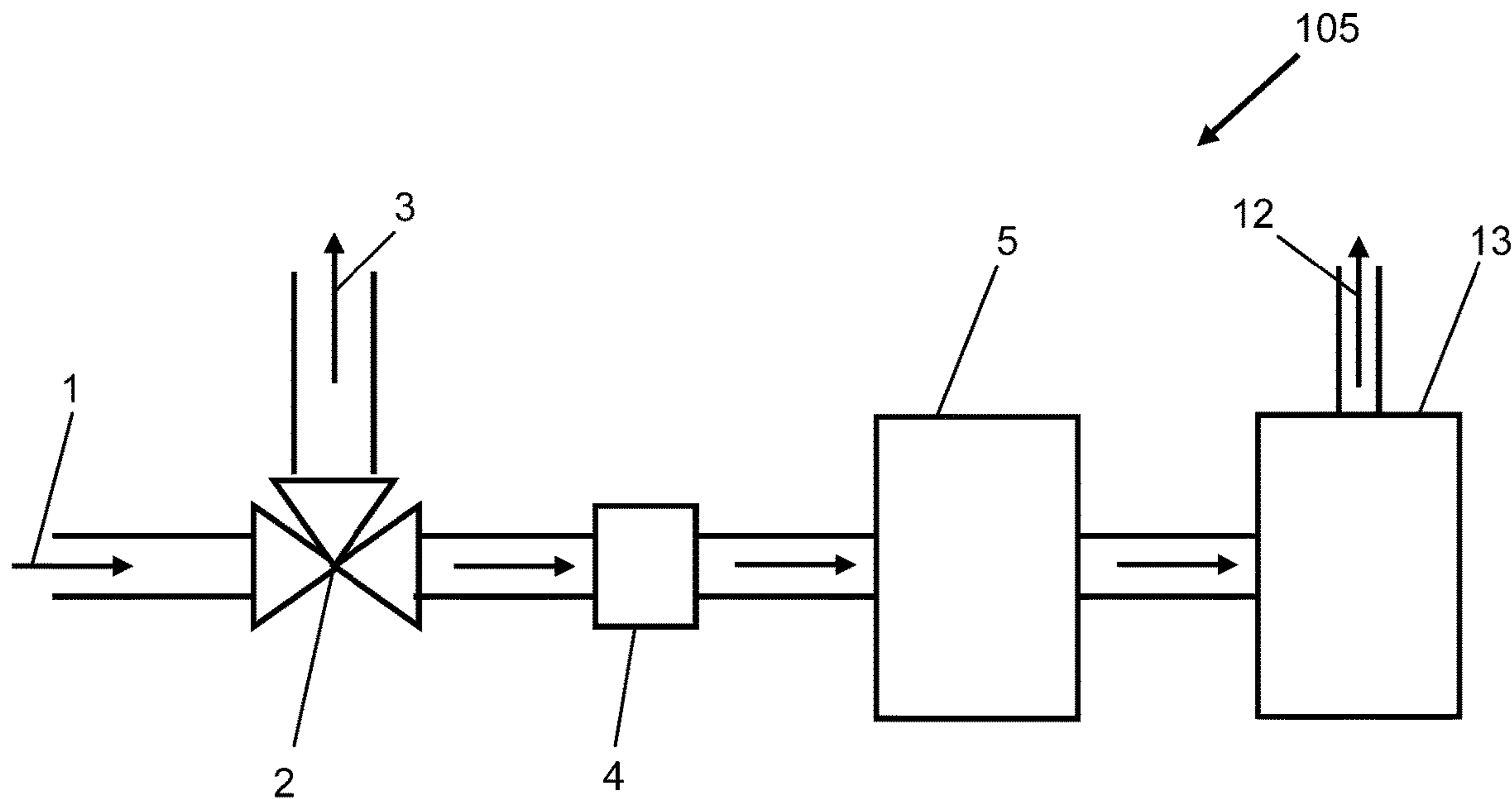


FIGURE 5

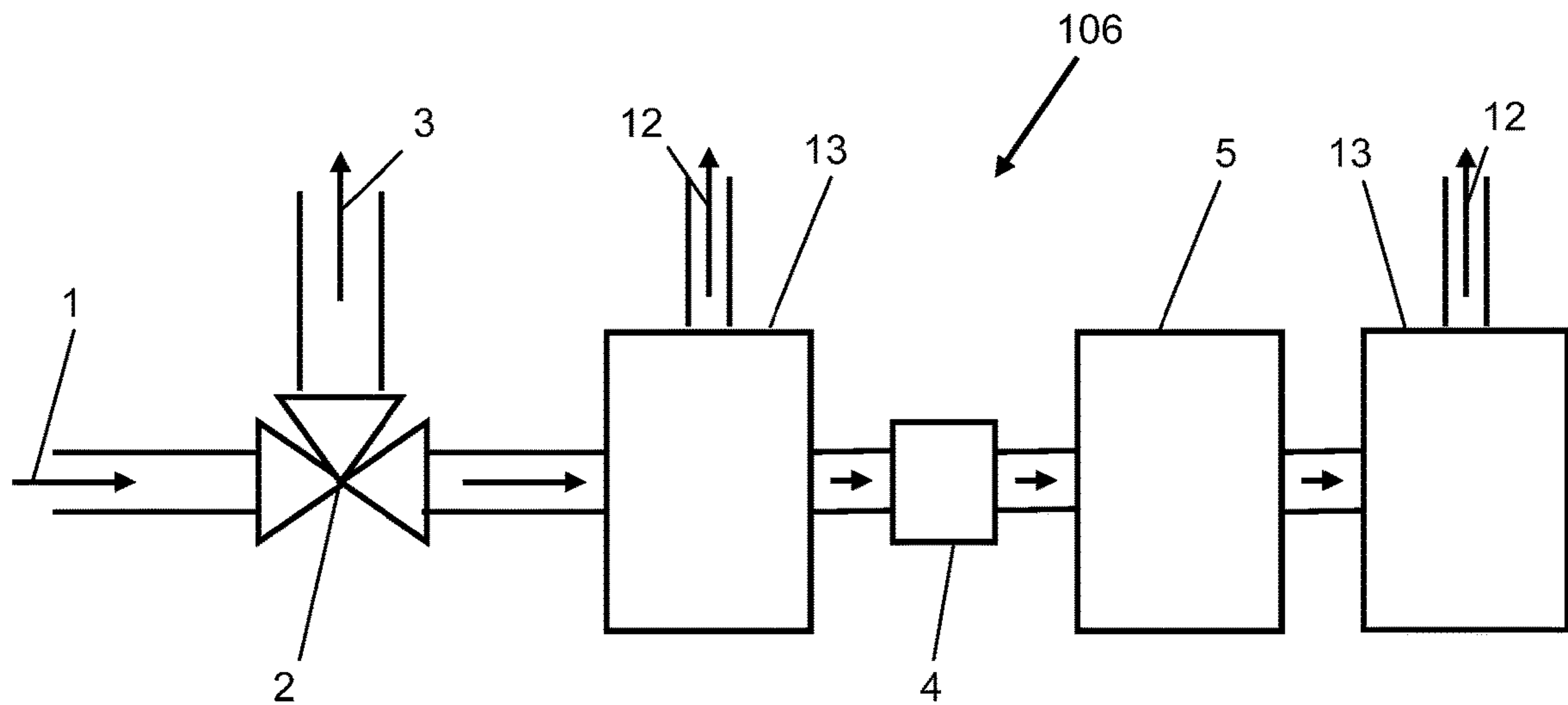


FIGURE 6

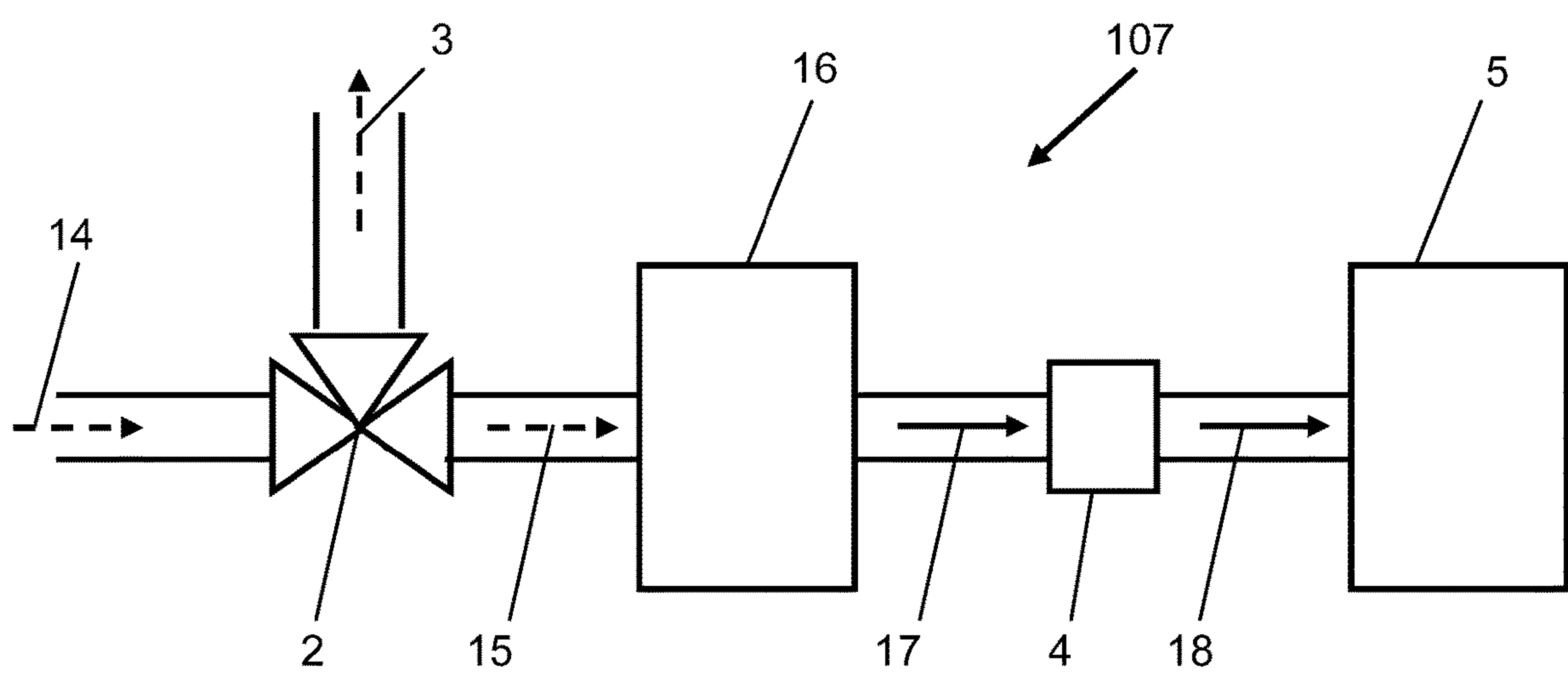


FIGURE 7

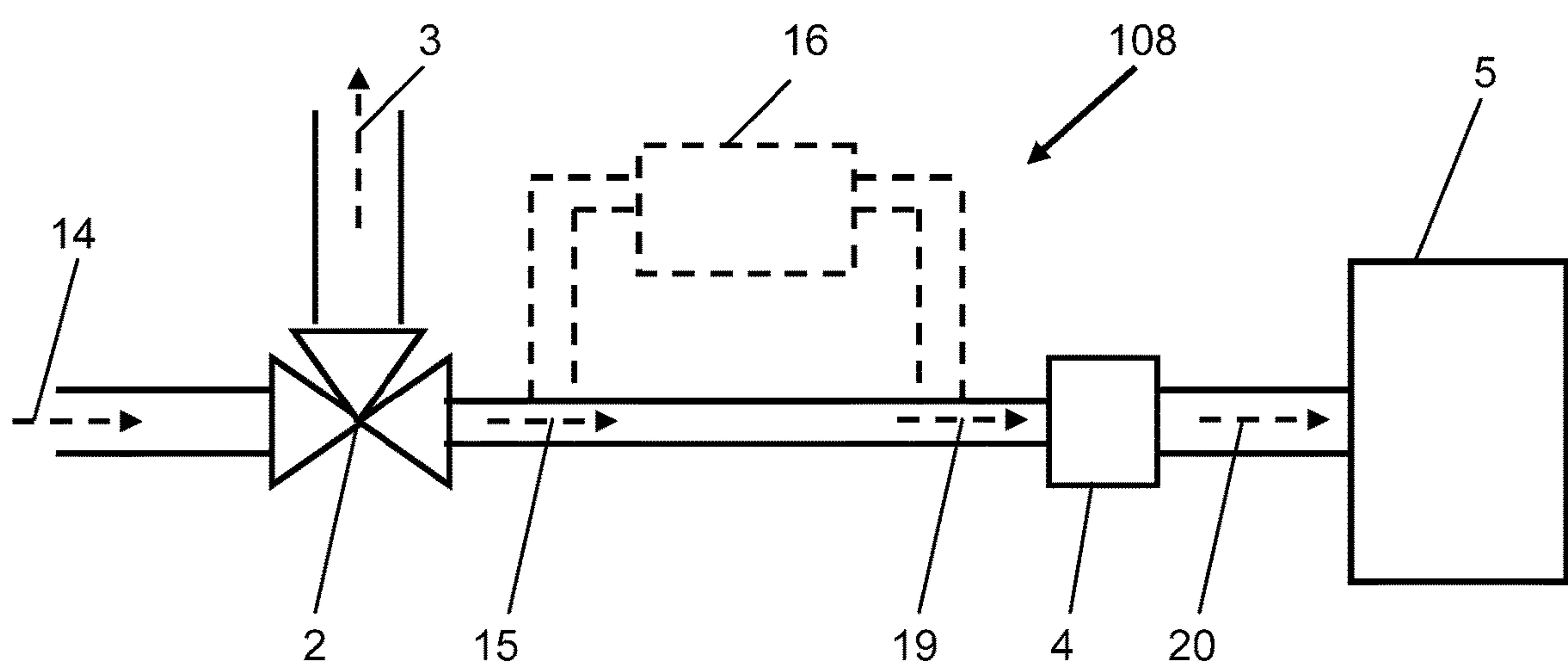


FIGURE 8

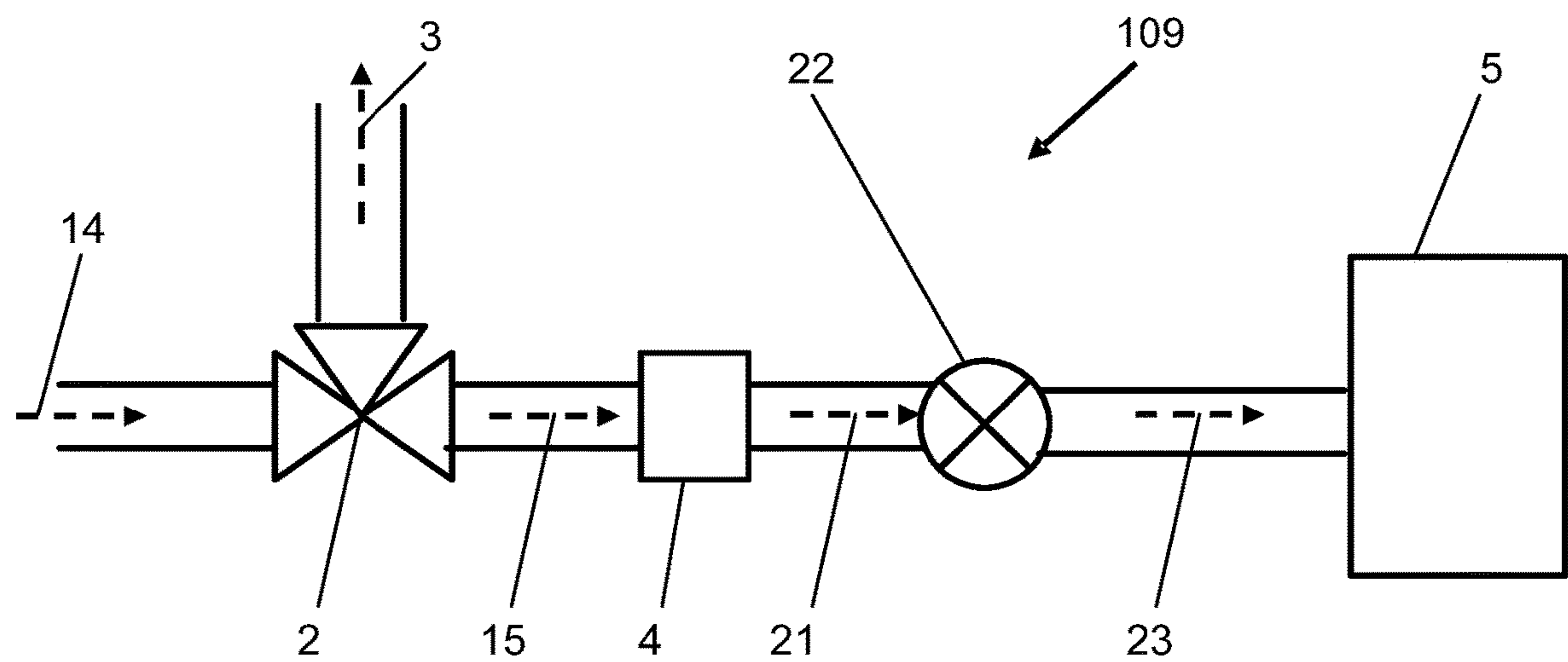


FIGURE 9

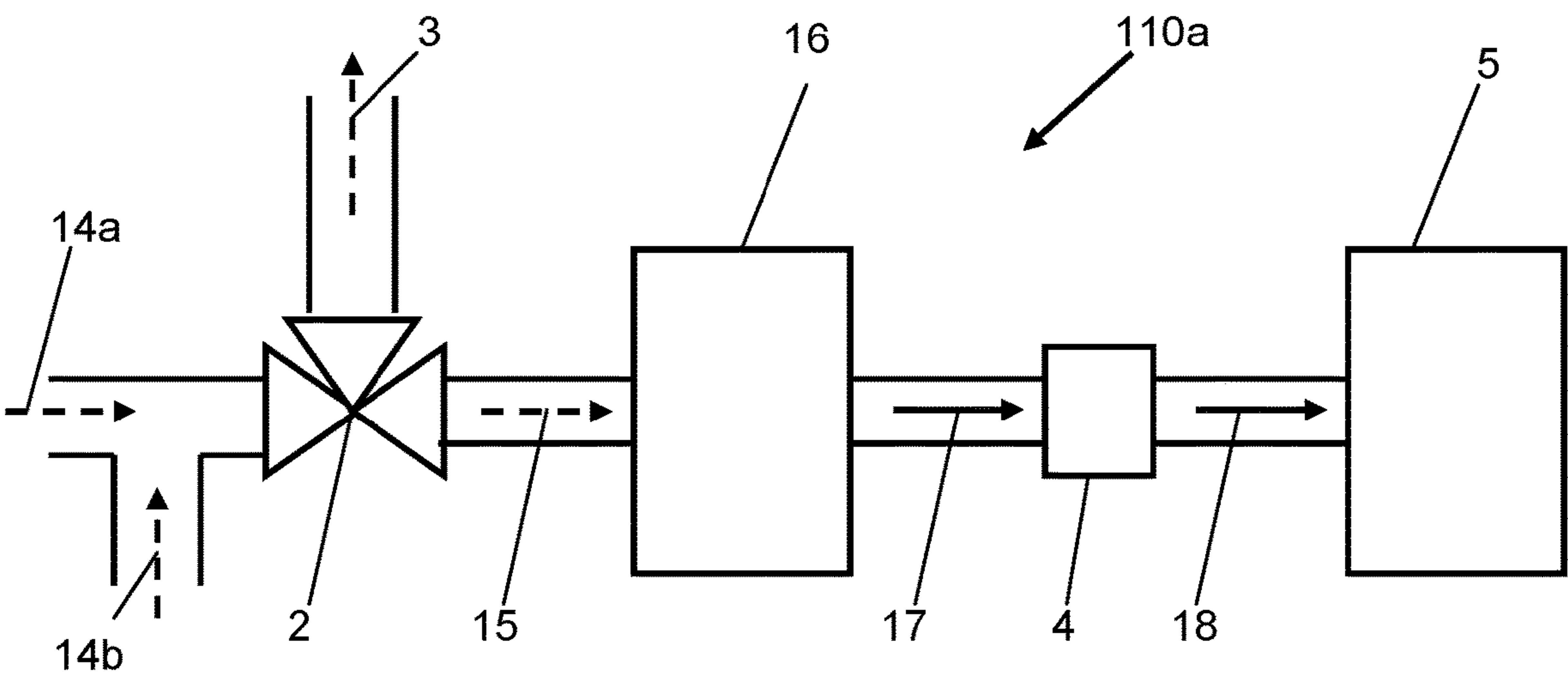


FIGURE 10a

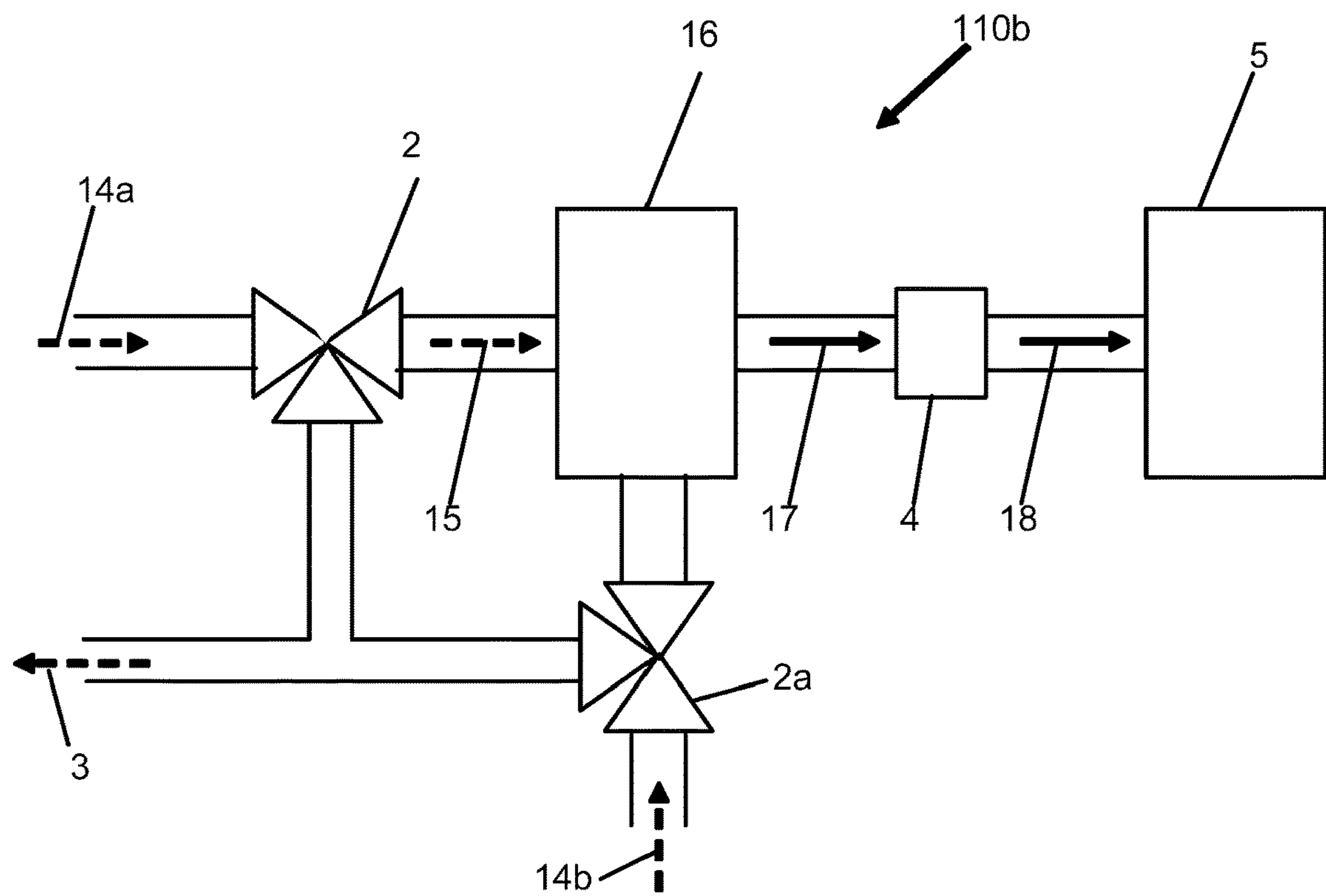


FIGURE 10b

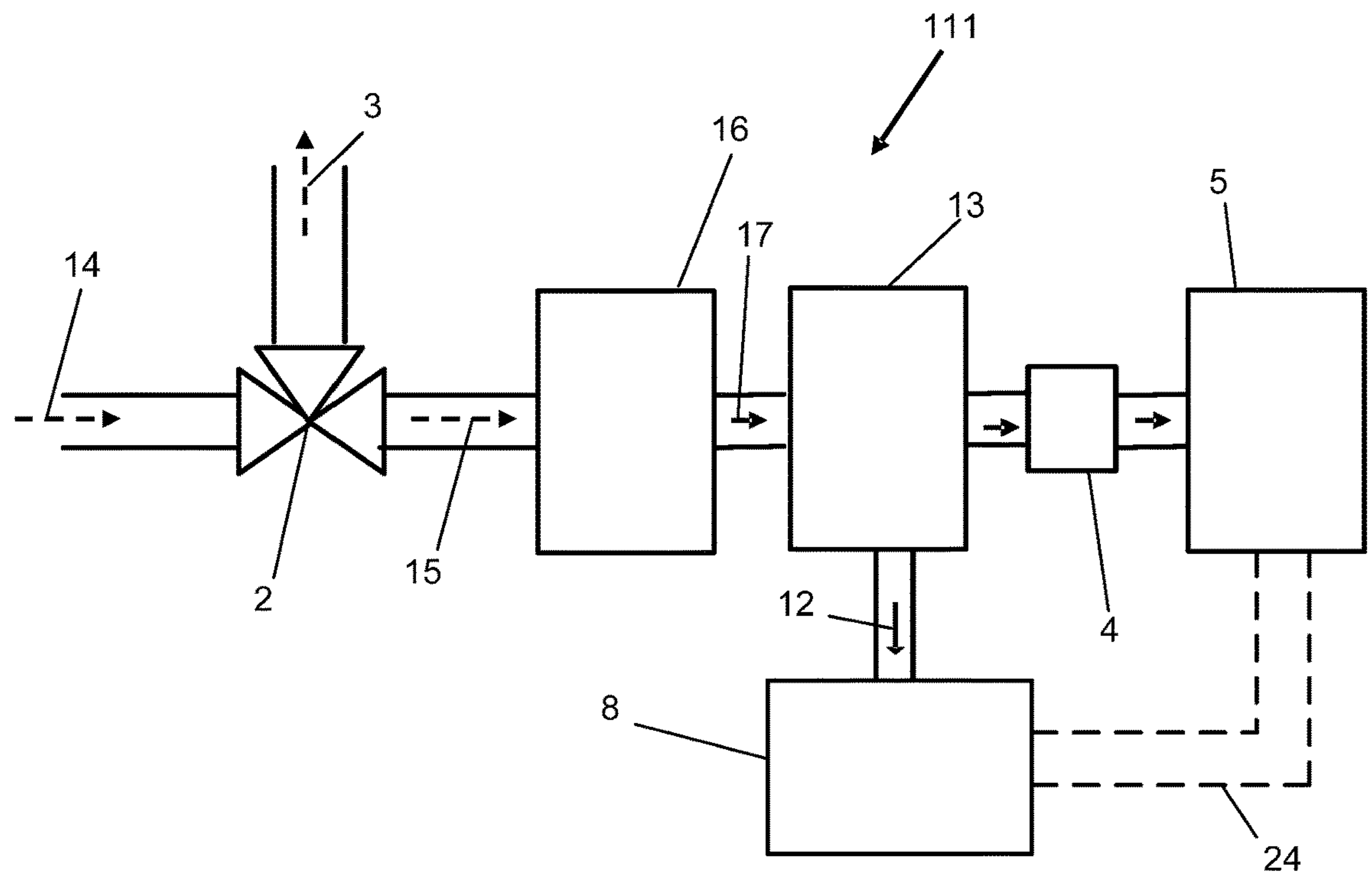


FIGURE 11

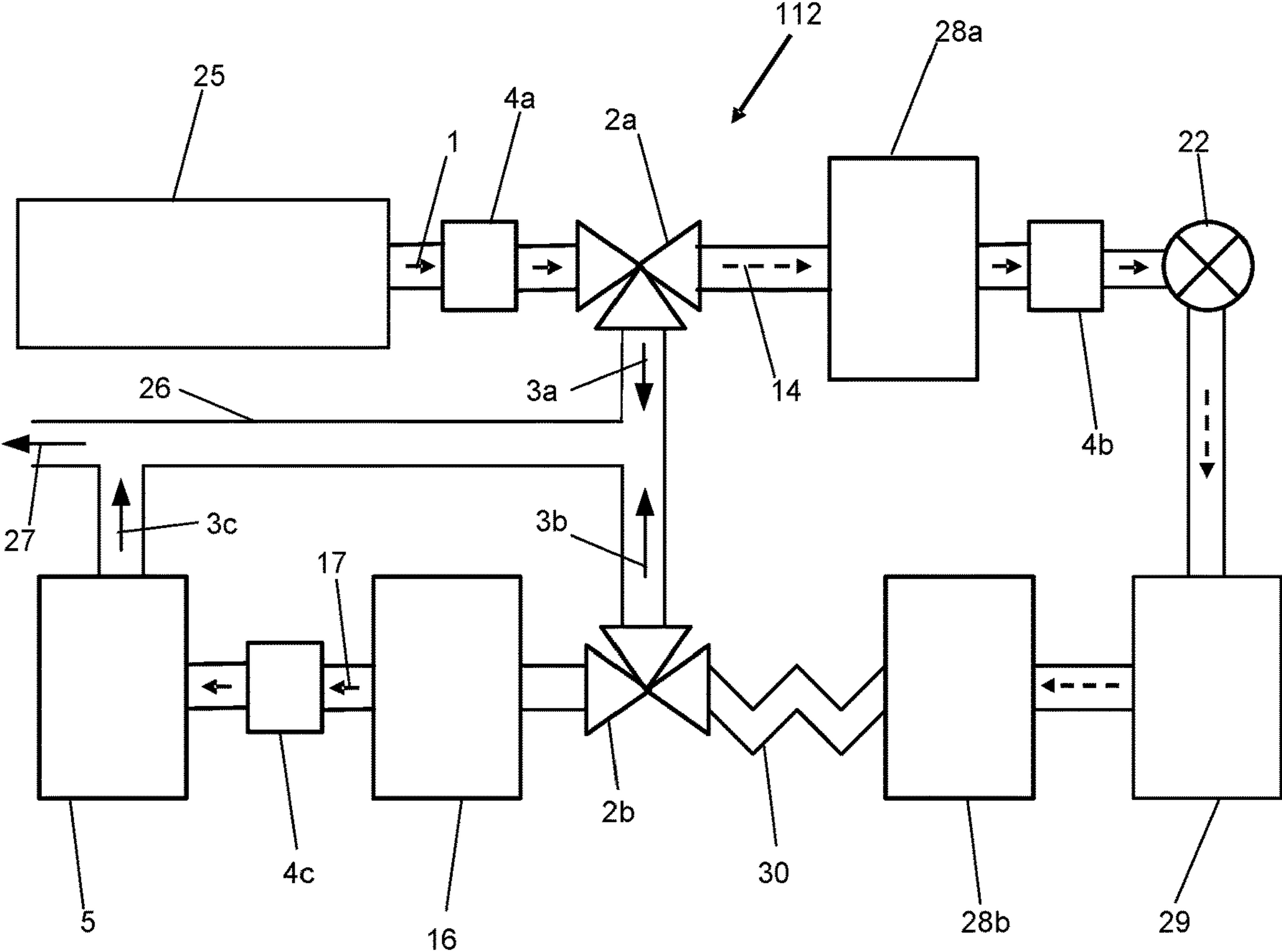


FIGURE 12

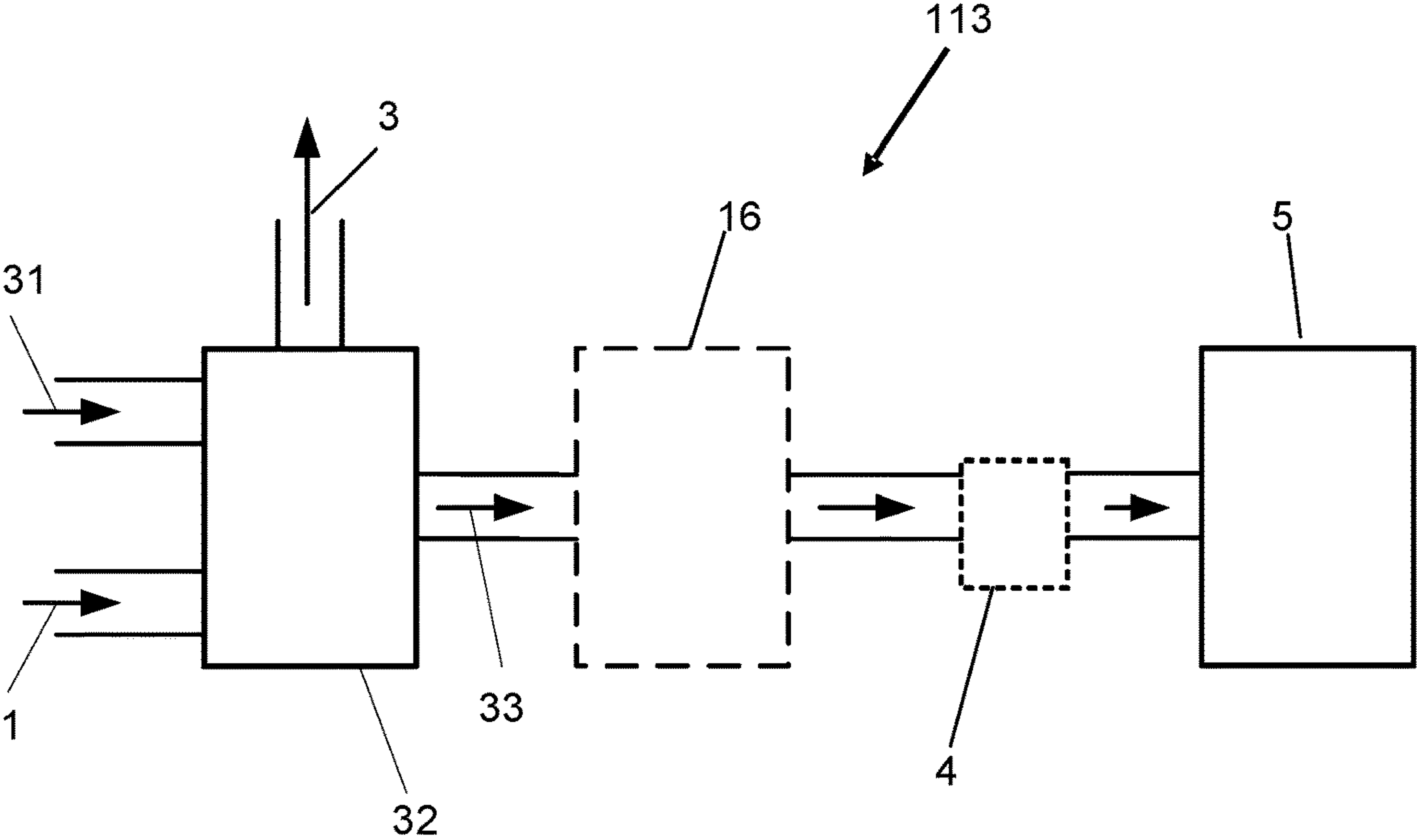


FIGURE 13

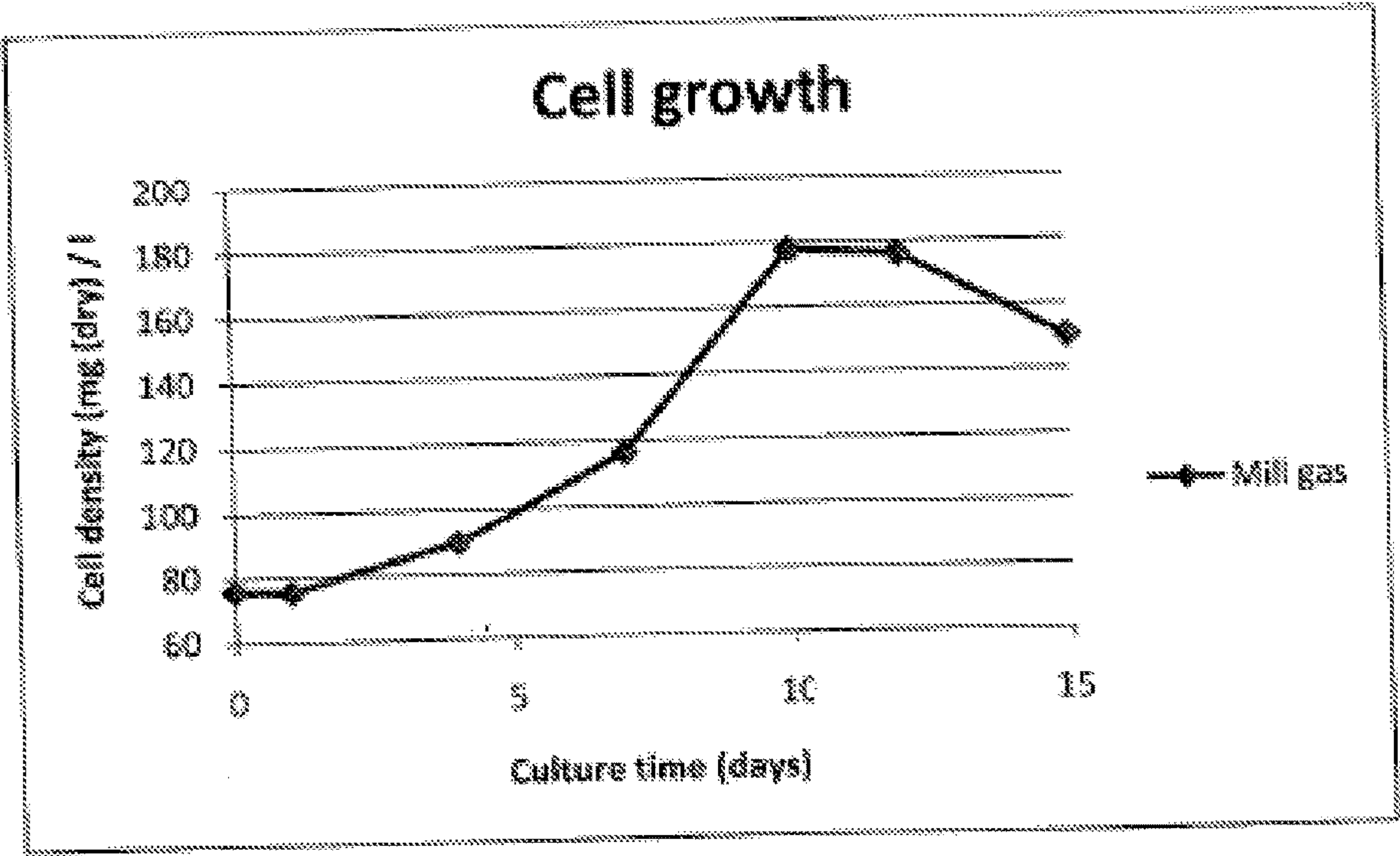


FIGURE 14

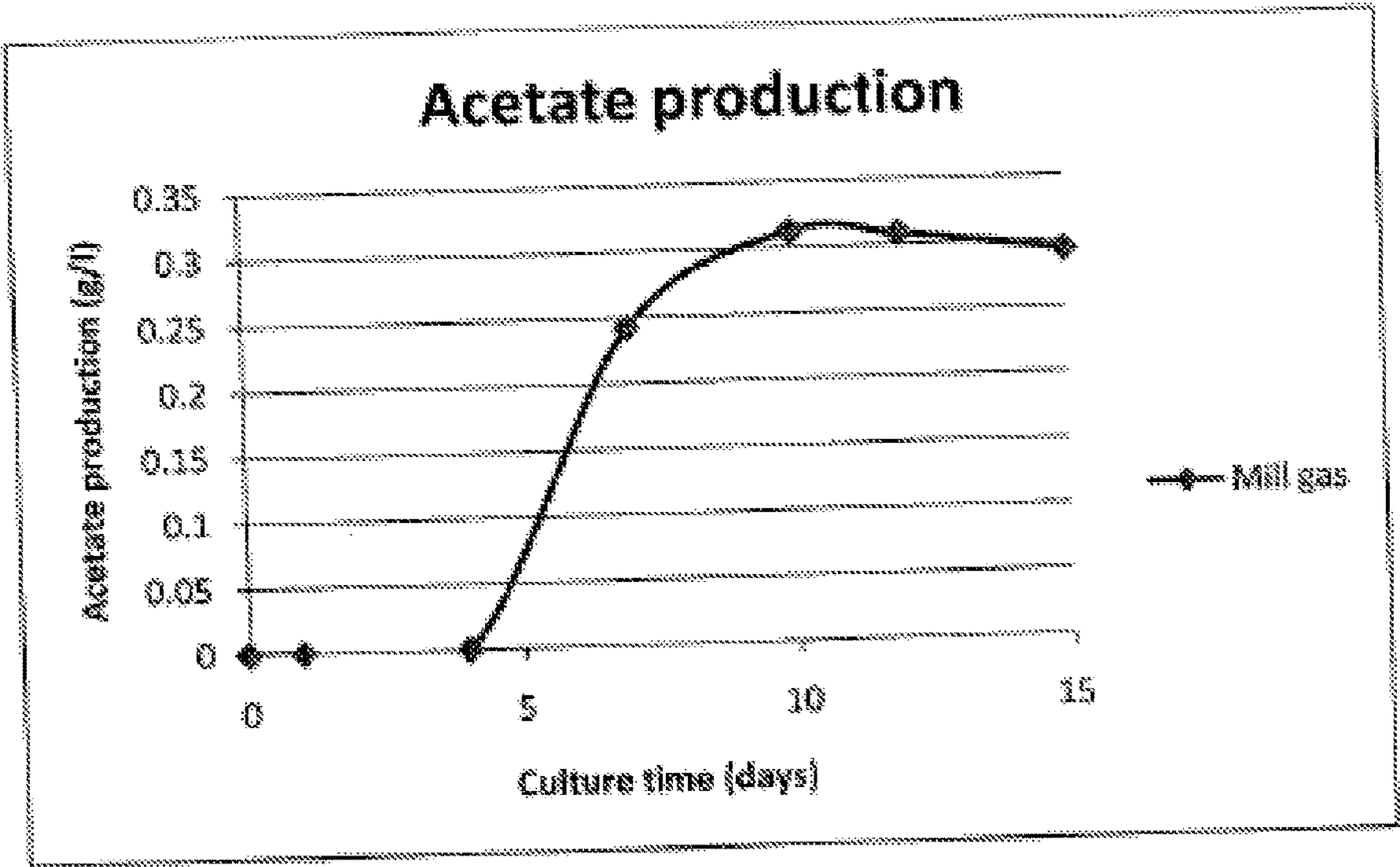


FIGURE 15

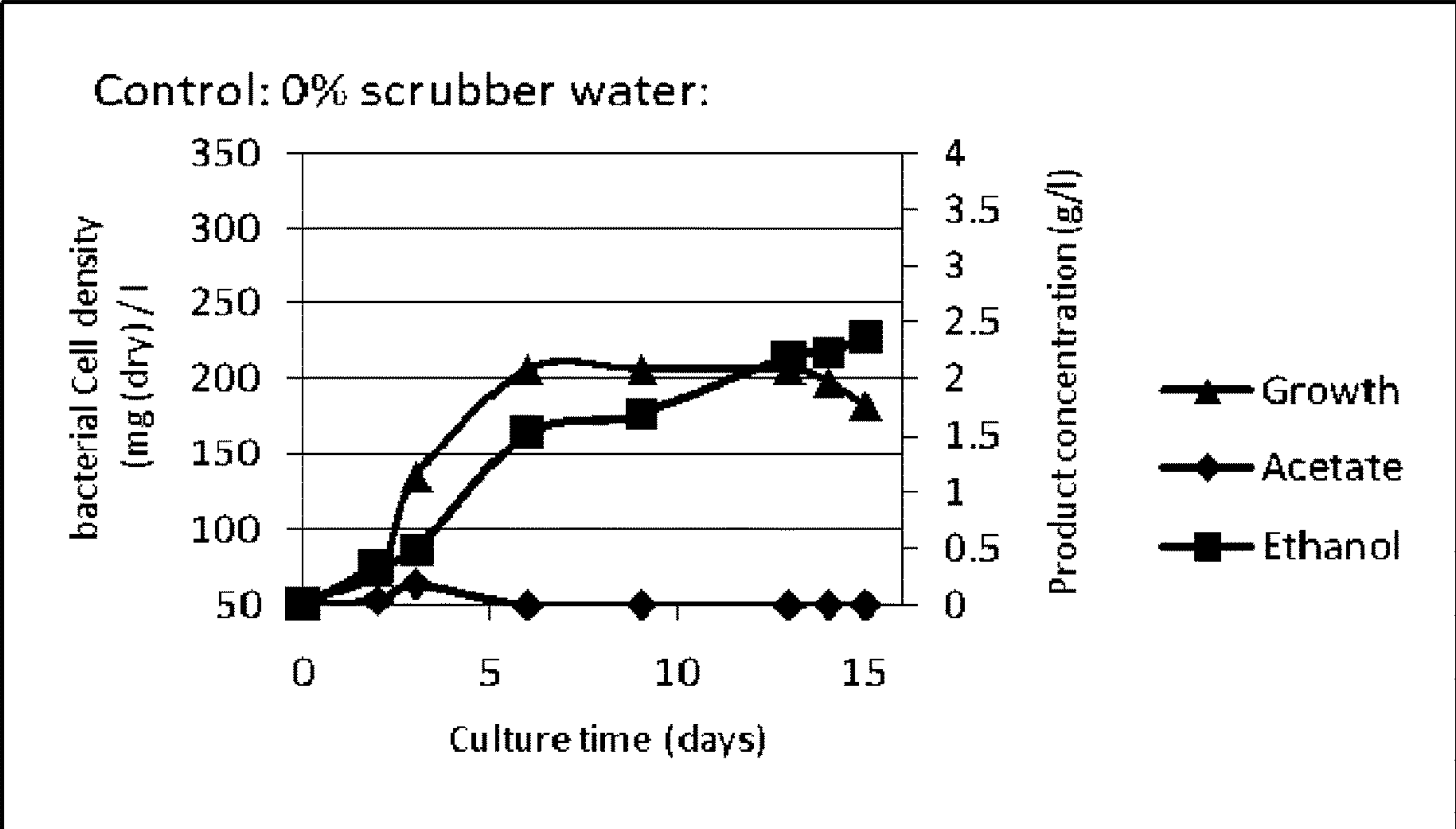


FIGURE 16

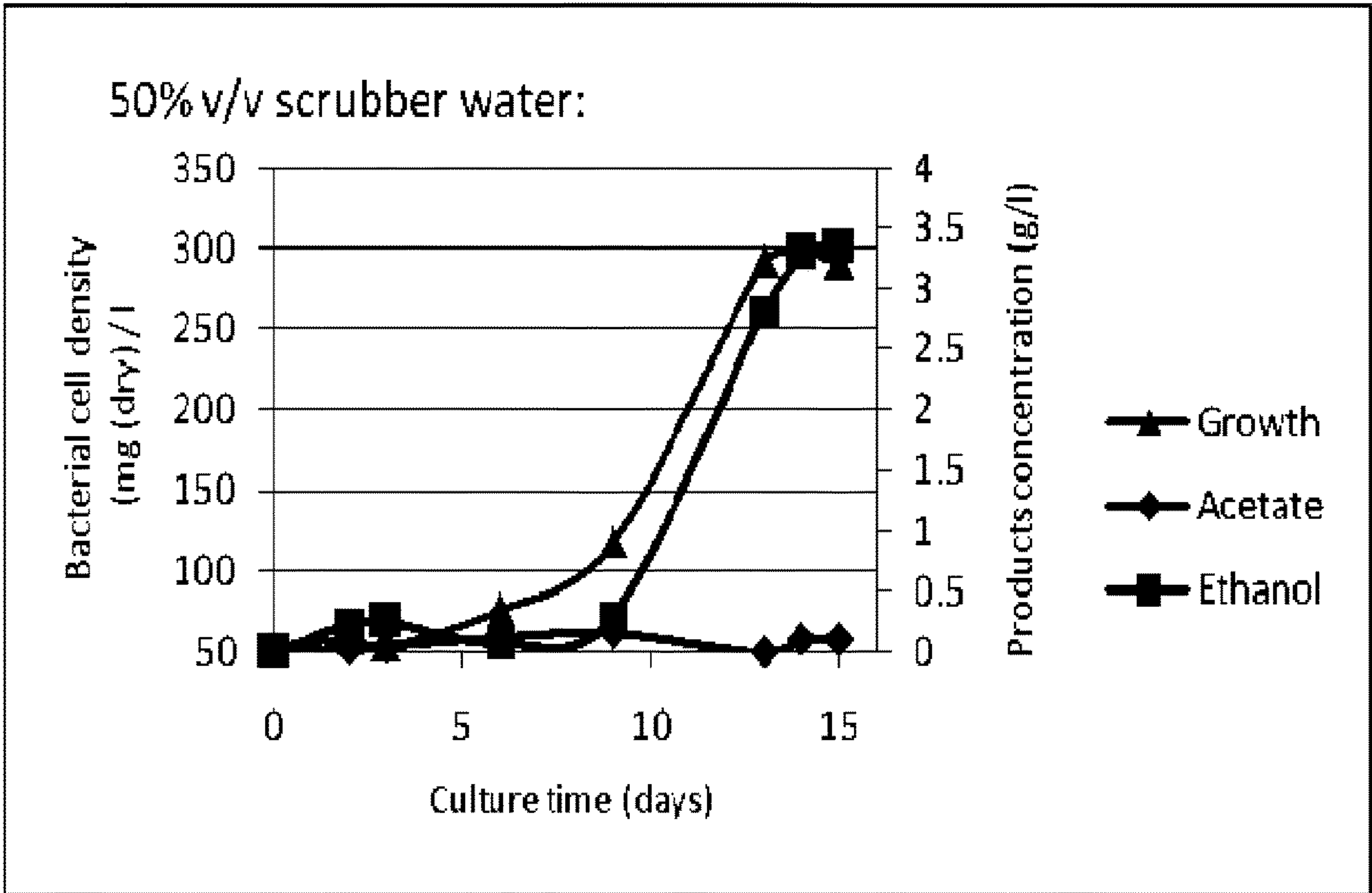


FIGURE 17

CARBON CAPTURE IN FERMENTATION FOR COMMODITY CHEMICALS

FIELD

[0001] This disclosure relates to systems and methods for improving overall carbon capture and/or improving overall efficiency in processes including microbial fermentation. In particular, the disclosure relates to improving carbon capture and/or improving efficiency in processes including microbial fermentation of a substrate comprising CO derived from an industrial source.

BACKGROUND

[0002] Ethanol is rapidly becoming a major hydrogen-rich liquid transport fuel around the world. Worldwide consumption of ethanol in 2005 was an estimated 12.2 billion gallons. The global market for the fuel ethanol industry has also been predicted to grow sharply in future, due to an increased interest in ethanol in Europe, Japan, the USA, and several developing nations.

[0003] For example, in the USA, ethanol is used to produce E10, a 10% mixture of ethanol in gasoline. In E10 blends the ethanol component acts as an oxygenating agent, improving the efficiency of combustion and reducing the production of air pollutants. In Brazil, ethanol satisfies approximately 30% of the transport fuel demand, as both an oxygenating agent blended in gasoline, and as a pure fuel in its own right. Also, in Europe, environmental concerns surrounding the consequences of Green House Gas (GHG) emissions have been the stimulus for the European Union (EU) to set member nations a mandated target for the consumption of sustainable transport fuels such as biomass derived ethanol.

[0004] The vast majority of fuel ethanol is produced via traditional yeast-based fermentation processes that use crop derived carbohydrates, such as sucrose extracted from sugarcane or starch extracted from grain crops, as the main carbon source. However, the cost of these carbohydrate feed stocks is influenced by their value as human food or animal feed, while the cultivation of starch or sucrose-producing crops for ethanol production is not economically sustainable in all geographies. Therefore, it is of interest to develop technologies to convert lower cost and/or more abundant carbon resources into fuel ethanol.

[0005] CO is a major, free, energy-rich by-product of the incomplete combustion of organic materials such as coal or oil and oil derived products. For example, the steel industry in Australia is reported to produce and release into the atmosphere over 500,000 tonnes of CO annually.

[0006] Catalytic processes may be used to convert gases consisting primarily of CO and/or CO and hydrogen (H₂) into a variety of fuels and chemicals. Micro-organisms may also be used to convert these gases into fuels and chemicals. These biological processes, although generally slower than chemical reactions, have several advantages over catalytic processes, including higher specificity, higher yields, lower energy costs and greater resistance to poisoning.

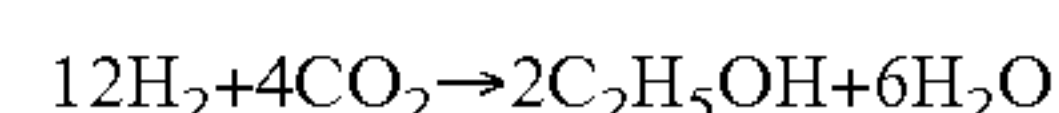
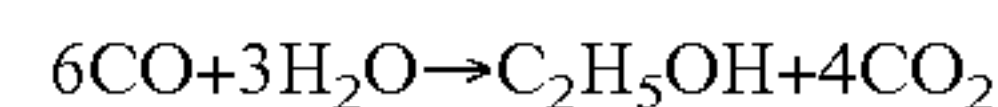
[0007] The ability of micro-organisms to grow on CO as a sole carbon source was first discovered in 1903. This was later determined to be a property of organisms that use the acetyl coenzyme A (acetyl CoA) biochemical pathway of autotrophic growth (also known as the Woods-Ljungdahl pathway and the carbon monoxide dehydrogenase/acetyl

CoA synthase (CODH/ACS) pathway). A large number of anaerobic organisms including carboxydophilic, photosynthetic, methanogenic and acetogenic organisms have been shown to metabolize CO to various end products, namely CO₂, H₂, methane, n-butanol, acetate and ethanol. While using CO as the sole carbon source, all such organisms produce at least two of these end products.

[0008] Anaerobic bacteria, such as those from the genus *Clostridium*, have been demonstrated to produce ethanol from CO, CO₂ and H₂ via the acetyl CoA biochemical pathway. For example, various strains of *Clostridium ljungdahlii* that produce ethanol from gases are described in WO 00/68407, EP 117309, U.S. Pat. Nos. 5,173,429, 5,593,886, and 6,368,819, WO 98/00558 and WO 02/08438. The bacterium *Clostridium autoethanogenum* sp is also known to produce ethanol from gases (Abrini et al., Archives of Microbiology 161, pp 345-351 (1994)).

[0009] However, ethanol production by micro-organisms by fermentation of gases is always associated with co-production of acetate and/or acetic acid. As some of the available carbon is converted into acetate/acetic acid rather than ethanol, the efficiency of production of ethanol using such fermentation processes may be less than desirable. Also, unless the acetate/acetic acid by-product can be used for some other purpose, it may pose a waste disposal problem. Acetate/acetic acid is converted to methane by micro-organisms and therefore has the potential to contribute to GHG emissions.

[0010] Microbial fermentation of CO in the presence of H₂ can lead to substantially complete carbon transfer into an alcohol. However, in the absence of sufficient H₂, some of the CO is converted into alcohol, while a significant portion is converted to CO₂ as shown in the following equations:



[0011] The production of CO₂ represents inefficiency in overall carbon capture and if released, also has the potential to contribute to Green House Gas emissions.

[0012] WO2007/117157, the disclosure of which is incorporated in its entirety herein by reference, describes a process that produces alcohols, particularly ethanol, by anaerobic fermentation of gases containing carbon monoxide. Acetate produced as a by-product of the fermentation process is converted into hydrogen gas and carbon dioxide gas, either or both of which may be used in the anaerobic fermentation process. WO2008/115080, the disclosure of which is incorporated in its entirety herein by reference, describes a process for the production of alcohol(s) in multiple fermentation stages. By-products produced as a result of anaerobic fermentation of gas(es) in a first bioreactor can be used to produce products in a second bioreactor. Furthermore, by-products of the second fermentation stage can be recycled to the first bioreactor to produce products.

[0013] U.S. Pat. No. 7,078,201 and WO 02/08438 also describe improving fermentation processes for producing ethanol by varying conditions (e.g. pH and redox potential) of the liquid nutrient medium in which the fermentation is performed. As disclosed in those publications, similar processes may be used to produce other alcohols, such as butanol.

[0014] Even minor improvements to a fermentation process for producing one or more acids and/or one or more

alcohols can have a significant impact on the efficiency, and more particularly, the commercial viability, of such a process.

[0015] For example, regardless of the source used to feed the fermentation reaction, problems can occur when there are breaks in the feed supply. More particularly, such interruptions can be detrimental to the efficiency of production by the micro-organisms used in the reaction, and in some cases, can be harmful thereto. For example, where CO gas in an industrial waste gas stream may be used in fermentation reactions to produce acids/alcohols, there may be times when the stream is not produced. During such times, the micro-organisms used in the reaction may go into an inactive, non-productive state or hibernation. When the stream is available again, there may then be a lag before the micro-organisms are fully productive at performing the desired reaction. Accordingly, there would be significant benefit if there were a means to reduce or eliminate this lag time.

[0016] As another example, in many industrial processes, scrubber systems or apparatus are used to reduce the concentration of particulates (such as dust) and other components that contaminate exhaust gases. Dry or wet scrubbing systems are known. In a wet scrubbing system, water or other liquids are used to “scrub” the contaminants from the gas stream. A typical wet scrubbing system is seen in steel mills, where water is used to clean flue gases generated at various stages of steel manufacture: for example gases generated by the coking ovens, the blast furnace, the basic oxygen furnace or the electric arc furnace. While scrubbing has the benefit of reducing the level of contaminants within exhaust gases, it by no means eliminates the contaminants altogether. The unwanted substances are simply removed from the gas into a solid or powder form or into the scrubber water or liquid. The water or liquid used in the scrubber system thus becomes a waste stream generated by this industry. The disposal of such waste represents an environmental hazard. The need to clean and dispose of such waste materials also represents a significant cost to the industry.

[0017] While conventional industrial scrubbers (such as at steel mills) remove a portion of the contaminants from industrial waste gas streams, it has been accepted in the art that additional scrubbing and/or treatment steps are required to be performed on the gases before they may be used to feed a fermentation reaction due to the perceived harmful effects of such gases on the micro-organisms used in the reaction. See, for example, Datar et al., *Fermentation of biomass-generated producer gas to ethanol*, 2004, *Biotechnology and Bioengineering* Vol. 86, pp 587-594. The use of additional scrubbing and/or treatment steps requires additional space in an industrial plant, which can be particularly problematic where the use of fermentation processes is added to an existing plant. Accordingly, there is a need for improved processes in which such additional scrubbing or other treatment steps are not required or are at least kept to minimum.

[0018] It is an object of the present disclosure to provide system(s) and/or method(s) that overcomes or ameliorates at least one disadvantage known in the art and provides the public with new methods for improved and/or increased production of a variety of useful products, including commodity chemicals.

SUMMARY

[0019] In a first aspect, there is provided a method of capturing carbon by microbial fermentation, the method comprising:

[0020] i. receiving off or waste gas stream(s) comprising CO from an industrial process;

[0021] ii. passing the gas stream(s) to a bioreactor containing a culture of one or more microorganisms; and

[0022] iii. fermenting the culture in the bioreactor to produce one or more products.

[0023] In some embodiments, the method comprises capturing at least a portion of a CO₂ content, using a CO₂ remover, from one or both of:

[0024] i. a stream prior to the stream entering the bioreactor; and

[0025] ii. a stream after the stream has exited the bioreactor.

[0026] In certain embodiments, the method comprises a first gas separation step, the first gas separation step comprising (i) receiving a gas stream; (ii) substantially separating at least one portion of the gas stream, wherein the portion comprises one or more components of the gas stream; and (iii) passing at least part of the separated portion(s) to the bioreactor. In one embodiment, the at least part of the separated portion(s) passed to the bioreactor comprises CO.

[0027] In particular embodiments, the method comprises a second gas separation step, the second gas separation step comprising (i) receiving a gas stream; (ii) substantially separating at least one portion of the gas stream, wherein the portion comprises one or more components of the gas stream; and (iii) passing at least part of the separated portion(s) to the CO₂ remover. In one embodiment, the gas separation step substantially separates CO₂ from the gas stream and passes the separated CO₂ to the CO₂ remover.

[0028] In particular embodiments, the method comprises buffering of a gas stream and passing at least a portion thereof to the bioreactor in a substantially continuous manner. In one embodiment, the step of buffering comprises (i) receiving an intermittent or non-continuous gas stream at storage means; and (ii) passing a substantially continuous stream to the bioreactor from the storage means.

[0029] In certain embodiments, the method comprises blending one or more gas streams with at least one other stream.

[0030] In certain embodiments, the method comprises adding scrubber water from an industrial process to the bioreactor.

[0031] In a second aspect, there is provided a system for capturing carbon by microbial fermentation, the system comprising an inlet for receiving off or waste gas from an industrial process, wherein, in use, the system is configured to pass at least a portion of the gas to a bioreactor to generate products by microbial fermentation.

[0032] In some embodiments, the system comprises a CO₂ remover configured to, in use, capture at least a portion of a CO₂ content from one or both of:

[0033] i. a stream prior to the stream entering the bioreactor; and

[0034] ii. a stream after the stream has exited the bioreactor.

[0035] In particular embodiments, the system comprises a first gas separator configured to in use: (i) receive a gas stream; (ii) substantially separate at least one portion of the

gas stream, wherein the portion comprises one or more components of the gas stream; (iii) pass at least part of the separated portion to the bioreactor. In one embodiment, the first gas separator is adapted to, in use, substantially separate CO from the gas stream and pass the separated CO to the bioreactor.

[0036] In particular embodiments, the system comprises a second gas separator configured to in use: (i) receive a gas stream; (ii) substantially separate at least one portion of the gas stream, wherein the portion comprises one or more components of the gas stream; and (iii) pass at least part of the separated portion to the CO₂ remover. In one embodiment, the second gas separator is adapted to separate CO₂ from the gas stream and pass the separated CO₂ to the CO₂ remover.

[0037] In some embodiments, the system comprises buffering means adapted to, in use, provide a substrate stream to the bioreactor in a substantially continuous stream. In particular embodiments, the buffering means comprises a buffer storage tank adapted to in use:

[0038] i. receive an intermittent or non-continuous gas/substrate stream; and

[0039] ii. pass a substantially continuous gas/substrate stream to the bioreactor.

[0040] In certain embodiments, the system comprises blending means adapted to, in use, combine a gas stream with at least one other stream prior to passing the combined stream to the bioreactor.

[0041] In particular embodiments, the system comprises at least one determining means to monitor the composition of at least one gas stream fed to and/or exhausted from the bioreactor. In certain embodiments, the system comprises control means for directing at least a portion of one or more gas/exhaust stream(s) to one or more of:

[0042] i. the bioreactor;

[0043] ii. the CO₂ remover;

[0044] iii. the first gas separator;

[0045] iv. the second gas separator;

[0046] v. the buffering means;

[0047] vi. the blending means; and

[0048] vii. an exhaust means,

the particular one(s) of destinations i to vii being selected at least in part based on the determination made by the determining means.

[0049] In a third aspect, there is provided a system for increasing overall carbon capture in a process of producing products in a bioreactor by microbial fermentation of a substrate, the system including a CO₂ remover configured to capture at least a portion of a CO₂ content from one or both of:

[0050] i. a stream prior to the stream entering the bioreactor; and

[0051] ii. a stream after the stream has exited the bioreactor.

[0052] In a fourth aspect, there is provided a system for increasing the efficiency of processes for producing products by microbial fermentation of gas(es) wherein supply of said gas(es) is intermittent, the system comprising buffering means adapted to receive and store at least a portion of the gas, and a bioreactor adapted to receive at least a portion of the gas from the buffering means.

[0053] In a fifth aspect, there is provided a system for increasing the efficiency of processes for producing products by microbial fermentation of gas(es), the system comprising

a gas separator configured to receive a gas stream and to pass at least a portion of said stream to a bioreactor.

[0054] In a sixth aspect, there is provided a steel mill adapted to produce alcohols by microbial fermentation of waste gas(es).

[0055] According to particular aspects, the systems and methods of the disclosure are adapted for use in a process of producing alcohols, more particularly ethanol and/or butanol, by anaerobic fermentation of gases containing carbon monoxide. Additionally or alternatively, acids, such as acetic acid or acetate, may be produced. However, the disclosure is not limited thereto and is intended to cover other fermentation reactions, including aerobic fermentations, those generating different products, such as isopropanol or H₂, and those not including fermentation of carbon containing gases.

[0056] Embodiments of the disclosure find particular application in the fermentation of a gaseous substrate comprising CO to produce acids and/or alcohols although particular aspects of the disclosure are not limited to substrates including CO. The gaseous substrate may comprise a gas obtained as a by-product of an industrial process. In certain embodiments, the industrial process is selected from the group consisting of ferrous metal products manufacturing, non-ferrous products manufacturing, petroleum refining processes, gasification of biomass, gasification of coal, electric power production, carbon black production, ammonia production, methanol production and coke manufacturing. Most preferably, the gaseous substrate comprises a gas obtained from a steel mill.

[0057] In certain preferred embodiments the gaseous substrate comprises from 20% CO to 100% CO by volume, such as from 50% CO to 95% CO by volume, such as from 50% to 70% CO by volume. Gaseous substrates having lower concentrations of CO, such as 6%, may also be appropriate, particularly when H₂ and CO₂ are also present.

[0058] In preferred embodiments, the fermentation reaction is carried out by one of more strains of carboxydophilic bacteria.

[0059] Preferably, the carboxydophilic bacterium is selected from *Clostridium*, *Moorella* and *Carboxydotherrmus*. Most preferably, the carboxydophilic bacterium is *Clostridium autoethanogenum*.

BRIEF DESCRIPTION OF THE DRAWINGS

[0060] The disclosure will now be described in detail with reference to the accompanying Figures in which:

[0061] FIG. 1: is a schematic representation of a system including a CO₂ remover downstream of a bioreactor according to certain embodiments of the disclosure.

[0062] FIG. 2: is a schematic representation of a system including a CO₂ remover upstream of a bioreactor according to certain embodiments of the disclosure.

[0063] FIG. 3: is a schematic representation of a system including a CO₂ remover downstream of a bioreactor and means for returning a substrate stream to the bioreactor.

[0064] FIG. 4: is a schematic representation of a system including a gas separator upstream of a bioreactor according to certain embodiments of the disclosure.

[0065] FIG. 5: is a schematic representation of a system including a gas separator downstream of a bioreactor according to certain embodiments of the disclosure.

[0066] FIG. 6: is a schematic representation of a system including two gas separators: one gas separator upstream of

a bioreactor and one gas separator downstream of a bioreactor according to certain embodiments of the disclosure.

[0067] FIG. 7: is a schematic representation of a system including a buffer storage tank according to certain embodiments of the disclosure.

[0068] FIG. 8: is a schematic representation of a system including an optional buffer storage tank according to certain embodiments of the disclosure.

[0069] FIG. 9: is a schematic representation of a system including a compressor according to certain embodiments of the disclosure.

[0070] FIG. 10a: is a schematic representation of a system including multiple substrate stream sources and a buffer storage tank according to certain embodiments of the disclosure.

[0071] FIG. 10b: is a schematic representation of a system including multiple substrate stream sources and a buffer storage tank according to certain embodiments of the disclosure.

[0072] FIG. 11: is a schematic representation of a system including a buffer storage tank, gas separator and CO₂ remover according to certain embodiments of the disclosure.

[0073] FIG. 12: is a schematic representation of a system adapted to capture carbon from a waste stream according to certain embodiments of the disclosure.

[0074] FIG. 13: is a schematic representation of a system including a blending means, according to certain embodiments of the disclosure.

[0075] FIG. 14: illustrates microbial growth with time when steel mill gases are used as a resource; and

[0076] FIG. 15: illustrates product synthesis, namely acetate production, with time when steel mill gases are used as a resource.

[0077] FIG. 16: illustrates bacterial growth and product synthesis with time in laboratory media; and,

[0078] FIG. 17: illustrates bacterial growth and product synthesis with time using laboratory media mixed with steel mill scrubber water in a ratio of 1:1.

DETAILED DESCRIPTION

[0079] In accordance with certain methods of the disclosure, waste or off gases from an industrial process may be used to supplement and/or support fermentation reactions with minimal additional processing or treatment steps on the gases prior to passing them to the bioreactor in which the fermentation process is performed. This is particularly surprising, because it is commonly believed that the waste or off gases contain contaminants that would be detrimental to the growth and/or survival of the micro-organisms used in the fermentation. The disclosure has particular applicability to waste or off gases produced during steel making processes, particularly those containing CO, which are used to produce alcohols (e.g. ethanol, butanol, isopropanol) and/or acids (e.g. butyric acid, acetic acid and/or acetate) and/or hydrogen. While there are a variety of different micro-organisms capable of performing such processes, the disclosure has particular applicability to fermentation processes involving the use of *Clostridium autoethanogenum*.

[0080] This aspect of the disclosure has significant value in that it reduces the number of or eliminates the pre-processing steps performed on waste gases prior to their use in a fermentation reaction. Thus, the disclosure provides for wider suitability and/or applicability of such fermentation processes, particularly in established industrial plants where

a limited, predetermined amount of space may be available for adding equipment to perform fermentations. Also, because no or limited scrubbing and/or pre-treatment processes are performed on the waste gases, embodiments of the disclosure may also ameliorate or reduce waste from industrial processes due to there being no need to process the waste or contaminants resulting from the scrubbing and/or pre-treatment processes.

[0081] Waste gases from industrial processes other than steel manufacturing may be used in a similar fashion in the methods of the disclosure. The disclosure is readily applicable to fermentation reactions that utilise gaseous substrates other than carbon monoxide as a source of carbon and energy, produce alcohols other than ethanol, produce hydrogen, and/or which utilise micro-organisms other than *Clostridium autoethanogenum*.

[0082] Substrates suitable for use in fermentation processes often also contain CO₂. In addition, in many fermentation reactions, for example where CO is converted to products including acids and/or alcohols, significant volumes of CO₂ can be produced. The present disclosure relates to methods, systems, and processes to improve the overall carbon capture in such fermentation reactions.

[0083] The substrate and/or C1-carbon source may be a waste gas obtained as a byproduct of an industrial process or from another source, such as combustion engine exhaust fumes, biogas, landfill gas, direct air capture, or from electrolysis. The substrate and/or C1-carbon source may be syngas generated by pyrolysis, torrefaction, or gasification. In other words, carbon in waste material may be recycled by pyrolysis, torrefaction, or gasification to generate syngas which is used as the substrate and/or C1-carbon source. The substrate and/or C1-carbon source may be a gas comprising methane, and in certain embodiments the substrate and/or C1-carbon source may be a non-waste gas.

[0084] In certain embodiments, the industrial process is selected from ferrous metal products manufacturing, such as a steel manufacturing, non-ferrous products manufacturing, petroleum refining, electric power production, carbon black production, paper and pulp manufacturing, ammonia production, methanol production, coke manufacturing, petrochemical production, carbohydrate fermentation, cement making, aerobic digestion, anerobic digestion, catalytic processes, natural gas extraction, cellulosic fermentation, oil extraction, industrial processing of geological reservoirs, processing fossil resources such as natural gas coal and oil, or any combination thereof. Examples of specific processing steps within an industrial process include catalyst regeneration, fluid catalyst cracking, and catalyst regeneration. Air separation and direct air capture are other suitable industrial processes. Specific examples in steel and ferroalloy manufacturing include blast furnace gas, basic oxygen furnace gas, coke oven gas, direct reduction of iron furnace top-gas, and residual gas from smelting iron. Other general examples include flue gas from fired boilers and fired heaters, such as naturel gas, oil, or coal fired boilers or heaters, and gas turbine exhaust. In these embodiments, the substrate and/or C1-carbon source may be captured from the industrial process before it is emitted into the atmosphere, using any known method

[0085] The substrate and/or C1-carbon source may be synthesis gas known as syngas, which may be obtained from reforming, partial oxidation, or gasification processes. Examples of gasification processes include gasification of

coal, gasification of refinery residues, gasification of petroleum coke, gasification of biomass, gasification of lignocellulosic material, gasification of waste wood, gasification of black liquor, gasification of municipal solid waste, gasification of municipal liquid waste, gasification of industrial solid waste, gasification of industrial liquid waste, gasification of refuse derived fuel, gasification of sewerage, gasification of sewerage sludge, gasification of sludge from wastewater treatment, gasification of biogas such as when biogas is added to enhance gasification of another material. Examples of reforming processes include, steam methane reforming, steam naphtha reforming, reforming of natural gas, reforming of biogas, reforming of landfill gas, naphtha reforming, and dry methane reforming. Examples of partial oxidation processes include thermal and catalytic partial oxidation processes, catalytic partial oxidation of natural gas, partial oxidation of hydrocarbons. Examples of municipal solid waste include tires, plastics, and fibers such as in shoes, apparel, and textiles. Municipal solid waste may be simply landfill-type waste and may be sorted or unsorted. Examples of biomass may include lignocellulosic material and microbial biomass. Lignocellulosic material may include agriculture waste and forest waste.

[0086] The substrate and/or C1-carbon source may be a gas stream comprising methane. Such a methane containing gas may be obtained from: fossil methane emissions such as during fracking, wastewater treatment, livestock, agriculture, and municipal solid waste landfills. It is also envisioned that the methane may be burned to produce electricity or heat and the C1 byproducts may be used as the substrate or carbon source.

[0087] In accordance with methods of the disclosure, the removal of CO₂ (or other gases) from a substrate stream will increase the CO concentration (or CO partial pressure in a gaseous substrate) and so increase the efficiency of fermentation reactions where CO is a substrate. Increasing CO partial pressure in a gaseous substrate increases CO mass transfer into a fermentation media. Furthermore, the composition of gas streams used to feed a fermentation reaction can have a significant impact on the efficiency and/or costs of that reaction. For example, O₂ may reduce the efficiency of an anaerobic fermentation process. In addition, processing of unwanted or unnecessary gases in stages of a fermentation process before or after fermentation can increase the burden on such stages (e.g. where the gas stream is compressed before entering a bioreactor, unnecessary energy may be used to compress gases that are not needed in the fermentation). Additionally or alternatively, the CO₂ component of a particular substrate stream may be of greater value if used in another reaction, including another fermentation reaction.

[0088] Furthermore, in accordance with methods of the disclosure, increasing the concentration of CO₂ in a stream, for example increasing CO₂ partial pressure in a gaseous stream, will increase the efficiency of processes that utilise CO₂, such as fermentation. Examples of processes that utilise CO₂, such as fermentations, are well known in the art. Examples of some such processes are described in detail in WO2006/108532 and are incorporated herein by reference.

[0089] Certain aspects of the disclosure generally relate to systems and methods for improving overall carbon capture in processes including microbial fermentation. In particular embodiments, the disclosure relates to the capture of CO₂ from substrate streams provided to a fermentation reaction.

Alternatively, or in addition, the disclosure relates to the capture of CO₂ from exhaust streams after the stream has exited the bioreactor. In particular embodiments of the disclosure, the substrate provided to the fermentation reaction comprises CO.

[0090] Furthermore, it may be desirable to remove and/or capture carbon containing components such as CO₂ and/or CH₄ to improve overall carbon capture of the process as described above. Additionally or alternatively, components of a particular gas may be of greater value if used elsewhere than in the fermentation reaction (e.g. H₂ is of considerable value for use as a fuel).

[0091] Certain aspects of the disclosure generally relate to systems and methods for improving the efficiency of processes of producing products by microbial fermentation of gases, particularly through use of at least one gas separation process of the gas stream used to feed the fermentation and/or the gas stream produced as a result of the fermentation. In one embodiment, a gas separator is configured to substantially separate at least one portion of the gas stream, wherein the portion comprises one or more components. For example, the gas separator may separate CO₂ from a gas stream comprising the following components: CO, CO₂, H₂, wherein the CO₂ may be passed to a CO₂ remover and the remainder of the gas stream (comprising CO and H₂) may be passed to a bioreactor.

[0092] The gas streams generated during industrial processes, such as batch processing of steel from iron in a steel mill, can be intermittent in nature, which may not be desirable when such gases are used for bio-conversion. In addition, the nature of the streams may be such that the gas composition varies in a cyclic nature during various phases of a particular industrial process. For example, in the steel making process, during periods when essentially oxygen free gases are generated, concentrations of CO are highest. Conversely, when the gases are largely CO free, significant levels of O₂ may be present. Many fermentation reactions require high CO concentrations that are essentially free of O₂, such as those involving anaerobic bacteria, particularly, carboxydophilic bacteria.

[0093] Certain aspects of the disclosure generally relate to systems and methods for improving the efficiency of processes for producing products by microbial fermentation of gases where the gas streams (or other sources such as dissolved gases and/or carbohydrates) used to feed the fermentation reaction are intermittent in nature. Particular embodiments of the disclosure are described in the context of the steel making industry, with carboxydophilic bacteria being used to produce acids and/or alcohols, particularly ethanol or butanol. Those skilled in the art will be aware, upon consideration of the instant disclosure, that the disclosure may be applied to different industries, as well as to various stages in the steel making process. Also, those skilled in the art will be aware, upon consideration of the instant disclosure, that the disclosure may be applied to other fermentation reactions, including those using the same or different micro-organisms. It is therefore intended that the scope of the disclosure is not limited to the particular embodiments and/or applications described but is instead to be understood to relate to any fermentation process in which at least one element used to feed the process is supplied in an intermittent manner, such as where waste gases from an industrial process are used, the gases being produced in an intermittent manner.

[0094] Certain aspects of the disclosure generally relate to systems and methods for improving the efficiency of processes for producing products and/or capturing carbon by microbial fermentation of substrate streams wherein the substrate streams are blended with additional streams to optimise the composition for microbial fermentation.

[0095] Surprisingly, when scrubber water from the basic oxygen steel making process or basic oxygen furnace off-gas stream from a steel mill is mixed with a standard microbial growth media in a fermentation reaction utilising gas containing carbon monoxide to produce ethanol in accordance with methods of the disclosure, growth of *Clostridium autoethanogenum*, as well as its ability to produce ethanol, is improved. This is particularly surprising, as the water was expected to contain contaminants that would be detrimental to the growth and survival of micro-organisms.

[0096] This finding has significant value in ameliorating or reducing waste from industrial processes, increasing the efficiency of fermentation reactions, reducing the level of media required to support fermentation reactions, and accordingly lowering operating costs. The disclosure can thus serve to decrease the level of acetate by-product formed during such fermentation reactions. This can be of benefit in situations where the acetate is not of use and would otherwise be discarded, increasing costs to the industry and posing an environmental problem.

[0097] Based on the results obtained, the disclosure enables the use of scrubber water as a primary feedstock for fermentation reactions. The scrubber water from industrial processes other than steel manufacturing may be used in a similar fashion. In addition, the disclosure is readily applicable to fermentation reactions which utilise gaseous substrates other than carbon monoxide as a source of carbon and energy, produce alcohols other than ethanol, produce hydrogen, and/or which utilise micro-organisms other than *Clostridium autoethanogenum*.

[0098] One or more features of each of any two or more of the aforementioned aspects may be combined and used within the same system with the attendant advantages thereof.

Definitions

[0099] Unless otherwise defined, the following terms as used throughout this specification are defined as follows:

[0100] The term “carbon capture” as used herein refers to the sequestration of carbon compounds including CO₂ and/or CO from a stream comprising CO₂ and/or CO and either:

[0101] converting the CO₂ and/or CO into products; or

[0102] converting the CO₂ and/or CO into substances suitable for long term storage; or

[0103] trapping the CO₂ and/or CO in substances suitable for long term storage;

or a combination of these processes.

[0104] The term “substrate comprising carbon monoxide” and like terms should be understood to include any substrate in which carbon monoxide is available to one or more strains of bacteria for growth and/or fermentation, for example.

[0105] “Gaseous substrates comprising carbon monoxide” include any gas which contains carbon monoxide. The gaseous substrate will typically contain a significant proportion of CO, preferably at least about 5% to about 100% CO by volume.

[0106] The term “bioreactor” includes a fermentation device consisting of one or more vessels and/or towers or

pipings arrangements, which includes the Continuous Stirred Tank Reactor (CSTR), Immobilized Cell Reactor (ICR), Trickle Bed Reactor (TBR), Bubble Column, Gas Lift Fermenter, Membrane Reactor such as Hollow Fibre Membrane Bioreactor (HFMBR), Static Mixer, or other vessel or other device suitable for gas-liquid contact.

[0107] The term “co-substrate” refers to a substance that, while not necessarily being the primary energy and material source for product synthesis, can be utilised for product synthesis when added to another substrate, such as the primary substrate.

[0108] The term “acid” as used herein includes both carboxylic acids and the associated carboxylate anion, such as the mixture of free acetic acid and acetate present in a fermentation broth as described herein. The ratio of molecular acid to carboxylate in the fermentation broth is dependent upon the pH of the system. In addition, the term “acetate” includes both acetate salt alone and a mixture of molecular or free acetic acid and acetate salt, such as the mixture of acetate salt and free acetic acid present in a fermentation broth as described herein.

[0109] The term “limiting concentration” means an initial concentration of a given component in a microbial fermentation medium that is sufficiently low to ensure that it will be depleted at some stage in the fermentation.

[0110] The term “intermittent stream” means not only streams which are not continuously available, but also streams which do not continuously have a desired composition.

[0111] The term “scrubber water” refers to water or other liquids resulting from the cleaning of gas streams generated during industrial processes such as ferrous metal products manufacturing, non-ferrous products manufacturing, petroleum refining processes, gasification of coal, gasification of biomass, electric power production, carbon black production, and coke manufacturing.

[0112] The term “directly”, as used in relation to the passing of industrial off or waste gases to a bioreactor, is used to mean that no or minimal processing or treatment steps, such as cooling and particulate removal are performed on the gases prior to them entering the bioreactor (note: an oxygen removal step may be required for anaerobic fermentation).

[0113] The term “desired composition” is used to refer to the desired level and types of components in a substance, such as, for example, of a gas stream. More particularly, a gas is considered to have a “desired composition” if it contains a particular component (e.g. CO and/or CO₂) and/or contains a particular component at a particular level and/or does not contain a particular component (e.g. a contaminant harmful to the micro-organisms) and/or does not contain a particular component at a particular level. More than one component may be considered when determining whether a gas stream has a desired composition.

[0114] The term “stream” is used to refer to a flow of material into, through and away from one or more stages of a process, for example, the material that is fed to a bioreactor and/or an optional CO₂ remover. The composition of the stream may vary as it passes through particular stages. For example, as a stream passes through the bioreactor, the CO content of the stream may decrease, while the CO₂ content may increase. Similarly, as the stream passes through the CO₂ remover stage, the CO₂ content will decrease.

[0115] Unless the context requires otherwise, the phrases “fermenting”, “fermentation process” or “fermentation reaction” and the like, as used herein, are intended to encompass both the growth phase and product biosynthesis phase of the process.

[0116] The terms “increasing the efficiency”, “increased efficiency” and the like, when used in relation to a fermentation process, include, but are not limited to, increasing one or more of: the rate of growth of micro-organisms in the fermentation, the volume or mass of desired product (such as alcohols) produced per volume or mass of substrate (such as carbon monoxide) consumed, the rate of production or level of production of the desired product, and the relative proportion of the desired product produced compared with other by-products of the fermentation, and further may reflect the value (which may be positive or negative) of any by-products generated during the process.

[0117] The term “commodity chemical” is used to refer to a group of chemicals that are made on a very large scale to satisfy global markets, where commodity chemicals are differentiated primarily by the bulk of their manufacture.

[0118] In one aspect, the disclosure relates to systems and methods for increasing overall carbon capture in processes of producing products by microbial fermentation of substrates, said systems and methods including at least one CO₂ removal process performed on substrates and/or streams before (i.e., upstream) or after (i.e., downstream) the fermentation reaction. In certain embodiments of the disclosure, the substrate comprises CO. Typically, the substrate will be gaseous; however, the disclosure is not limited thereto.

[0119] In one aspect, the disclosure relates to systems and methods for increasing the efficiency of processes of producing products by microbial fermentation of gases, said systems and methods including at least one gas separation process performed on the gases before (i.e., upstream) or after (i.e., downstream) the fermentation reaction. As noted above, in particular embodiments, the substrate gas used in microbial fermentation comprises CO; however, the disclosure is not limited thereto.

[0120] In another particular aspect the disclosure relates to systems and methods for increasing the efficiency of processes of producing products by microbial fermentation of gases, particularly where the supply of the gases is intermittent in nature. In particular embodiments, the substrate gas used in microbial fermentation comprises CO; however, the disclosure is not limited thereto.

[0121] The disclosure further provides methods and systems for the production of alcohol using microbial fermentation. These methods and systems involve the use of waste gases from an industrial process, such as the manufacture of steel, in the fermentation reaction, wherein no or only minimal additional processing steps are performed on the gases prior to such use. In certain embodiments, waste gases from one or more industrial processes and/or alternative sources are combined or blended to provide a stream with a desirable or optimised composition to the fermentation reaction.

[0122] The disclosure also provides methods and systems for optimising the composition of a substrate stream comprising CO derived, at least in part, from an industrial process such as the manufacture of steel.

[0123] The disclosure also provides methods for the production of alcohol using microbial fermentation and meth-

ods for increasing the efficiency of alcohol production using microbial fermentation. In one embodiment, these methods involve utilising scrubber water from an industrial process in the fermentation reaction.

[0124] While certain embodiments of the disclosure, namely those that include the production of ethanol by anaerobic fermentation using CO as the primary substrate, are readily recognized as being valuable improvements to technology of great interest today, it should be appreciated that the disclosure is applicable to production of alternative products such as other alcohols and the use of alternative substrates, particularly gaseous substrates, as will be known by persons of ordinary skill in the art to which the disclosure relates upon consideration of the instant disclosure. For example, gaseous substrates containing carbon dioxide and hydrogen may be used in particular embodiments of the disclosure. Further, the disclosure may be applicable to fermentations to produce acetate, butyrate, propionate, caproate, ethanol, propanol, and butanol, and hydrogen. By way of example, these products may be produced by fermentation using microbes from the genus *Moorella*, *Clostridia*, *Ruminococcus*, *Acetobacterium*, *Eubacterium*, *Butyribacterium*, *Oxobacter*, *Methanosarcina*, *Methanosarcina*, and *Desulfotomaculum*.

[0125] Certain embodiments of the disclosure are adapted to use gas streams produced by one or more industrial processes. Such processes include steel making processes, particularly processes which produce a gas stream having a high CO content or a CO content above a predetermined level (i.e., 5%). According to such embodiments, carboxydophilic bacteria are preferably used to produce acids and/or alcohols, particularly ethanol or butanol, within one or more bioreactors. Those skilled in the art will be aware upon consideration of the instant disclosure that the disclosure may be applied to various industries or waste gas streams, including those of vehicles with an internal combustion engine. Also, those skilled in the art will be aware upon consideration of the instant disclosure that the disclosure may be applied to other fermentation reactions including those using the same or different micro-organisms. It is therefore intended that the scope of the disclosure is not limited to the particular embodiments and/or applications described but is instead to be understood in a broader sense; for example, the source of the gas stream is not limiting, other than that at least a component thereof is usable to feed a fermentation reaction. The disclosure has particular applicability to improving the overall carbon capture and/or production of ethanol and other alcohols from gaseous substrates such as automobile exhaust gases and high volume CO-containing industrial flue gases.

Fermentation

[0126] Processes for the production of ethanol and other alcohols from gaseous substrates (such as those described in the background section above) are known. Exemplary processes include those described for example in WO 2007/117157 and WO 2008/115080, as well as U.S. Pat. Nos. 6,340,581, 6,136,577, 5,593,886, 5,807,722 and 5,821,111, each of which is incorporated herein by reference.

[0127] A number of anaerobic bacteria are known to be capable of carrying out the fermentation of CO to alcohols, including n-butanol and ethanol, and acetic acid, and are suitable for use in the process of the present disclosure. Examples of such bacteria that are suitable for use in the

disclosure include those of the genus *Clostridium*, such as strains of *Clostridium ljungdahlii*, including those described in WO 00/68407, EP 117309, U.S. Pat. Nos. 5,173,429, 5,593,886, and 6,368,819, WO 98/00558 and WO 02/08438, *Clostridium carboxydivorans* (Liou et al., International Journal of Systematic and Evolutionary Microbiology 33: pp 2085-2091) and *Clostridium autoethanogenum* (Abrini et al., Archives of Microbiology 161: pp 345-351). Other suitable bacteria include those of the genus *Moorella*, including *Moorella* sp HUC22-1 (Sakai et al., Biotechnology Letters 29: pp 1607-1612), and those of the genus *Carboxydotherrmus* (Svetlichny, V. A., et al. (1991), Systematic and Applied Microbiology 14: 254-260). The disclosures of each of these publications are incorporated herein by reference. In addition, other carboxydophilic anaerobic bacteria can be used in the processes of the disclosure by a person of skill in the art. It will also be appreciated upon consideration of the instant disclosure that a mixed culture of two or more bacteria may be used in processes of the present disclosure.

[0128] Culturing of the bacteria used in a method of the disclosure may be conducted using any number of processes known in the art for culturing and fermenting substrates using anaerobic bacteria. Exemplary techniques are provided in the “Examples” section below. By way of further example, those processes generally described in the following articles using gaseous substrates for fermentation may be utilised: (i) K. T. Klasson, et al. (1991). Bioreactors for synthesis gas fermentations resources. Conservation and Recycling, 5; 145-165; (ii) K. T. Klasson, et al. (1991). Bioreactor design for synthesis gas fermentations. Fuel. 70. 605-614; (iii) K. T. Klasson, et al. (1992). Bioconversion of synthesis gas into liquid or gaseous fuels. Enzyme and Microbial Technology. 14; 602-608; (iv) J. L. Vega, et al. (1989). Study of Gaseous Substrate Fermentation: Carbon Monoxide Conversion to Acetate. 2. Continuous Culture. Biotech. Bioeng. 34. 6. 785-793; (vi) J. L. Vega, et al. (1989). Study of gaseous substrate fermentations: Carbon monoxide conversion to acetate. 1. Batch culture. Biotechnology and Bioengineering. 34. 6. 774-784; (vii) J. L. Vega, et al. (1990). Design of Bioreactors for Coal Synthesis Gas Fermentations. Resources, Conservation and Recycling. 3. 149-160; all of which are incorporated herein by reference.

[0129] One exemplary micro-organism suitable for use in the present disclosure is *Clostridium autoethanogenum*. In one embodiment, the *Clostridium autoethanogenum* is a *Clostridium autoethanogenum* having the identifying characteristics of the strain deposited at the German Resource Centre for Biological Material (DSMZ) under the identifying deposit number 19630. In another embodiment, the *Clostridium autoethanogenum* is a *Clostridium autoethanogenum* having the identifying characteristics of DSMZ deposit number DSMZ 10061.

[0130] The fermentation may be carried out in any suitable bioreactor. In some embodiments of the disclosure, the bioreactor may comprise a first, growth reactor in which the micro-organisms are cultured, and a second, fermentation reactor, to which fermentation broth from the growth reactor is fed and in which most of the fermentation product (e.g. ethanol and acetate) is produced.

[0131] According to various embodiments of the disclosure, the carbon source for the fermentation reaction is a gaseous substrate containing CO. The gaseous substrate may be a CO-containing waste gas obtained as a by-product of an

industrial process, or from some other source such as from automobile exhaust fumes. In certain embodiments, the industrial process is selected from the group consisting of ferrous metal products manufacturing, such as is conducted in a steel mill, non-ferrous products manufacturing, petroleum refining processes, gasification of coal, electric power production, carbon black production, ammonia production, methanol production and coke manufacturing. In these embodiments, the CO-containing gas may be captured from the industrial process before it is emitted into the atmosphere, using any convenient method. Depending on the composition of the gaseous CO-containing substrate, it may also be desirable to treat it to remove any undesired impurities, such as dust particles before introducing it to the fermentation. For example, the gaseous substrate may be filtered or scrubbed using known methods.

[0132] The CO-containing gaseous substrate will ideally contain a significant proportion of CO, such as at least 5% to about 100% CO by volume, or from 20% to 95% CO by volume, or from 40% to 95% CO by volume, or from 60% to 90% CO by volume or from 70% to 90% CO by volume. Gaseous substrates having lower concentrations of CO, such as 6%, may also be appropriate, particularly when H₂ and CO₂ are also present.

[0133] While it is not necessary for the gaseous substrate to contain any hydrogen, the presence of hydrogen will generally not be detrimental to product formation in accordance with methods of the disclosure. However, in certain embodiments of the disclosure, the gaseous substrate is substantially hydrogen free (less than 1%). The gaseous substrate may also contain some CO₂, such as about 1% to about 30% by volume, or such as about 5% to about 10% CO₂.

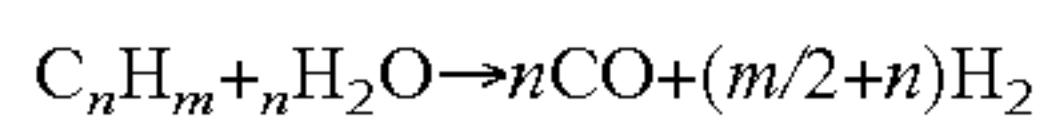
[0134] As noted previously, the presence of hydrogen in the substrate stream can lead to an improvement in efficiency of overall carbon capture and/or ethanol productivity. For example, WO0208438 describes the production of ethanol using gas streams of various compositions. In one preferred embodiment, a substrate stream comprising 63% H₂, 32% CO and 5% CH₄ was provided to a culture of *C. ljungdahlii* in a bioreactor to promote microbial growth and ethanol production. When the culture reached a steady state and microbial growth was no longer the main objective, the substrate stream was switched to 15.8% H₂, 36.5% CO, 38.4% N₂ and 9.3% CO₂ in order to provide CO in a slight excess and promote ethanol production. This document also describes gas streams with higher and lower CO and H₂ concentrations.

[0135] Accordingly, it may be necessary to alter the composition of the substrate stream in order to improve alcohol production and/or overall carbon capture. Additionally or alternatively, the composition may be altered (i.e. CO, CO₂ and/or H₂ levels adjusted) to optimise the efficiency of the fermentation reaction and ultimately improve alcohol production and/or overall carbon capture.

[0136] In some embodiments, the CO-containing gaseous substrate may be sourced from the gasification of organic matter such as methane, ethane, propane, coal, natural gas, crude oil, low value residues from oil refinery (including petroleum coke or petcoke), solid municipal waste or biomass. Biomass includes by-products obtained during the extraction and processing of foodstuffs, such as sugar from sugarcane, or starch from maize or grains, or non-food biomass waste generated by the forestry industry. Any of

these carbonaceous materials can be gasified, i.e. partially combusted with oxygen, to produce synthesis gas (syngas comprising significant amounts of H₂ and CO). Gasification processes typically produce a synthesis gas with a molar ratio of H₂ to CO of about 0.4:1 to 1.2:1, together with lesser amounts of CO₂, H₂S, methane and other inert substances. The ratio of the gas produced can be varied by means known in the art and are described in detail in WO200701616. However, by way of example, the following gasifier conditions can be altered to adjust the CO:H₂ product ratio: feedstock composition (particularly C:H ratio), operating pressure, temperature profile (influencing quench of product mix) and oxidant employed (air, oxygen enriched air, pure O₂ or steam; wherein steam tends to result in higher CO:H₂ ratios). Accordingly, the operating conditions of the gasifier can be adjusted to provide a substrate stream with a desirable composition for fermentation or blending with one or more other streams to provide an optimised or desirable composition for increased alcohol productivity and/or overall carbon capture in a fermentation process.

[0137] In other embodiments, the substrate comprising CO can be derived from the steam reforming of hydrocarbons. Hydrocarbons, such as natural gas hydrocarbons can be reformed at high temperature to yield CO and H₂ according to the following:



[0138] By way of example, steam methane reforming involves reacting steam with methane to produce CO and H₂ at elevated temperature (700-1100° C.) in the presence of a nickel catalyst. The resulting stream (comprising 1 mol CO and 3 mol H₂ for every mol CH₄ converted) can be passed directly to the fermenter or blended with a substrate stream from another source to increase ethanol productivity and/or overall carbon capture in a fermentation process. Alcohols such as methanol can also be reformed to produce CO₂ and H₂ that may be used in a similar manner.

[0139] In another embodiment, the substrate comprising CO is derived from the steel manufacturing process. In the steel making process, iron ore is crushed and pulverised, subjected to pre-treatments such as sintering or pelletizing, and then passed to a blast furnace (BF), where it is smelted. In the smelting process, coke serves as the source of carbon, which works as a reducing agent to reduce the iron ore. Coke acts as the heat source for heating and melting the materials. The hot metal is decarburised in a basic oxygen furnace (BOF) by injecting a high-velocity jet of pure oxygen against the surface of the hot metal. The oxygen reacts directly with carbon in the hot metal to produce carbon monoxide (CO). Thus, a gas stream with a high CO content is exhausted from the BOF. According to certain embodiments of the disclosure, this stream is used to feed one or more fermentation reactions. However, as would be apparent to one of skill in the art, CO may be produced elsewhere within the steel making process, and according to various embodiments of the disclosure, such alternative sources may be used instead of or in combination with exhaust gases from the BOF. Depending on the source (i.e., the particular stage within the steel making process), the CO content of the gases exhausted thereby may vary. Also, there may be periods when there are breaks in one or more of such streams, particularly in batch processing plants.

[0140] Typically, streams exhausted from the steel mill decarburisation process comprise a high concentration of

CO and low concentrations of H₂. While such streams can be directly passed to the bioreactor with little or no further treatment, it may be desirable to optimise the composition of the substrate stream in order to achieve higher efficiency of alcohol production and/or overall carbon capture. For example, the concentration of H₂ in the substrate stream may be increased before the stream is passed to the bioreactor.

[0141] According to particular embodiments of the disclosure, streams from two or more sources can be combined and/or blended to produce a desirable and/or optimised substrate stream. For example, a stream comprising a high concentration of CO, such as the exhaust from a steel mill converter, can be combined with a stream comprising high concentrations of H₂, such as the off-gas from a steel mill coke oven.

[0142] An early stage of the steel making process typically involves the reduction of iron ore using coke. Coke is a solid carbon fuel source used to melt and reduce iron ore and is typically produced on-site at a steel mill. In the coke-making process, bituminous coal is fed into a series of ovens, which are sealed and heated at high temperatures in the absence of oxygen, typically in cycles lasting 14 to 36 hours. The solid carbon remaining in the oven is coke. It is taken to the quench tower, where it is cooled with a watery spray or by circulating an inert gas (nitrogen), then screened and sent to the blast furnace.

[0143] The volatile compounds produced during this process are generally processed to remove tar, ammonia, naphthalene, phenol, light oils and sulphur before the gas is used as fuel to heat ovens. Gas produced as a result of coke production typically has a high H₂ content (typical composition: 55% H₂, 25% CH₄, 6% CO, 3% N₂, 2% other hydrocarbons). As such, at least a portion of the coke oven gas may be diverted to the fermentation process for blending with a stream comprising CO, to improve alcohol productivity and/or overall carbon capture. It may be necessary to treat the coke oven gas prior to passing it to the fermenter to remove by-products that may be toxic to the culture.

[0144] Alternatively or additionally, an intermittent stream comprising CO, such as an exhaust stream from the converter, may be combined with and/or blended with a substantially continuous stream comprising CO and optionally H₂, such as syngas produced in a gasification process as described previously. In certain embodiments, this would maintain the provision of a substantially continuous substrate stream to the bioreactor. In a particular embodiment, the stream produced by the gasifier may be increased and/or decreased in accordance with the intermittent production of CO from an industrial source in order to maintain a substantially continuous substrate stream with a desirable or optimised composition. In another embodiment, the gasifier conditions may be altered as described previously in order to increase or decrease the CO:H₂ ratio, in accordance with the intermittent production of CO from an industrial source, in order to maintain a substantially continuous substrate stream with a desirable or optimised CO and H₂ composition.

[0145] Typically, the substrate streams used in the disclosure will be gaseous; however, the disclosure is not limited thereto. For example, the carbon monoxide may be provided to a bioreactor in a liquid. For example, a liquid may be saturated with a carbon monoxide containing gas and then that liquid added to a bioreactor. This may be achieved using standard methodology. By way of example, a microbubble dispersion generator (Hensirisak et al., Scale-up of

microbubble dispersion generator for aerobic fermentation; Applied Biochemistry and Biotechnology Volume 101, Number 3, October, 2002) could be used for this purpose.

[0146] It will be appreciated that for growth of the bacteria and CO-to-ethanol fermentation to occur, in addition to the CO-containing substrate gas, a suitable liquid nutrient medium will need to be fed to the bioreactor. A nutrient medium will contain vitamins and minerals sufficient to permit growth of the micro-organism used. Anaerobic media suitable for the fermentation of ethanol using CO as the sole carbon source are known in the art. For example, suitable media are described in U.S. Pat. Nos. 5,173,429 and 5,593,886 and WO 02/08438, WO2007/115157 and WO2008/115080, referred to above. The “Examples” herein provide other exemplary media.

[0147] The fermentation should desirably be carried out under appropriate conditions for the desired fermentation to occur (e.g. CO-to-alcohol). Reaction conditions that should be considered include pressure, temperature, gas flow rate, liquid flow rate, media pH, media redox potential, agitation rate (if using a continuous stirred tank reactor), inoculum level, maximum gas substrate concentrations to ensure that CO in the liquid phase does not become limiting, and maximum product concentrations to avoid product inhibition.

[0148] The optimum reaction conditions will depend partly on the particular micro-organism used. However, in general, it may be preferable that the fermentation be performed at a pressure higher than ambient pressure. Operating at increased pressures allows a significant increase in the rate of CO transfer from the gas phase to the liquid phase where it can be taken up by the micro-organism as a carbon source for the production of ethanol. This in turn means that the retention time (defined as the liquid volume in the bioreactor divided by the input gas flow rate) can be reduced when bioreactors are maintained at elevated pressure rather than atmospheric pressure.

[0149] Also, because a given CO-to-ethanol conversion rate is in part a function of the substrate retention time, and achieving a desired retention time in turn dictates the required volume of a bioreactor, the use of pressurized systems can greatly reduce the volume of the bioreactor required, and consequently the capital cost of the fermentation equipment. According to examples given in U.S. Pat. No. 5,593,886, reactor volume can be reduced in linear proportion to increases in reactor operating pressure, i.e. bioreactors operated at 10 atmospheres of pressure need only be one tenth the volume of those operated at 1 atmosphere of pressure.

[0150] The benefits of conducting a gas-to-ethanol fermentation at elevated pressures have also been described elsewhere. For example, WO 02/08438 describes gas-to-ethanol fermentations performed under pressures of 30 psig and 75 psig, giving ethanol productivities of 150 g/l/day and 369 g/l/day respectively. However, example fermentations performed using similar media and input gas compositions at atmospheric pressure were found to produce between 10 and 20 times less ethanol per litre per day.

[0151] It is also desirable that the rate of introduction of the CO-containing gaseous substrate is such as to ensure that the concentration of CO in the liquid phase does not become limiting. This is because a consequence of CO-limited conditions may be that the ethanol product is consumed by the culture.

Product Recovery

[0152] The products of the fermentation reaction can be recovered using known methods. Exemplary methods include those described in WO2007/117157, WO2008/115080 and U.S. Pat. Nos. 6,340,581, 6,136,577, 5,593,886, 5,807,722 and 5,821,111. However, briefly and by way of example only, ethanol may be recovered from the fermentation broth by methods such as fractional distillation or evaporation, and extractive fermentation.

[0153] Distillation of ethanol from a fermentation broth yields an azeotropic mixture of ethanol and water (i.e. 95% ethanol and 5% water). Anhydrous ethanol can subsequently be obtained through the use of molecular sieve ethanol dehydration technology, which is also well known in the art.

[0154] Extractive fermentation procedures involve the use of a water-miscible solvent that presents a low toxicity risk to the fermentation organism, to recover the ethanol from the dilute fermentation broth. For example, oleyl alcohol is a solvent that may be used in this type of extraction process. In this process, oleyl alcohol is continuously introduced into a fermenter, whereupon this solvent rises forming a layer at the top of the fermenter which is continuously extracted and fed through a centrifuge. Water and cells are then readily separated from the oleyl alcohol and returned to the fermenter while the ethanol-laden solvent is fed into a flash vaporization unit. Most of the ethanol is vaporized and condensed while the non-volatile oleyl alcohol is recovered for re-use in the fermentation.

[0155] Acetate may also be recovered from the fermentation broth using methods known in the art. For example, an adsorption system involving an activated charcoal filter may be used. In this case, microbial cells are typically first removed from the fermentation broth using a suitable separation method. Numerous filtration-based methods of generating a cell free fermentation broth for product recovery are known in the art. The cell free ethanol—and acetate—containing permeate is then passed through a column containing activated charcoal to adsorb the acetate. Acetate in the acid form (acetic acid) rather than the salt (acetate) form is more readily adsorbed by activated charcoal. It is therefore preferred that the pH of the fermentation broth be reduced to less than about 3 before it is passed through the activated charcoal column, to convert the majority of the acetate to the acetic acid form.

[0156] Acetic acid adsorbed to the activated charcoal may be recovered by elution using methods known in the art. For example, ethanol may be used to elute the bound acetate. In certain embodiments, ethanol produced by the fermentation process itself may be used to elute the acetate. Because the boiling point of ethanol is 78.8° C. and that of acetic acid is 107° C., ethanol and acetate can readily be separated from each other using a volatility-based method such as distillation.

[0157] Other methods for recovering acetate from a fermentation broth are known in the art and may be used in processes of the present disclosure. For example, U.S. Pat. Nos. 6,368,819 and 6,753,170 describe a solvent and cosolvent system that can be used for extraction of acetic acid from fermentation broths. As with the oleyl alcohol-based system described above for the extractive fermentation of ethanol, the systems described in U.S. Pat. Nos. 6,368,819 and 6,753,170 describe a water immiscible solvent/co-solvent that can be mixed with the fermentation broth in either the presence or absence of the fermented micro-organisms to

extract the acetic acid. The solvent/co-solvent containing the acetic acid is then separated from the broth by distillation. A second distillation step may then be used to purify the acetic acid from the solvent/co-solvent system.

[0158] The products of the fermentation reaction (for example ethanol and acetate) may be recovered from the fermentation broth by continuously removing a portion of the broth from the fermentation bioreactor, separating microbial cells from the broth (conveniently by filtration), and recovering one or more products from the broth simultaneously or sequentially. Ethanol may conveniently be recovered by distillation, and acetate may be recovered by adsorption on activated charcoal, using the methods described above. The separated microbial cells can be returned to the fermentation bioreactor. The cell free permeate remaining after the ethanol and acetate have been removed can also be returned to the fermentation bioreactor. Additional nutrients (such as B vitamins) may be added to the cell free permeate to replenish the nutrient medium before it is returned to the bioreactor. Also, if the pH of the broth was adjusted as described above to enhance adsorption of acetic acid to the activated charcoal, the pH should be re-adjusted to a similar pH to that of the broth in the fermentation bioreactor, before being returned to the bioreactor.

CO₂ Removal

[0159] According to certain embodiments of the disclosure, the system used for CO₂ removal includes a means for selectively separating CO₂ from a mixed stream and a means for converting the CO₂ to products and/or preparing the CO₂ for storage or further use. Alternatively, the process includes a means for converting the CO₂ in a stream directly to products and/or substances suitable for storage or further use.

[0160] In one embodiment, CO₂ is selectively separated from a mixed gas stream using any separation means known in the art such as the exemplary methods provided below. Other methods of CO₂ separation that may be used in embodiments of the disclosure include extraction with a metal oxide, such as CaO, and use of porous carbon or selective solvent extraction such as amine extraction.

[0161] Amines such as aqueous monoethanolamine (MEA), diglycolamine (DGA), diethanolamine (DEA), diisopropanolamine (DIPA) and methyldiethanolamine (MDEA) are widely used industrially for removing CO₂ and hydrogen sulfide from natural gas streams and refinery process streams.

[0162] The CO₂ separated in such processes may be permanently stored. Many examples of permanent CO₂ storage are known in the art, such as geological storage (geo-sequestration), ocean storage and mineral storage (e.g. conversion to metal carbonates).

[0163] Geological storage involves injecting carbon dioxide, generally in supercritical form, directly into underground geological formations. Oil fields, gas fields, saline formations, unmineable coal seams, and saline-filled basalt formations have been suggested as storage sites. Various physical (e.g., highly impermeable caprock) and geochemical trapping mechanisms can be used to prevent the CO₂ from escaping to the surface. For well-selected, designed and managed geological storage sites, the Intergovernmental Panel on Climate Change estimates that CO₂ could be

trapped for millions of years, and the sites are likely to retain over 99% of the injected CO₂ over 1,000 years.

[0164] Several options for ocean storage have been proposed: (i) 'dissolution' injection of CO₂ by ship or pipeline into the water at depths of 1000 m or more, and the CO₂ subsequently dissolves; (ii) 'lake' deposition of CO₂ directly onto the sea floor at depths greater than 3000 m, where CO₂ is denser than water and is expected to form a 'lake' that would delay dissolution of CO₂ into the environment; (iii) conversion of the CO₂ to bicarbonates (using limestone); and (iv) storage of the CO₂ in solid clathrate hydrates already existing on the ocean floor, or use in growing more solid clathrate.

[0165] In mineral storage, CO₂ is exothermically reacted with abundantly available metal oxides to produce stable carbonates. This process occurs naturally over many years and is responsible for much of the surface limestone. The reaction rate can be made faster, for example by reacting at higher temperatures and/or pressures, or by pre-treatment of the minerals, although this method can require additional energy.

[0166] Alternatively, the separated CO₂ may be used to make products, such as direct or indirect conversion to hydrocarbons. A well-known process to produce a hydrocarbon is the process for making methanol from CO₂ and H₂. Catalytic or electrochemical dissociation of water to produce oxygen and hydrogen ions, wherein the hydrogen ions can be used to convert CO₂ to hydrocarbons is also known in the art. If CO₂ is heated to 2400° C., it splits into carbon monoxide and oxygen.

[0167] The Fischer-Tropsch process can then be used to convert the CO into hydrocarbons. In such processes, the CO may be returned to the fermentation process. By way of example, the required temperature can be achieved by using a chamber containing a mirror to focus sunlight on the gas.

[0168] Alternatively, the separated CO₂ may be used in further fermentation(s) to produce products. Those skilled in the art will appreciate there are many examples of microbial fermentation reactions that convert CO₂ into products. For example, CO₂ may be converted into methane by anaerobic fermentation using methanogenic microbes. Examples of this and other related fermentation processes are disclosed in the aforementioned WO2006/108532. Further examples of fermentation reactions using CO₂ to produce products are provided in the aforementioned WO2007/117157 and WO2008/115080.

[0169] CO₂ is also a desirable feedstock in syngas production. CO₂ can be supplied to the reformer (gasifier) to reduce methane consumption and improve/increase the H₂:CO ratio. Accordingly, in one embodiment, at least a portion of the separated CO₂ may be supplied to a gasifier integrated into the fermentation process.

[0170] In another embodiment of the disclosure, the separated CO₂ may be converted to products such as concrete cement. In a process mimicking marine cement produced by coral when making their shells and reefs, magnesium and/or calcium can be combined with CO₂ to produce carbonates.

[0171] CO₂ is also readily absorbed by algae in a photosynthetic process, which can be used to capture carbon from waste streams. Algae rapidly grow in the presence of CO₂ and sunlight and can be harvested and converted into products such as biodiesel and/or alcohol.

[0172] Alternatively, the CO₂ may be directly captured from a stream without the need of an additional separation

step. For example, in a particular embodiment, a stream, preferably a gaseous stream, comprising CO₂ may be passed through a second fermentation process to convert CO₂ to products.

Gas Separation

[0173] According to certain embodiments of the disclosure, the process used for gas separation comprises one or more steps of cryogenic fractionation, molecular sieving, adsorption, pressure swing adsorption, or absorption. Whatever process is used, gas separation can be performed to isolate at least a portion of one or more of the following components: H₂, O₂, CO₂ and CO, from the gas stream. Additionally or alternatively, gas separation according to embodiments of the disclosure may be used to remove one or more portions from the gas stream (e.g. N₂, O₂) so that the remainder may be more efficiently used, such as in the bioreactor.

[0174] Adsorption is the accumulation of gases, liquids or solutes on the surface of a solid or liquid. Absorption is the process by which one substance, such as a solid or liquid, takes up another substance, such as a liquid or gas, through minute pores or spaces between its molecules.

[0175] Pressure swing adsorption (PSA) is an adiabatic process which may be used for the purification of gases to remove accompanying impurities by adsorption through suitable adsorbents in fixed beds contained in pressure vessels under high pressure. Regeneration of adsorbents is accomplished by countercurrent depressurization and by purging at low pressure with previously recovered near product quality gas. To obtain a continuous flow of product, preferably at least two adsorbers are provided, such that at least one adsorber is receiving a gas stream (such as a waste/exhaust/biogas gas stream) and actually produces a product of desired purity. Simultaneously, the subsequent steps of depressurization, purging and repressurization back to the adsorption pressure are executed by the other adsorber (s). Common adsorbents may readily be selected by one of skill in the art dependent on the type of impurity to be adsorbed and removed. Suitable adsorbents include zeolitic molecular sieves, activated carbon, silica gel or activated alumina. Combinations of adsorbent beds may be used on top of one another, thereby dividing the adsorber contents into a number of distinct zones. Pressure swing adsorption involves a pendulating swing in parameters such as pressure, temperature, flow and composition of gaseous and adsorbed phase.

[0176] Purification or separation of gases using PSA normally takes place at near ambient feed gas temperatures, whereby the components to be removed are selectively adsorbed. Adsorption should ideally be sufficiently reversible to enable regeneration of adsorbents at similar ambient temperature. PSA may be used for treatment and/or purification of most common gases including CO, CO₂ and H₂. Examples of Pressure Swing Adsorption techniques are described in detail in Ruthven, Douglas M. et al., 1993 Pressure Swing Adsorption, John Wiley and Sons.

[0177] A molecular sieve is a material containing tiny pores of a precise and uniform size that is used as an adsorbent for gases and liquids. Molecules that are small enough to pass through the pores are adsorbed while larger molecules are not. A molecular sieve is similar to a common filter but operates on a molecular level. Molecular sieves often consist of aluminosilicate minerals, clays, porous

glasses, microporous charcoals, zeolites, active carbons, or synthetic compounds that have open structures through which small molecules, such as nitrogen and water, can diffuse. Methods for regeneration of molecular sieves include pressure changing (e.g. in oxygen concentrators) and heating and purging with a carrier gas.

[0178] Membranes may be used, for example, to separate hydrogen from gases like nitrogen and methane, to recover hydrogen, to separate methane from biogas, or to remove water vapour, CO₂, H₂S or volatile organic liquids. Different membranes, including porous and non-porous membranes, may be selected to serve the desired purpose as would be apparent to one of skill in the art upon consideration of the instant disclosure. For example, a Palladium membrane permits transport solely of H₂. In a particular embodiment, CO₂ can be separated from a stream, using a CO₂ permeable membrane. The CO₂ separated from the stream can be passed to a CO₂ remover such as the gasifier discussed previously.

[0179] Cryogenic fractionation involves compressing the gas stream and cooling it to a temperature low enough to allow separation by distillation. It may be used, for example, to remove CO₂. Certain components (e.g. water) are typically removed from the stream prior to performing cryogenic fractionation.

[0180] The same techniques can also be used to remove oxygen from a gaseous stream to produce CO and/or CO₂-rich anaerobic streams. In addition, oxygen can be removed biologically, by, for instance, passing the combustion exhaust gas into a sealed fermenter containing facultative aerobic micro-organisms, a reduced carbon substrate, and the necessary nutrients for the micro-organisms. The facultative aerobic micro-organisms can consume oxygen to create CO and/or CO₂-rich anaerobic streams.

[0181] Alternative methods for separating or removing O₂ from a gaseous stream are also well known in the art. However, by way of example, oxygen can be simply reduced and/or removed using hot copper or a catalytic converter.

[0182] Tailoring the gas separation process to a particular source of gas can make an otherwise non-commercially viable bioconversion process commercially viable. For example, with appropriate separation of CO from a car exhaust stream, a usable energy source may be obtained from the stream and unwanted gas emissions can be reduced. According to one embodiment of the disclosure, the gaseous substrate comprises Syngas containing CO and H₂, and gas separation is performed to remove hydrogen from the stream so that it may be isolated and used as a fuel outside of the fermentation process. The CO may be used to feed the fermentation reaction.

Intermittent Gas Streams

[0183] According to various aspects of the disclosure, the fermentation substrate is derived from an industrial source. Typically, substrates derived from industrial sources are gaseous and such gases may vary in composition and/or pressure and in some instances may be intermittent in nature. In certain embodiments, the disclosure provides means to improve or “smooth” supply of a gaseous substrate to a bioreactor for fermentation to produce products, particularly in instances where the substrate supply is intermittent or non-continuous in nature. Any known means for improving continuity or “smoothing” of a gaseous substrate stream may be used; however, particular embodiments of the disclosure

include processes or systems that include at least one buffering means adapted to receive an intermittent substrate stream, and to deliver a substantially continuous substrate stream to a bioreactor.

[0184] In particular embodiments, the buffering means includes a storage tank adapted to receive intermittent gas streams. The intermittent stream may be compressed prior to entering the storage tank; alternatively, the storage tank may be configured to expand as it receives the substrate stream. For example, the buffer storage tank may include a ‘floating roof’ adapted to rise and fall to accommodate a gaseous substrate. Floating roof type storage tanks are known in the art, such as those used to accommodate supply and demand fluctuations in gas supply. The storage tank may be adapted to supply a substantially continuous substrate stream to a fermentation bioreactor, and as such may include a means for controlling the rate of flow of the stream exiting the tank.

[0185] In such embodiments, the storage tank serves as a substrate reservoir. However, according to an alternative embodiment, the buffer storage tank may be substituted by an alternative form of storage that performs the same function. For example, alternative forms may include one or more of absorption, adsorption, and pressure and/or temperature swings. Additionally or alternatively, the substrate may be dissolved in a liquid in the reservoir or held in a matrix, such as a porous solid material, until it is required. In particular embodiments of the disclosure, the substrate may be dissolved in a liquid in the storage tank and delivered directly to the bioreactor in solution when required.

[0186] Alternatively, the bioreactor itself may be configured such that the headspace above a fermentation liquid nutrient medium acts as a buffer for the intermittent stream. For example, the system may include a means to compress the gaseous substrate stream (when available) and pass it to the bioreactor. The pressure in the headspace in the bioreactor will increase when additional substrate is provided. The substrate is thus continuously available for conversion to products by microbial fermentation.

[0187] In another embodiment, the system may be adapted to receive gaseous substrate streams from multiple intermittent sources. Such a system may include means to combine and/or switch between streams to provide a substantially continuous substrate stream to the bioreactor.

[0188] Micro-organisms used in the fermentation reaction typically have an allowable temperature range, above or below which the reaction rate slows significantly. As such, the system may include cooling means, wherein when availability of the substrate stream is limited, the media in the bioreactor can be cooled to slow down the fermentation reaction and reduce demand for the substrate. Conversely, when the availability of the substrate stream increases, the temperature inside the bioreactor can be increased toward the upper end of the temperature range to increase the reaction rate.

[0189] Alternatively or additionally, a cooling means may be configured to even the cooling load so as to reduce the peak cooling load on a fermentation system. For example, assume that the cooling load required to cope with heat within a gas feed stream and/or a fermentation exotherm in a predetermined period (while gas is being processed) is 2 MW. To maintain the contents of the fermentation tank at a constant temperature during this period, heat must be removed at this rate to maintain a constant temperature within the tank. Conversely, during periods when there is no

gas being processed and the exotherm essentially ceases, the cooling load will be zero. Thus, particularly for large-scale industrial applications, there will be periods when the cooling load is very high, which imposes significant constraints on the system. By levelling the cooling load, the maximum required cooling rate is reduced. Thus, it is possible to operate with a smaller scale cooling system, although on a continuous (or more continuous) basis.

[0190] Using the parameters of the previous example but assuming that the periods when gas is, and those when gas is not, processed are of equal duration, then heat may be removed from the fermentation tank continuously at 1 MW. Under these conditions, the heat removal rate when gas is being processed will not keep up with the heat input/generation, and the temperature within the fermentation tank will rise. When the gas is stopped, but cooling continues, the temperature within the fermentation tank will drop. In this way, a cooling system sized for 1 MW continuous load is required rather than a system sized for a 2 MW load that only runs half of the time. However, the temperature rise and subsequent drop must be limited to maintain the temperature inside the tank within the allowable range for the micro-organisms. Thus, according to particular embodiments, while not constant, the cooling load may be “smoothed,” so that variations therein may be more gradual and/or more limited, in that there is a smaller difference between the maximum and minimum cooling loads.

Industrial Off Gas as a Resource for Fermentation

[0191] In accordance with other aspects of the disclosure, industrial waste gases are used in a fermentation reaction with no or only minimal additional scrubbing or pre-treatment steps being used to make the gases suitable therefor.

[0192] The waste gases may result from any number of industrial processes. The disclosure has particular applicability to supporting the production of ethanol from gaseous substrates such as high volume CO-containing industrial flue gases. Examples include gases produced during ferrous metal products manufacturing, non-ferrous products manufacturing, petroleum refining processes, gasification of coal, gasification of biomass, electric power production, carbon black production, ammonia production, methanol production and coke manufacturing. In a particular embodiment of the disclosure, the waste gases are generated during a process for making steel. For example, those skilled in the art will appreciate the waste gases produced during various stages of the steel making process have high CO and/or CO₂ concentrations. In particular, the waste gas produced during the decarburisation of steel in various methods of steel manufacturing, such as in an oxygen converter (e.g. BOF or KOBM), has a high CO content and low O₂ content making it a suitable substrate for anaerobic carboxydophilic fermentation.

[0193] Waste gases produced during the carburisation of steel are optionally passed through water to remove particulate matter before passing to a waste stack or flue for directing the waste gas into the atmosphere. Typically, the gases are driven into the waste stack with one or more fans.

[0194] In particular embodiments of the disclosure, at least a portion of the waste gas produced during the decarburisation of steel is diverted to a fermentation system by suitable conduit means. By way of example, piping or other transfer means can be connected to the waste gas stack from a steel mill to divert at least a portion of the waste gas to a

fermentation system. Again, one or more fans can be used to divert at least a portion of the waste gas into the fermentation system. In particular embodiments of the disclosure, the conduit means is adapted to provide at least a portion of the waste gas produced during the decarburisation of steel to a fermentation system. The control of and means for feeding gases to a bioreactor will be readily apparent to those of ordinary skill in the art to which the disclosure relates.

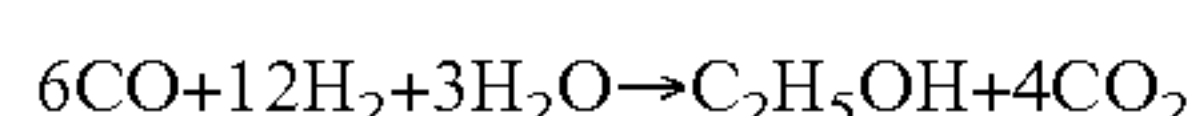
[0195] While steel mills can be adapted to substantially continuously produce steel and subsequently waste gases, particular aspects of the process may be intermittent. Typically the decarburisation of steel is a batch process lasting several minutes to several hours. As such, the conduit means may be adapted to divert at least a portion of the waste gas, such as the gas produced during the decarburisation of steel, to the fermentation system if it is determined the waste gas has a desirable composition.

[0196] The pH of the contents of the bioreactor used in the fermentation process may be adjusted as required. The appropriate pH will be dependent on the conditions required for a particular fermentation reaction having regard to the nutrient media and micro-organisms used, as will be appreciated by persons of ordinary skill in the art to which the disclosure relates. In one preferred embodiment, in fermentation of a gaseous substrate containing CO utilising *Clostridium autoethanogenum*, the pH may be adjusted to approximately 5.5 to 6.5, most preferably to approximately 5.5. Further examples include pH 5.5 to 6.5 using *Moorella thermoacetica* for the production of acetic acid, pH 4.5 to 6.5 using *Clostridium acetobutylicum* for the production of butanol, and pH 7 using *Carboxydotherrmus hydrogenafornans* for the production of hydrogen. Those skilled in the art will be aware of suitable means for maintaining the bioreactor at the required pH. However, by way of example, aqueous bases such as NaOH and aqueous acids such as H₂SO₄ can be used to raise and lower the pH of the fermentation medium and maintain the desired pH.

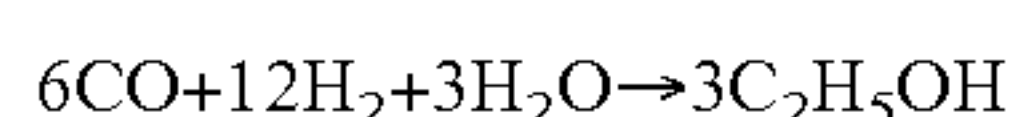
[0197] An additional benefit of the disclosure is that because there is no or only minimal scrubbing and/or other treatment processes performed on the waste gases prior to their use in a fermentation reaction, the gases will contain additional material resulting from the industrial process, which additional material may be used, at least in part, as a feedstock for the fermentation reaction.

Blending of Streams

[0198] As noted previously, it may be desirable to blend an industrial waste stream with one or more further streams in order to improve efficiency, alcohol production and/or overall carbon capture of the fermentation reaction. Without wishing to be bound by theory, in some embodiments of the present disclosure, carboxydophilic bacteria convert CO to ethanol according to the following:



However, in the presence of H₂, the overall conversion is as follows:



[0199] Accordingly, where industrial streams have a high CO content, but include minimal or no H₂, it may be desirable to blend one or more streams comprising H₂ with the waste stream comprising CO, prior to providing the blended substrate stream to the fermenter. The overall effi-

ciency, alcohol productivity and/or overall carbon capture of the fermentation will be dependent on the stoichiometry of the CO and H₂ in the blended stream. However, in particular embodiments the blended stream may substantially comprise CO and H₂ in the following molar ratios: 20:1, 10:1, 5:1, 3:1, 2:1, 1:1 or 1:2.

[0200] In addition, it may be desirable to provide CO and H₂ in particular ratios at different stages of the fermentation. For example, substrate streams with a relatively high H₂ content (such as 1:2 CO:H₂) may be provided to the fermentation stage during start up and/or phases of rapid microbial growth. However, when the growth phase slows, such that the culture is maintained at a substantially steady microbial density, the CO content may be increased (such as at least 1:1 or 2:1 or higher, wherein the H₂ concentration may be greater or equal to zero).

[0201] Blending of streams may also have further advantages, particularly in instances where a waste stream comprising CO is intermittent in nature. For example, an intermittent waste stream comprising CO may be blended with a substantially continuous stream comprising CO and optionally H₂ and provided to the fermenter. In particular embodiments of the disclosure, the composition and flow rate of the substantially continuous stream may be varied in accordance with the intermittent stream in order to maintain provision of a substrate stream of substantially continuous composition and flow rate to the fermenter.

[0202] Blending of two or more streams to achieve a desirable composition may involve varying flow rates of all streams, or one or more of the streams may be maintained constant while other stream(s) are varied in order to 'trim' or optimise the substrate stream to the desired composition. For streams that are processed continuously, little or no further treatment (such as buffering) may be necessary and the stream can be provided to the fermenter directly. However, it may be necessary to provide buffer storage for streams where one or more is available intermittently, and/or where streams are available continuously, but are used and/or produced at variable rates.

[0203] Those skilled in the art will appreciate it will be necessary to monitor the composition and flow rates of the streams prior to blending. Control of the composition of the blended stream can be achieved by varying the proportions of the constituent streams to achieve a target or desirable composition. For example, a base load gas stream may be predominantly CO, and a secondary gas stream comprising a high concentration of H₂ may be blended to achieve a specified H₂:CO ratio. The composition and flow rate of the blended stream can be monitored by any means known in the art. The flow rate of the blended stream can be controlled independently of the blending operation; however the rates at which the individual constituent streams can be drawn must be controlled within limits. For example, a stream produced intermittently, drawn continuously from buffer storage, must be drawn at a rate such that buffer storage capacity is neither depleted nor filled to capacity.

[0204] At the point of blending, the individual constituent gases will enter a mixing chamber, which will typically be a small vessel, or a section of pipe. In such cases, the vessel or pipe may be provided with static mixing devices, such as baffles, arranged to promote turbulence and rapid homogenisation of the individual components.

[0205] Buffer storage of the blended stream can also be provided if necessary, in order to maintain provision of a substantially continuous substrate stream to the bioreactor.

[0206] A processor adapted to monitor the composition and flow rates of the constituent streams and control the blending of the streams in appropriate proportions, to achieve the required or desirable blend may optionally be incorporated into the system. For example, particular components may be provided in an as required or an as available manner in order to optimise the efficiency of alcohol productivity and/or overall carbon capture.

[0207] It may not be possible or cost effective to provide CO and H₂ at a particular ratio all the time. As such, a system adapted to blend two or more streams as described above may be adapted to optimise the ratio with the available resources. For example, in instances where an inadequate supply of H₂ is available, the system may include means to divert excess CO away from the system in order to provide an optimised stream and achieve improved efficiency in alcohol production and/or overall carbon capture. In certain embodiments of the disclosure, the system is adapted to continuously monitor the flow rates and compositions of at least two streams and combine them to produce a single blended substrate stream of optimal composition and means for passing the optimised substrate stream to the fermenter. In particular embodiments employing carboxydophilic microbes to produce alcohol, the optimum composition of the substrate stream comprises at least 0% H₂ and up to about 1:2 CO:H₂.

[0208] By way of non-limiting example, particular embodiments of the disclosure involve the utilisation of converter gas from the decarburisation of steel as a source of CO. Typically, such streams contain little or no H₂, therefore it may be desirable to combine the stream comprising CO with a stream comprising H₂ in order to achieve a more desirable CO:H₂ ratio. H₂ is often produced in large quantities at a steel mill in the coke oven. Accordingly, a waste stream from the coke oven comprising H₂ can be blended with a converter waste stream comprising CO to achieve a desirable composition.

[0209] Additionally, or alternatively, a gasifier may be provided to produce CO and H₂ from a variety of sources. The stream produced by the gasifier may be blended with a stream comprising CO to achieve a desirable composition. Those skilled in the art will appreciate that gasifier conditions can be controlled to achieve a particular CO:H₂ ratio. Furthermore, the gasifier may be ramped up and down to increase and decrease the flow rate of the stream comprising CO and H₂ produced by the gasifier. Accordingly, a stream from a gasifier may be blended with a substrate stream comprising CO to optimise the CO:H₂ ratio in order to increase alcohol productivity and/or overall carbon capture. Furthermore, the gasifier may be ramped up and down to provide a stream of varying flow and/or composition that may be blended with an intermittent stream comprising CO to achieve a substantially continuous stream of desirable composition.

[0210] Other sources of CO and/or H₂ that may be blended with a substrate stream comprising CO include reformation of hydrocarbons, such as natural gas and/or methane and reformation of methanol.

Addition of Scrubber Water

[0211] In accordance with the disclosure, scrubber water is utilised in the fermentation reaction to increase the efficiency of growth and product production.

[0212] The scrubber water may be from any appropriate industrial source, as described hereinbefore. In a particular embodiment of the disclosure, the scrubber water is obtained from the process of cleaning one or more exhaust gases generated during steel manufacture. By way of example, the scrubber water is obtained from cleaning off gases from coking ovens, the blast furnace, the basic oxygen furnace, and/or the electric arc furnace.

[0213] In certain embodiments, the scrubber water is sourced from the same industrial process from which the (gaseous) fermentation substrate is sourced; for example, both the scrubber water and substrate (CO containing exhaust gases) are derived from the same steel mill.

[0214] The scrubber water may be used in raw form direct from the scrubbing system or apparatus of the industrial process. However, the scrubber water may be processed to remove or at least reduce the level of residual particulate matter therein. Methods for processing the scrubber water will be appreciated by those of skill in the art to which the disclosure relates. However, by way of example, the scrubber water may be filtered, centrifuged or allowed to settle prior to introduction into the fermenter.

[0215] As discussed above, the pH of the scrubber water may be adjusted prior to use. The appropriate pH will be dependent on the conditions required for a particular fermentation reaction having regard to the nutrient media and micro-organisms used, as will be appreciated by persons of ordinary skill in the art to which the disclosure relates. In one preferred embodiment, in fermentation of a gaseous substrate containing CO utilising *Clostridium autoethanogenum* the pH may be adjusted to approximately 5.5 to 6.5, most preferably approximately 5.5. Further examples include pH 5.5 to 6.5 using *Moorella thermoacetica* for the production of acetic acid, pH 4.5 to 6.5 using *Clostridium acetobutylicum* for the production of butanol and pH 7 using *Carboxydotherrmus hygroenaformans* for the production of hydrogen.

[0216] The scrubber water may be added to the fermentation reaction using any appropriate means. By way of example, it may be directly fed from a scrubbing apparatus into a bioreactor in which fermentation is occurring or is to occur. Alternatively, it may be collected from a scrubbing apparatus and stored in an appropriate chamber which feeds the bioreactor, or it may be collected from a scrubbing apparatus, stored and manually fed into the bioreactor. Addition of the scrubber water to the bioreactor can be continuous, or the scrubber water can be added at certain time points in the fermentation reaction, or on demand, as circumstances require.

[0217] In one embodiment of the disclosure, the scrubber water is mixed with nutrient media to be used in the fermentation reaction and then added to the bioreactor by any one of the foregoing means. Accordingly, the disclosure also provides a composition comprising a mixture of an appropriate nutrient media and scrubber water. Persons of ordinary skill in the art to which the disclosure relates will appreciate appropriate nutrient media for use in microbial fermentation. However, by way of example, such media may contain sources of nitrogen, phosphate, potassium, sodium,

sulphur, a range of metal ions and B vitamins, and the like. An exemplary media is provided herein after in the section entitled "Examples".

[0218] The scrubber water may be used in an amount of up to approximately 1:9 nutrient media to scrubber water. In one preferred embodiment of the disclosure, the scrubber water is used at a ratio of approximately 1:1 nutrient media to scrubber water.

[0219] As will be appreciated, the scrubber water resulting from particular processes may contain components that are toxic or harmful to particular micro-organisms. Thus, the disclosure does not exclude all pre-treatment processes but avoids such additional processes if possible. Alternatively or in addition, the ratio of the content of scrubber water in a bioreactor may be controlled such that potentially toxic or harmful components are maintained below acceptable concentrations.

Scrubber Water as Primary Feedstock

[0220] In another embodiment of the disclosure, the fermentation reaction is conducted using only scrubber water as the feedstock. In other words, the scrubber water is the primary source of carbon for the fermentation reaction. In this embodiment, the fermentation reaction may be carried out substantially as hereinbefore described, but it is not necessary to provide or capture CO containing gases or provide an alternative carbon source.

[0221] Scrubber water may be fed to a bioreactor in which fermentation will occur as described hereinbefore. In one embodiment, it is fed directly and continuously to a bioreactor from a scrubbing system or apparatus at an appropriate level to maintain optimal conditions for the fermentation reaction.

[0222] In a related embodiment, the scrubber water is stored and then fed to the bioreactor at times when an alternative feedstock or substrate is unavailable. For example, exhaust gases generated in certain steel manufacturing processes are not constant but intermittent. When these gases are unavailable to feed the fermentation reaction, scrubber water is fed to the bioreactor to maintain alcohol production and increase the overall efficiency of the reaction. Such a process, whereby scrubber water is supplemented with fermentation media and fed to a bioreactor, can be conducted in using a cell retention system (e.g. cross flow membrane filtration, continuous centrifugation or an immobilised cell system). In this embodiment, the mixture of scrubber water and fermentation media can flow through the reactor, providing nutrition to the bacteria. An advantage of this system is that the scrubber water contains high levels of dissolved carbon monoxide. Because a major process operating cost associated with the use of gaseous substrates for fermentation is the purchase and operation of equipment to allow CO gas mass-transfer from the gas phase to the liquid phase, the use of liquid streams that already contain CO will significantly reduce this cost.

General

[0223] Embodiments of the disclosure are described by way of example. However, it should be appreciated that particular steps or stages necessary in one embodiment may not be necessary in another. Conversely, steps or stages included in the description of a particular embodiment can

be optionally advantageously utilised in embodiments where they are not specifically mentioned.

[0224] While the disclosure is broadly described with reference to any type of stream that may be moved through or around the system(s) by any known transfer means, in certain embodiments, the substrate and/or exhaust streams are gaseous. Those skilled in the art will appreciate that particular stages may be coupled by suitable conduit means or the like, configurable to receive or pass streams throughout a system. A pump or compressor may be provided to facilitate delivery of the streams to particular stages. Furthermore, a compressor can be used to increase the pressure of gas provided to one or more stages, for example the bioreactor. As discussed hereinabove, the pressure of gases within a bioreactor can affect the efficiency of the fermentation reaction performed therein. Thus, the pressure can be adjusted to improve the efficiency of the fermentation. Suitable pressures for common reactions are known in the art.

[0225] In addition, the systems or processes of the disclosure may optionally include means for regulating and/or controlling other parameters to improve overall efficiency of the process. One or more processors may be incorporated into the system to regulate and/or control particular parameters of the process. For example particular embodiments may include determining means to monitor the composition of substrate and/or exhaust stream(s). In addition, particular embodiments may include a means for controlling the delivery of substrate stream(s) to particular stages or elements within a particular system if the determining means determines the stream has a composition suitable for a particular stage. For example, in instances where a gaseous substrate stream contains low levels of CO or high levels of O₂ that may be detrimental to a fermentation reaction, the substrate stream may be diverted away from the bioreactor. In particular embodiments of the disclosure, the system includes means for monitoring and controlling the destination of a substrate stream and/or the flow rate, such that a stream with a desired or suitable composition can be delivered to a particular stage.

[0226] In addition, it may be necessary to heat or cool particular system components or substrate stream(s) prior to or during one or more stages in the process. In such instances, known heating or cooling means may be used. For example, heat exchangers may be employed to heat or cool the substrate streams.

[0227] Furthermore, the system may include one or more pre/post treatment steps to improve the operation or efficiency of a particular stage. For example, a pre-treatment step may include means for removing particulate matter and/or long chain hydrocarbons or tars from a gaseous substrate stream. Other pre- or post-operations that may be conducted include separation of desired product(s) from particular stages, such as, for example, the bioreactor production stage (e.g. removal of ethanol by distillation).

[0228] Various embodiments of systems of the disclosure are described in the accompanying Figures. The alternative embodiments described in FIGS. 1-13 comprise features in common with one another and the same reference numbers have been used to denote the same or similar features in the various figures. Only the new features (relative to FIG. 1) of FIGS. 2 to 13 are described, and so these Figures should be considered in conjunction with the description of FIG. 1.

[0229] FIG. 1 is a schematic representation of a system 101 according to one embodiment of the disclosure. Input substrate stream 1 enters system 101 via a suitable conduit. Input substrate stream 1 comprises CO and optionally CO₂ and, in certain embodiments, the substrate stream is a waste gas stream from an industrial process, such as that released during the carburisation of steel in a basic oxygen furnace. The levels of components within gas stream 1 may fluctuate. Optional valve 2 may be included to divert stream 1 elsewhere (indicated by stream 3) if it is determined that stream 1 does not have a desired composition. For example, where it is desirable to obtain CO from stream 1, a minimum CO content may be set for stream 1, whereby the stream is diverted away from further processing in system 101 if the minimum CO content is not met. Such a threshold may be set to avoid uneconomic or non-viable processing of a stream. Any known means may be used to determine whether a gas has a desirable composition. Also, “desirable composition” refers not only to substances desired to be included in stream 1, but also to undesired components. For example, stream 1 may be diverted if a particular contaminant is present in stream 1.

[0230] As will be appreciated by one of skill in the art, valve 2 may be positioned elsewhere within system 101. For example, it may be placed after processing by bioreactor 5.

[0231] If a determination is made that stream 1 has a desired composition, it is passed to optional pre-treat 4. Pre-treat 4 may be used to control various aspects of the stream, including temperature and levels of contaminants or other undesired components or constituents. It may also be used to add components to the stream. This will depend on the particular source of gas stream 1 and/or the particular fermentation reaction and/or the micro-organisms selected therefor.

[0232] Pre-treat 4 may be positioned elsewhere within system 101 or may be omitted, or multiple pre-treats 4 may be provided at various points in system 101. This will depend on the particular source of gas stream 1 and/or the particular fermentation reaction and/or the micro-organisms selected therefor. For example, additional pre-treat(s) may be provided upstream of CO₂ remover 8 to control aspects of the stream entering the CO₂ remover 8.

[0233] Following optional pre-treatment the stream may be passed to bioreactor 5 by any known transfer means. For example, the stream may be driven through the system with one or more fans and/or pumps. Bioreactor 5 is configured to perform the desired fermentation reaction to produce products. According to certain embodiments, bioreactor 5 is configured to process a CO containing substrate so as to produce one or more acids and/or one or more alcohols. In a particular embodiment, bioreactor 5 is used to produce ethanol and/or butanol. Bioreactor 5 may comprise more than one tank, each tank being configured to perform the same reaction and/or different stages within a particular fermentation process and/or different reactions, including different reactions for different fermentation processes which may include one or more common stages.

[0234] Bioreactor 5 may be provided with cooling means for controlling the temperature therein within acceptable limits for the micro-organisms used in the particular fermentation reaction to be performed.

[0235] The products produced in the bioreactor 5 may be recovered by any recovery process known in the art. However, in some embodiments of the disclosure, at least a

portion of the product may exit the bioreactor 5 in a stream 7, comprising components such as CO₂ and optionally unconverted CO. Such streams can be optionally treated in product remover 6 to remove any product before the substantially product free stream 7 is passed to CO₂ remover 8.

[0236] CO₂ remover 8 is configured to receive stream 7, wherein at least a portion of CO₂ present in stream 7 is removed therefrom, leaving a remaining waste stream 9. In certain embodiments, the CO₂ remover 8 is configured to separate at least a portion of the CO₂ component from the stream 7 and is adapted to capture the separated CO₂ and/or convert it into products suitable for further use or storage. Alternatively, the CO₂ remover 8 may be configured to capture CO₂ directly and/or convert it into products, from stream 7.

[0237] Where bioreactor 5 comprises a plurality of stages or separate tanks, streams from at least a subset of the stages may be received by CO₂ remover 8. Also, more than one downstream CO₂ remover 8 may be provided so that the same stream undergoes a plurality of CO₂ removal steps, or the same or a different removal step may be performed on streams from different fermentation stages or tanks.

[0238] According to an alternative embodiment represented in FIG. 2, CO₂ remover 8 is positioned upstream of bioreactor 5 (c.f. downstream in FIG. 1). Thus, according to the embodiment of FIG. 2, CO₂ remover 8 may be used to capture CO₂ from the substrate stream before it is passed to bioreactor 5. Optional valve 2 may be configured such that, if it is determined that the CO₂ content is too low for efficient and/or effective CO₂ capture, the stream can be passed directly to the bioreactor 5. Alternatively, stream 3 may be directed away from the system altogether, for example, in circumstances where a stream is unsuitable for CO₂ removal or fermentation.

[0239] According to the embodiment of FIG. 3, CO₂ remover 8 is provided downstream of bioreactor 5 and valve 10 is configured to direct stream 9 back into bioreactor 5 if it is determined that sufficient CO remains in stream 9 for further fermentation to products. However, if it is determined that the CO content of the stream is below a desired level, the stream can be directed elsewhere (as indicated by stream 11). The embodiment of FIG. 3 also includes the attendant advantages of both the embodiment of FIG. 1 and that of FIG. 2.

[0240] FIG. 4 is a schematic representation of a system 104 according to a further embodiment of the disclosure. Input gas stream 1 enters system 104 via a suitable conduit. Input gas stream 1 may be a waste gas stream from an industrial process, such as that released during the carburisation of steel in a basic oxygen furnace. Input gas stream 1 preferably comprises at least one carbon-based gas. In particular embodiments, stream 1 comprises CO and/or CO₂. The levels of components within gas stream 1 may fluctuate. Optional valve 2 may be included to divert stream 1 elsewhere if it is determined that stream 1 does not have a desired composition (as indicated by stream 3). For example, where it is desirable to obtain CO from stream 1, a minimum CO content may be set for stream 1, whereby the stream is diverted away from further processing in system 104 if the minimum content is not met. Such a threshold may be set to avoid uneconomic or non-viable processing of a stream. Any known means may be used to determine whether a gas has a desirable composition. As noted above, “desirable composition” may refer not only to substances

desired to be included in stream 1, but also to undesired components. For example, stream 1 may be diverted if a particular contaminant is present in stream 1.

[0241] As will be appreciated by one of skill in the art upon consideration of the instant disclosure, valve 2 may be positioned elsewhere within system 104. For example, it may be placed in the system after processing by gas separator 13.

[0242] If a determination is made that stream 1 has a desired composition, it is passed to gas separator 13. At least a first component of gas stream 1 is separated therefrom, leaving a remaining component. Either the at least a first component or the remaining component may be diverted as stream 12, with the other component being passed to optional pre-treat 4 and bioreactor 5. Thus, where CO is required in a gas stream to feed a fermentation reaction, the CO may be separated from the remainder of the stream, with only the CO (or a CO-enriched stream) being passed to bioreactor 5. Alternatively, one or more components of the stream (e.g. O₂ and/or H₂) may be separated so that they are at least partly removed, with the remainder of the stream being passed to bioreactor 5.

[0243] As would be apparent to one of skill in the art upon consideration of the instant disclosure, gas separator 13 may comprise one or a plurality of stages or separate units, with one or more gases being separated at each stage.

[0244] Further description of processes and arrangements for gas separation are provided hereinbelow.

[0245] As noted above, pre-treat 4 may be positioned elsewhere within system 104 or may be omitted, or multiple pre-treats 4 may be provided at various points in system 104. Use of pre-treat 4 may depend on the particular source of gas stream 1 and/or the particular fermentation reaction and/or the micro-organisms selected therefor.

[0246] Bioreactor 5 is configured to perform the desired fermentation reaction. According to certain embodiments, bioreactor 5 is configured to process a CO containing substrate so as to produce one or more acids and/or one or more alcohols. In particular embodiments, bioreactor 5 is used to produce ethanol and/or butanol. Bioreactor 5 may comprise more than one tank, each tank being configured to perform the same reaction and/or different stages within a particular fermentation process and/or different reactions, including different reactions for different fermentation processes which may include one or more common stages.

[0247] Bioreactor 5 may be provided with cooling means for controlling the temperature therein within acceptable limits for the micro-organisms used in the particular fermentation reaction to be performed.

[0248] According to the alternative embodiment of FIG. 5, gas separator 13 is positioned downstream of bioreactor 5 (c.f. upstream in FIG. 4). Thus, according to the embodiment of FIG. 5, gas separator 13 may be used to separate one or more components of gases produced by the fermentation reaction in bioreactor 5, and/or separate gases that have been fed to but not used in bioreactor 5. Where bioreactor 5 comprises a plurality of stages or separate tanks, gases from at least a subset of the stages may be received by gas separator 13. Also, more than one downstream gas separator 13 may be provided, so that the same gas stream undergoes a plurality of separations, or the same or a different separation may be performed on gas streams from different fermentation stages or tanks.

[0249] According to the embodiment of FIG. 6, gas separators 13 are provided upstream and downstream of bioreactor 5 with the attendant advantages of both the embodiment of FIG. 4 and FIG. 5.

[0250] A pump or compressor (not shown) may be provided upstream of bioreactor 5 so that the pressure of gas within bioreactor 5 is increased. As discussed hereinabove, the pressure of gases within a bioreactor can affect the efficiency of the fermentation reaction performed therein. Thus, the pressure can be adjusted to improve the efficiency of the fermentation. Suitable pressures for common reactions are known in the art.

[0251] Gas stream 1 may comprise a plurality of different streams. Separate processing elements may be provided for different streams, with only a subset of the elements being common. For example, a first stream may be received by a first gas separator and a second stream may be received by a second separator. Outputs from both the first and second separators may then be passed to a common bioreactor. Other levels of commonality or difference are included within the scope of the disclosure.

[0252] FIG. 7 is a schematic representation of a system 107 according to one embodiment of the disclosure. Waste gas stream 14 enters system 107 via a suitable conduit from an industrial process (e.g. carburisation of steel in a BOF). Stream 14 is intermittent in nature as indicated by the broken line. Stream 14 may be a constant stream in the sense that it is constantly supplied, but the content of particular gases within the stream may vary over time. For example, CO within stream 14 may vary between high and low levels over time. Regardless of whether stream 14 is in fact constantly or intermittently produced, during times when the levels of a desired gas are too low to support a fermentation reaction (or levels of an unwanted gas (e.g. O₂) are too high), valve 2 may be used to divert the stream 14 elsewhere, including into the atmosphere (as indicated by stream 3). At times when stream 14 comprises a desired gas at a desired concentration, valve 2 passes resultant stream 15 to buffer storage tank 16. Stream 15 is also shown in broken line due to its possibly intermittent nature.

[0253] Buffer storage tank 16 acts as a reservoir which feeds gas to bioreactor 5 after any pre-treatment of the gas at pre-treat 4. Pre-treat 4 may be positioned elsewhere within system 1 or even omitted, depending on the particular source of gas stream 14 and/or the particular fermentation reaction and/or the micro-organisms selected therefor.

[0254] Buffer storage tank 16 preferably releases a steady stream 17 of gas, which is passed to pre-treat 4 and then to bioreactor 5 as steady stream 18. Streams 17 and 18 are shown as a continuous line to reflect their substantially continuous nature. Gas may be compressed within buffer storage tank 16 to reduce the space required therefor. A valve (not shown) or other means may be used to set the rate of flow of gas from buffer storage tank 16. The rate is preferably constant and selected such that buffer storage tank 16 always has a supply of gas and does not become depleted. According to one embodiment, control means (not shown) may control the valve to vary the rate of flow of gas 15 depending on the amount of gas contained therein. More particularly, when the gas stored in buffer storage tank 16 falls below a predetermined level, the rate of flow of gas from buffer storage tank 16 may be reduced so that, while optimal levels of gas are not passed to bioreactor 5, reduced levels are provided which may at least mitigate the effects on

the productivity of bioreactor **5** by providing improved conditions for the micro-organisms inside bioreactor **5**.

[0255] Thus, the intermittent nature of stream **14** is mitigated in the embodiment shown in FIG. 7 due to buffering of gas in storage tank **16**.

[0256] As would be apparent to one of skill in the art upon consideration of the instant disclosure, buffer storage tank **16** preferably includes an exhaust port for removing waste gases of the fermentation process. Bioreactor **5** may also be provided with cooling means for controlling the temperature therein within acceptable limits for the micro-organisms.

[0257] According to an alternative embodiment of system **107**, buffer storage tank **16** is substituted by an alternative form of storage that performs the same or a similar function. These may include one or more of absorption, adsorption, and pressure and/or temperature swings. According to one embodiment, the gas is stored in solution with the solution then being fed to bioreactor **5**. Such an arrangement may reduce the processing time within bioreactor **5**, because the required gases are already dissolved before they reach the bioreactor **5**.

[0258] In the arrangement shown in FIG. 8, buffer storage tank **16** is optional, as indicated by the broken line. In embodiments omitting buffer storage tank **16**, stream **14** is passed to bioreactor **5** as and when available and having an acceptable composition, resulting in streams **19** and **20** being intermittent in nature. As aforementioned, this may not be ideal for particular micro-organisms or processes. Where buffer storage tank **16** is included, a portion of stream **15** may be diverted thereto so that when stream **15** is available, gas is passed both to bioreactor **5** and buffer storage tank **16**. Gas passed to buffer storage tank **16** may be stored until such times when stream **15** is not available. At least a low-level stream of gas may then be passed from buffer storage tank **16** to bioreactor **5**.

[0259] As would be apparent to one of skill in the art, waste gas stream **14** from an industrial process may be at a high temperature. Allowable temperature ranges for micro-organisms vary but are of the order of 30° C. to 50° C. for anaerobic bacteria typically used to produce alcohols such as ethanol. Gas stream **14** can cause the temperature within bioreactor **5** to rise, which is exacerbated by the exothermic nature of the fermentation processes, resulting in a need for cooling measures to be included within the system. According to one embodiment, the intermittent nature of stream **14** is taken into account when configuring the cooling means for bioreactor **5**. More particularly, during times when stream **14** is not available or not of the desired composition, the temperature inside bioreactor **5** may be reduced to that of the lower end of the allowable temperature range for the micro-organisms used (e.g. towards 30° C.). Then, when gas stream **14** is available with the desired composition, a temperature rise inside bioreactor **5** is allowed, thereby reducing the requirements of the cooling means that are provided when gas is fed to bioreactor **5**. Thus, for anaerobic bacteria typically used to produce alcohols such as ethanol, the temperature inside bioreactor **5** may be allowed to approach 50° C. According to one embodiment, in the event of bioreactor **5** approaching a maximum allowable temperature, gas stream **14** may be inhibited from entering bioreactor **5** even if it is available at the desired composition, so as to enable the temperature within bioreactor **5** to be more readily controlled. In such cases, the gas may be stored for later use or diverted elsewhere, where it may undergo

additional processing steps as would be apparent to one of skill in the art upon consideration of the instant disclosure. Particular embodiments of the disclosure provide for leveling of the cooling load.

[0260] FIG. 9 is a schematic representation of a system **109** according to another embodiment of the disclosure. Compressor **22** serves to compress intermittent stream **21**, when available, delivering compressed stream **23** to bioreactor **5**. Thus, according to the embodiment of FIG. 9, bioreactor **5** effectively functions as both a fermentation tank and a storage tank by holding gas therein at an elevated pressure when stream **14** is available and has the desired composition. During breaks in stream **14** or when stream **14** does not have the desired composition, waste gases may be slowly exhausted from bioreactor **5** such that the pressure of gas within bioreactor **5** drops, but such that sufficient levels of any desired gases are continuously maintained or sufficiently well maintained to avoid significant periods of time during which the micro-organisms are deprived of the gases.

[0261] FIGS. 10a and 10b are schematic representations of systems **110a** and **110b** according to another embodiment of the disclosure in which multiple intermittent gas streams **14a** and **14b** are used to feed the fermentation reaction within bioreactor **5**. Thus, when stream **14a** is unavailable or does not have the desired composition, bioreactor **5** may alternatively be fed by stream **14b**. As would be apparent to one of skill in the art, more than two gas stream sources may be available. Also, the amount of commonality between processing steps of the streams may vary depending on particular compositions of each stream. The arrangement shown in FIGS. 10a and 10b may be implemented in a steel mill with different streams originating from different stages in the steel making process. Additionally or alternatively, other gas sources may be used. For example, in fermentations using anaerobic bacteria to produce alcohols such as ethanol, conventional sources may be used to provide the stream (e.g. biomass).

[0262] FIG. 11 is a schematic representation of a system **111** according to another embodiment of the disclosure, incorporating several of the stages previously described herein. Intermittent stream **14** is converted into a substantially continuous stream **17** as previously described with reference to FIG. 7. Substantially continuous stream **17** is passed to gas separator **13**, which is adapted to separate CO₂ from other components of the substrate stream, such as CO. The separated stream **12** comprising CO₂, is passed to CO₂ remover **8**, where it can be converted into products suitable for further use or stored. The remainder of the stream comprising CO is passed to optional pre-treat **4** and then on to bioreactor **5**. Optional conduit **24** may be provided to pass a stream comprising CO₂ exiting the bioreactor **5** back to CO₂ remover **8**, where it can be converted into products suitable for further use or stored.

[0263] FIG. 12 is a schematic representation of a system **112** according to a further embodiment of the disclosure. Blown oxygen furnace **25** may be part of an industrial process such as the decarburisation of steel and produces waste stream **1**. In particular embodiments, waste stream **1** comprises CO and/or CO₂. Waste stream **1** is passed through optional pre-treat **4a**. Typically pre-treat **4a** will be a scrubber or water bath adapted to remove particulate matter from stream **1**. Valve **2a** is adapted to divert at least a portion of stream **1** into waste stack **26** when it is determined the stream does not have the desired composition. The diverted

stream is represented by arrow **3a**. Typically the stream diverted into the waste stack **26** will exit into the atmosphere as represented by arrow **27**. Typically, the stream is gaseous and can be driven into the waste stack and optionally throughout the system **113** by one or more fans and/or pumps.

[0264] If it is determined that stream **1** has a desired composition, it can be diverted to optional heat exchanger **28a** as stream **14**. Typically, stream **14** will be intermittent in nature and may need cooling. Heat exchanger **28a** may be any heat exchanging means known in the art. However, by way of example, it is a shell tube heat exchanger. Optional pre-treat **4b** may be used to remove residual particulate matter from the stream if necessary. By way of example, a membrane filter can be used to remove residual particulate matter from the stream. Pre-treat **4b** may also include means to remove condensed water from the optionally cooled stream, such as a knock-out pot or other suitable moisture collection means known in the art.

[0265] The stream can be pressurised by any suitable means, such as gas compressor **22**, before passing to oxygen removal stage **29**. Any means suitable for removal of oxygen may be used, however by way of example; oxygen removal stage **29** includes a hot copper catalyst or a catalytic converter. The stream can be cooled using optional heat exchanger **28b**, before passing into holding tube **30**. Holding tube **30** is of sufficient length such that the composition of the stream can be determined by any appropriate determining means (not shown) before the stream reaches valve **2b**. If it is determined that the stream has a desirable composition it can be diverted to buffer storage means **16** by valve **2b**. If the composition is not suitable for fermentation, for example the oxygen content is too high, valve **2b** can divert the stream into waste stack **26** (as represented by arrow **3b**). Buffering means **16** passes substantially continuous substrate stream **17** to the bioreactor **5**, via optional pre-treat **4c**. Optional pre-treat **4c** can be used to remove unwanted contaminants such as microbes from stream **17**. By way of example, a sterilisation filter or membrane can be used to remove unwanted bacteria from the stream. Waste stream **3c** exiting the bioreactor **5** can also be passed to waste stack **26**.

[0266] Means for determining the composition of the stream may be optionally included at any stage of the system. By way of example means for determining the O_2 , CO and/or CO_2 composition can be included upstream of valve **2a**, upstream of holding tube **30** or valve **2b** and/or upstream of bioreactor **5**. Furthermore, due to the potentially flammable nature of the streams, safety equipment such as flame arresters can also be included at any stage of the system.

[0267] FIG. 13 is a schematic representation of a system **113** according to a further embodiment of the disclosure. Waste stream **1** and stream **31**, either or both of which may be intermittent in nature, are directed into blender **32**. Blender **32** is adapted to control the flow of at least two streams (such as stream **1** and **31**) and mix the streams to achieve a stream with a desired composition (stream **33**). Unwanted streams, such as streams with an undesirable composition may be diverted away from the system **113** as indicated by arrow **3**, while streams with a desirable composition **33**, may be directed to optional buffer **16**, optional pretreat **4** then bioreactor **5** for conversion to products. The

composition and flow rates of streams **1**, **3**, **31** and **33** may be monitored continuously or otherwise by any means known in the art.

[0268] Stream **1** and/or stream **31** may additionally or alternatively be diverted away from blender **32** before entry therein based on their individual compositions. Such an arrangement enables one of streams **1**, **31** to be used where only one has an undesirable composition.

[0269] In particular embodiments, the blender **32** includes a mixing chamber which will typically comprise a small vessel or a section of pipe. In such cases, the vessel or pipe may be provided with mixing means, such as baffles, adapted to promote turbulence and rapid homogenisation of the individual components.

[0270] In certain embodiments of the disclosure, the blender **32** includes means for controlling the blending of two or more streams to achieve a desirable optimised substrate stream **33**. For example, the blender **32** may include means to control the flow rates of each of the streams **1** and **31** entering the blender **32** such that a desirable composition of stream **33** is achieved (e.g. desirable $CO:H$ ratio). The blender also preferably includes monitoring means (continuous or otherwise) downstream of the mixing chamber. In particular embodiments, the blender includes a processor adapted to control the flow rates and/or compositions of the various streams as a result of feedback from the monitoring means.

EXAMPLES

[0271] The disclosure will now be further described in more detail with reference to the following non-limiting examples.

[0272] The media and solutions used in the fermentations described in these examples contains the following components, unless otherwise noted.

Media:

[0273]

TABLE 1

composition of media LM23 and LM33		
Media Component	Concentration per 1.0 L of Media (LM23)	Concentration per 1.0 L of Media (LM33)
$MgCl_2 \cdot 6H_2O$	0.5 g	0.5 g
NaCl	0.2 g	0.2 g
$CaCl_2$	0.2 g	0.2 g
100 mM sodium phosphate buffer (pH 6.0)*	160 ml	—
NaH_2PO_4	—	2.04 g
NH_4Cl	0.6 g	2.5 g
85% H_3PO_4	0.05 ml	—
KCl	0.15 g	0.15 g
Composite trace metal solution (LSO6)	10 mL	10 mL
Composite B vitamin Solution (LSO3)	10 mL	10 mL
Resazurin (1000 mg/L stock)	1 mL	2 mL
$FeCl_3$	0.0025 g	0.01 g
Cysteine HCl monohydrate	0.75 g	0.5 g
Agarose (optional)	15 g	15 g
Distilled water	To 1 litre	To 1 litre

Combine NaH_2PO_4 (13.2 g) and $Na_2HPO_4 \cdot 7H_2O$ (1.1 g) in H_2O (1 L).

TABLE 2

Composition of vitamin solution (LS03) and trace metal solution (LS06)			
Composite B vitamin Solution (LS03)	per L of Stock	Composite trace metal solution (LS06)	per L of stock
Biotin	20.0 mg	Nitrilotriacetic Acid	1.5 g
Folic acid	20.0 mg	MgSO ₄ •7H ₂ O	3.0 g
Pyridoxine hydrochloride	10.0 mg	MnSO ₄ •H ₂ O	0.5 g
Thiamine•HCl	50.0 mg	NaCl	1.0 g
Riboflavin	50.0 mg	FeSO ₄ •7H ₂ O	0.1 g
Nicotinic acid	50.0 mg	Fe(SO ₄) ₂ (NH ₄) ₂ •6H ₂ O	0.8 g
Calcium D-(*)- pantothenate	50.0 mg	CoCl ₂ •6H ₂ O	0.2 g
Vitamin B12	50.0 mg	ZnSO ₄ •7H ₂ O	0.2 g
p-Aminobenzoic acid	50.0 mg	CuCl ₂ •2H ₂ O	0.02 g
Thioctic acid	50.0 mg	AlK(SO ₄) ₂ •12H ₂ O	0.02 g
Distilled water	To 1 Litre	H ₃ BO ₃	0.30 g
		NaMoO ₄ •2H ₂ O	0.03 g
		Na ₂ SeO ₃	0.02 g
		NiCl ₂ •6H ₂ O	0.02 g
		Na ₂ WO ₄ •6H ₂ O	0.02 g
		Distilled water	To 1 Litre

Methodology

Media

[0274] Media solutions LM23 and LM33 were prepared at pH 5.5 as follows. All ingredients with the exception of cysteine HCL were mixed in 400 ml dH₂O. This solution was made anaerobic by heating to boiling and allowing it to cool to room temperature under a constant flow of 95% CO₂, 5% CO₂ gas. Once cool, the cysteine HCL was added and the pH of the solution adjusted to 5.5 before making the volume up to 1000 ml (for example 1) or 500 mL (for example 2). Anaerobicity was maintained throughout the experiments.

Steel Mill Off Gas

[0275] Steel mill off gas was obtained from the New Zealand Steel Glenbrook facility in Glenbrook, New Zealand. More particularly, it was captured and stored in gas impermeable bags or pressurised in steel gas cylinders at 100-130 bar. Steel mill off gases stored within the bags were accessed via a gas impermeable butyl rubber septa. The steel mill off gas composition varies over time depending on the stage of steel production. However, the gas was collected during the decarburisation process and such gases typically contain CO: 43-50%; CO₂: 17-20%; H₂: 2-3%; N₂: 27-34%.

Steel Mill Scrubber Water

[0276] Water, used to scrub (cleanse) the KOBM off-gas stream at the New Zealand Steel Glenbrook facility in Glenbrook, New Zealand, was filtered once using a Buchner funnel and vacuum line through S95 filter paper. The pH of the filtered water was adjusted to 5.5 and 95% CO₂, 5% CO₂ gas was bubbled through for 45 minutes prior to further use.

Bacteria

[0277] *Clostridium autoethanogenum* were obtained from the German Resource Centre for Biological Material (DSMZ). The accession number given to the bacteria is DSMZ 10061. Alternatively, the *Clostridium autoethanoge-*

num used is that deposited at the German Resource Centre for Biological Material (DSMZ) and allocated the accession number 19630.

Sampling and Analytical Procedures

[0278] Media samples were taken at intervals over a 5-day period. Each time the media was sampled care was taken to ensure that no gas was allowed to enter into or escape from the reactors/serum bottles.

[0279] All samples were used to establish the absorbency at 600 nm (spectrophotometer) to determine cell density and the level of substrates and products were determined by HPLC and GC. HPLC was routinely used to quantify the level of acetate, ethanol. GC was used to quantify the percentage of gas (v/v) of carbon monoxide, carbon dioxide, hydrogen and nitrogen.

HPLC

[0280] HPLC System Agilent 1100 Series. Mobile Phase: 0.0025N Sulphuric Acid. Flow and pressure: 0.800 mL/min. Column: Alltech IOA; Catalog #9648, 150×6.5 mm, particle size 5 µm. Temperature of column: 60° C. Detector: Refractive Index. Temperature of detector: 45° C.

[0281] Method for sample preparation: 400 µL of sample+ 50 µL of 0.15M ZnSO₄+50 µL of 0.15M Ba(OH)₂ into an Eppendorf tube. Centrifuge 10 min at 12,000 rpm, 4° C. Transfer 200 µL supernatant into an HPLC vial and inject into the HPLC instrument 5 µL.

Gas Chromatography

[0282] A Gas Chromatograph CP-4900 Micro-GC, dual channel was used: CP-4900 channel Molecular Sieve 5A PLOT, 10 m, 0.25 mm ID, 4.2 seconds backflush, 70 C injector and column temperature, Argon carrier gas 200 kPa, 40 millisecond injection. CP-4900 channel PoraPLOT Q, 10 m, 0.25 mm ID, 70 C injector temperature and 90 C column temperature, Helium carrier gas 150 kPa, 40 millisecond injection. 20 seconds sampling time. 2 minutes method run time. The sample line was heated to 70 C and linked with a Nafion dryer.

Example 1 Fermentation Using Steel Mill Off Gas

Example 1a (Serum Bottle)

[0283] Incubation was performed in 250 ml sealed serum bottles each containing 50 ml of the media. The headspace of each serum bottle was first flushed three times with CO₂, before being evacuated and filled with the collected steel mill off gas to a final pressure of 25 psig. Each bottle was inoculated with 1 ml of a *Clostridium autoethanogenum* culture. A shaking incubator was used and the reaction temperature was maintained at 37° C.

[0284] Media samples were taken at intervals over a 15-day period. Each time the media was sampled. Care was taken to ensure that no gas was allowed to enter into or escape from the serum bottle.

[0285] All samples were used to establish the cell density and the level of acetate.

[0286] As can be seen from FIGS. 14 and 15, cell growth and acetate production rose during the first 10 days before subsequently slowly tailing off. Thus, cell growth and acetate production were readily supported using the steel

mill off gas even though no additional treatment steps were performed on the gas prior to its use in the fermentation reaction.

Example 1b (Serum Bottle)

[0287] Incubation was performed in 234 ml sealed serum bottles each containing 50 ml of the media LM33. The 184 ml headspace of each serum bottle was first flushed three times with steel mill off gas, before being evacuated and filled to an overpressure of 30 psig. Each bottle was inoculated with 2 ml of a *Clostridium autoethanogenum* culture. A shaking incubator was used and the reaction temperature was maintained at 37° C. The results of the experiments are provided in Table 3.

TABLE 3

Serum bottle (30 psig; 50% CO ₂ ; 18% CO ₂ ; 3% H ₂ ; 29% N ₂)			
Day	0	1	2
Biomass (g/L)	0.08	0.22	0.19
Acetate (g/L)	0.4	1.4	3.2
Ethanol (g/L)	0	0	0.3
Overpressure (psig)	30	28	18

Example 1c (10 L Continuous Stirred Tank Reactor)

[0288] A Bioflo 3000 bioreactor was filled with 5 L of the media LM33 without Cysteine nor vitamins solution (LS03) and autoclaved for 30 minutes at 121° C. While cooling down, the media was sparged with N₂ and the LS03 solution as well as Cysteine was added. The gas was switched to steel mill off gas prior to inoculation with 150 ml of a *Clostridium autoethanogenum* culture. The bioreactor was maintained at 37° C. and stirred at 200 rpm at the start of the culture with a gas flow of 60 ml/min. During the growth phase, the agitation was increased to 400 rpm and the gas flow was set to 100 ml/min. The pH was set to 5.5 and maintained by automatic addition of 5 M NaOH. The results of this experiment, including gas consumption are provided in Table 4.

TABLE 4

CSTR fermentation supplied with steel mill gas					
Day	1	2	3	4	5
Biomass (g/L)	0.05	0.08	0.30	0.28	0.20
Acetate (g/L)	0.40	0.78	2.76	3.10	4.03
Ethanol (g/L)	0.00	0.00	0.00	0.48	0.71
Gas Flow in (mL/min)	60	60	100	100	100
CO [%] gas in	49.5	49.5	49.5	49.5	49.5
CO ₂ [%] gas in	17.5	17.5	17.5	17.5	17.5
H ₂ [%] gas in	2.9	2.9	2.9	2.9	2.9
N ₂ [%] gas in	27.7	27.7	27.7	27.7	27.7
Gas flow out (mL/min)	55	55	73	94	92
CO [%] gas out	49.2	35.8	16.4	46.1	45.5
CO ₂ [%] gas out	16.9	30.6	44.6	21.1	21.1
H ₂ [%] gas out	3.2	3.2	1.1	3.2	3.2
N ₂ [%] gas out	30.4	30.4	37.8	29.5	30.2

Example 1d (50 L Gas Lift Reactor)

[0289] A 50 L gas-lift reactor (2900 mm height×150 mm diameter and draft tube 2000 mm height×95 mm diameter) was filled with 37 L of the media LM33 filter sterilised

through a 0.2 micrometer pore filter (Pall KA2 DFL P2 filter). The media was sparged for 18 h with nitrogen before switched to steel mill off gas prior to inoculation with 5 L of a *Clostridium autoethanogenum* culture. The gas-lift reactor was maintained at 37° C. and mixing was achieved by recirculating the headspace gas at 15 L/min. The Gas flow into the reactor was 500 ml/min at the start. The headspace overpressure was maintained at 8 psig. During the growth phase, the gas flow into the reactor was increased to 1000 ml/min. The pH was set to 5.5 and maintained by automatic addition of 5 M NaOH. The results of this experiment, including gas consumption are provided in Table 5.

TABLE 5

Gas-Lift reactor fermentation supplied with steel mill gas					
Day	1	2	3	4	5
Biomass (g/L)	0.15	—	—	0.36	0.36
Acetate (g/L)	0.87	3.93	7.90	9.44	10.86
Ethanol (g/L)	0.00	0.00	0.21	0.38	0.48
Gas Flow in (mL/min)	500	500	1000	1000	1000
CO [%] gas in	43.6	43.6	43.6	44.3	44.3
CO ₂ [%] gas in	19.7	19.7	19.7	20.0	20.0
H ₂ [%] gas in	2.8	2.8	2.8	2.8	2.8
N ₂ [%] gas in	33.9	33.9	33.9	32.8	32.8
Gas flow out (mL/min)	341	—	—	690	716
CO [%] gas out	29.1	—	—	33.1	41.3
CO ₂ [%] gas out	29.5	—	—	28.5	22.7
H ₂ [%] gas out	2.9	—	—	3.0	2.6
N ₂ [%] gas out	38.4	—	—	34.8	33.5

[0290] Particular embodiments of the disclosure increase the applicability of microbial fermentation reactions using industrial off gases, particularly for the production of ethanol by *Clostridium autoethanogenum*. Waste gases comprising CO obtained directly from industrial processes can be used as a substrate in fermentation reactions to produce products such as acetate and/or ethanol. This will result in capturing carbon in waste gases thus ameliorating or reducing waste produced by industrial processes, particularly steel manufacture, and lowering the costs associated with operating fermentation reactions.

Example 2: Fermentation Using Scrubber Water

[0291] Under a continuous gas flow of 95% CO₃ 5% CO₂, 25 ml of LM23 media was dispensed into 250 ml serum bottles and mixed with either 25 ml of dH₂O (control bottles) or 25 ml of pH adjusted anaerobic scrubber water (experimental bottles). All bottles were stopped with gas impermeable butyl rubber septa and crimp sealed before autoclaving at 121° C. for 20 minutes.

[0292] Once cool, all serum bottles were inoculated with 1 ml of a *Clostridium autoethanogenum* culture that was actively growing on 95% CO₃ 5% CO₂. Headspace gas was pressurised to 35 psig with 95% CO₃ 5% CO₂. An initial media sample was taken aseptically from each bottle. Bottles were placed on a shaking incubator at 37° C.

[0293] Media samples were taken at regular intervals over a 15-day period. Each time the media was sampled the headspace of each bottle was flushed three times with 95% CO₃ 5% CO₂ gas before pressurising with this gas to 35 psig.

[0294] All samples were used to establish the cell concentration plus the level of ethanol and acetate in each culture.

Results

[0295] As can be seen from FIGS. 16 and 17, the impact of adding 50% scrubber water to the media included:

[0296] 1. Lower levels of acetate production;

[0297] 2. 45% increase in final bacterial density; and

[0298] 3. 44% increase in ethanol production level

[0299] As such, particular embodiments of the disclosure increase the efficiency of growth and alcohol production in fermentation reactions, particularly production of ethanol by *Clostridium autoethanogenum*. Waste material from industrial processes can be used to supplement the nutrient media used in fermentation reactions. In particular, scrubber water can be used as an alternative primary feedstock or substrate for microbial fermentation reactions. This will result in amelioration or reduction of waste produced by industrial processes, particularly steel manufacture, reduction of the level of media required to support fermentation reactions, and reduction of the level of acetate by-product in production of ethanol by fermentation, thus improving overall efficiency of fermentation reactions and lowering the costs associated with operating such reactions.

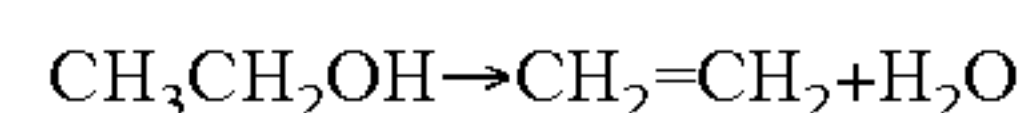
[0300] The microorganism may be cultured with the gas stream to produce one or more products. For instance, the microorganism may produce or may be engineered to produce ethanol (WO 2007/117157), acetate (WO 2007/117157), 1-butanol (WO 2008/115080, WO 2012/053905, and WO 2017/0066498), butyrate (WO 2008/115080), 2,3-butanediol (WO 2009/151342 and WO 2016/094334), lactate (WO 2011/112103), butene (WO 2012/024522), butadiene (WO 2012/024522), methyl ethyl ketone (2-butanone) (WO 2012/024522 and WO 2013/185123), ethylene (WO 2012/026833), acetone (WO 2012/115527), isopropanol (WO 2012/115527), lipids (WO 2013/036147), 3-hydroxypropionate (3-HP) (WO 2013/180581), terpenes, including isoprene (WO 2013/180584), fatty acids (WO 2013/191567), 2-butanol (WO 2013/185123), 1,2-propanediol (WO 2014/036152), 1 propanol (WO 2014/0369152 and WO 2017/0066498), 1 hexanol (WO 2017/0066498), 1 octanol (WO 2017/0066498), chorismate-derived products (WO 2016/191625), 3 hydroxybutyrate (WO 2017/066498), 1,3 butanediol (WO 2017/0066498), 2-hydroxyisobutyrate or 2-hydroxyisobutyric acid (WO 2017/0066498), isobutylene (WO 2017/0066498), adipic acid (WO 2017/0066498), 1,3 hexanediol (WO 2017/0066498), 3-methyl-2-butanol (WO 2017/0066498), 2-buten-1-ol (WO 2017/0066498), isovalerate (WO 2017/0066498), isoamyl alcohol (WO 2017/0066498), and monoethylene glycol (WO 2019/126400), the disclosures of which are herein incorporated in their entirety by reference. In certain embodiments, microbial biomass itself may be considered a product. These products may be further converted to produce one or more articles of manufacture or as at least one component of diesel, jet fuel, and/or gasoline. Additionally, the microbial biomass may be further processed to produce a single cell protein (SCP).

Ethanol and Derivatives

[0301] In one embodiment, ethanol or ethyl alcohol produced according to the method of the disclosure may be used in numerous product applications, including antiseptic hand rubs (WO 2014/100851), therapeutic treatments for methylene glycol and methanol poisoning (WO 2006/088491), as a pharmaceutical solvent for applications such as pain

medication (WO 2011/034887) and oral hygiene products (U.S. Pat. No. 6,811,769), as well as an antimicrobial preservative (U.S. Patent Application No. 2013/0230609), engine fuel (U.S. Pat. No. 1,128,549), rocket fuel (U.S. Pat. No. 3,020,708), plastics, fuel cells (U.S. Pat. No. 2,405,986), home fireplace fuels (U.S. Pat. No. 4,692,168), as an industrial chemical precursor (U.S. Pat. No. 3,102,875), *cannabis* solvent (WO 2015/073854), as a winterization extraction solvent (WO 2017/161387), as a paint masking product (WO 1992/008555), as a paint or tincture (U.S. Pat. No. 1,408,091), purification and extraction of DNA and RNA (WO 1997/010331), and as a cooling bath for various chemical reactions (U.S. Pat. No. 2,099,090). In addition to the foregoing, the ethanol generated by the disclosed method may be used in any other application for which ethanol might otherwise be applicable.

[0302] A further embodiment comprises converting the ethanol generated by the method into ethylene. This can be accomplished by way of an acid catalyzed dehydration of ethanol to give ethylene according to the following formula:



[0303] The ethylene generated in this way may be used for a variety of applications on its own or can be used as a raw material for more refined chemical products. Specifically, ethylene alone may be used as an anaesthetic, as part of a mixture with nitrogen to control ripening of fruit, as a fertilizer, as an element in the production of safety glass, as part of an oxy-fuel gas in metal cutting, welding and high velocity thermal spraying, and as a refrigerant.

[0304] As a raw material, ethylene can be used in the manufacture of polymers such as polyethylene (PE), polyethylene terephthalate (PET) and polyvinyl chloride (PVC) as well as fibres and other organic chemicals. These products are used in a wide variety of industrial and consumer markets such as the packaging, transportation, electrical/electronic, textile and construction industries as well as consumer chemicals, coatings and adhesives.

[0305] Ethylene can be chlorinated to ethylene dichloride (EDC) and can then be cracked to make vinyl chloride monomer (VCM). Nearly all VCM is used to make polyvinyl chloride which has its main applications in the construction industry.

[0306] Other ethylene derivatives include alpha olefins which are used in Linear low-density polyethylene (LLDPE) production, detergent alcohols and plasticizer alcohols; vinyl acetate monomer (VAM) which is used in adhesives, paints, paper coatings and barrier resins; and industrial ethanol which is used as a solvent or in the manufacture of chemical intermediates such as ethyl acetate and ethyl acrylate.

[0307] Ethylene may further be used as a monomer base for the production of various polyethylene oligomers by way of coordination polymerization using metal chloride or metal oxide catalysts. The most common catalysts consist of titanium (III) chloride, the so-called Ziegler-Natta catalysts. Another common catalyst is the Phillips catalyst, prepared by depositing chromium (VI) oxide on silica.

[0308] Polyethylene oligomers so produced may be classified according to its density and branching. Further, mechanical properties depend significantly on variables such as the extent and type of branching, the crystal struc-

ture, and the molecular weight. There are several types of polyethylene which may be generated from ethylene, including

Ultra-high-molecular-weight polyethylene (UHMWPE)
Ultra-low-molecular-weight polyethylene (ULMWPE or PE-WAX)

High-molecular-weight polyethylene (HMWPE)

High-density polyethylene (HDPE)

High-density cross-linked polyethylene (HDXLPE)

Cross-linked polyethylene (PEX or XLPE)

Medium-density polyethylene (MDPE)

Linear low-density polyethylene (LLDPE)

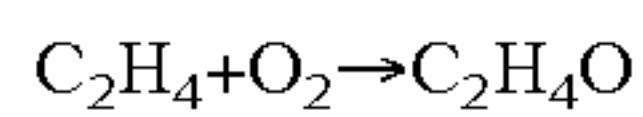
Low-density polyethylene (LDPE)

Very-low-density polyethylene (VLDPE)

Chlorinated polyethylene (CPE)

[0309] Low density polyethylene (LDPE) and linear low-density polyethylene (LLDPE) mainly go into film applications such as food and non-food packaging, shrink and stretch film, and non-packaging uses. High density polyethylene (HDPE) is used primarily in blow molding and injection molding applications such as containers, drums, household goods, caps and pallets. HDPE can also be extruded into pipes for water, gas and irrigation, and film for refuse sacks, carrier bags and industrial lining.

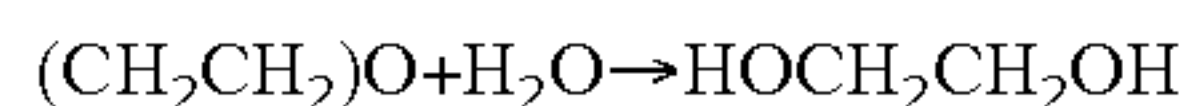
[0310] According to one embodiment, the ethylene formed from the ethanol described above may be converted to ethylene oxide via direct oxidation according to the following formula:



[0311] The ethylene oxide produced thereby is a key chemical intermediate in a number of commercially important processes including the manufacture of monoethylene glycol. Other EO derivatives include ethoxylates (for use in shampoo, kitchen cleaners, etc), glycol ethers (solvents, fuels, etc) and ethanolamines (surfactants, personal care products, etc).

Monoethylene Glycol and Derivatives

[0312] According to one embodiment of the disclosure, the ethylene oxide produced as described above may be used to produce commercial quantities of monoethylene glycol by way of the formula:



[0313] According to another embodiment, the claimed microorganism can be modified in order to directly produce monoethylene glycol. As described in WO 2019/126400, the disclosure of which is incorporated by reference herein, the microorganism further comprises one or more of an enzymes capable of converting acetyl-CoA to pyruvate; an enzyme capable of converting pyruvate to oxaloacetate; an enzyme capable of converting pyruvate to malate; an enzyme capable of converting pyruvate to phosphoenolpyruvate; an enzyme capable of converting oxaloacetate to citryl-CoA; an enzyme capable of converting citryl-CoA to citrate; an enzyme capable of converting citrate to aconitate and aconitate to iso-citrate; an enzyme capable of converting phosphoenolpyruvate to oxaloacetate; an enzyme capable of converting phosphoenolpyruvate to 2-phospho-D-glycerate; an enzyme capable of converting 2-phospho-D-glycerate to 3-phospho-D-glycerate; an enzyme capable of converting 3-phospho-D-glycerate to 3-phosphonooxypyruvate; an enzyme capable of converting 3-phospho-

nooxypyruvate to 3-phospho-L-serine; an enzyme capable of converting 3-phospho-L-serine to serine; an enzyme capable of converting serine to glycine; an enzyme capable of converting 5,10-methylenetetrahydrofolate to glycine; an enzyme capable of converting serine to hydroxypyruvate; an enzyme capable of converting D-glycerate to hydroxypyruvate; an enzyme capable of converting malate to glyoxylate; an enzyme capable of converting glyoxylate to glycolate; an enzyme capable of converting hydroxypyruvate to glycolaldehyde; and/or an enzyme capable of converting glycolaldehyde to ethylene glycol.

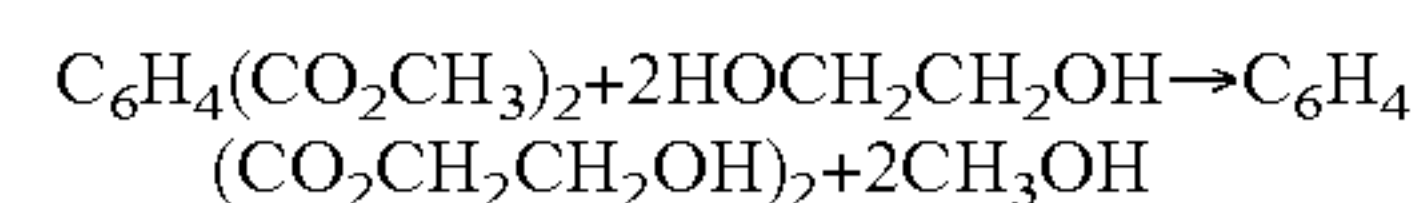
[0314] Monoethylene glycol produced according to either of the described methods may be used as a component of a variety of products including as a raw material to make polyester fibers for textile applications, including nonwovens, cover stock for diapers, building materials, construction materials, road-building fabrics, filters, fiberfill, felts, transportation upholstery, paper and tape reinforcement, tents, rope and cordage, sails, fish netting, seatbelts, laundry bags, synthetic artery replacements, carpets, rugs, apparel, sheets and pillowcases, towels, curtains, draperies, bed ticking, and blankets.

[0315] MEG may be used on its own as a liquid coolant, antifreeze, preservative, dehydrating agent, drilling fluid or any combination thereof. The MEG produced may also be used to produce secondary products such as polyester resins for use in insulation materials, polyester film, de-icing fluids, heat transfer fluids, automotive antifreeze and other liquid coolants, preservatives, dehydrating agents, drilling fluids, water-based adhesives, latex paints and asphalt emulsions, electrolytic capacitors, paper, and synthetic leather.

[0316] Importantly, the monoethylene glycol produced may be converted to the polyester resin polyethylene terephthalate ("PET") according to one of two major processes. The first process comprises transesterification of the monoethylene glycol utilizing dimethyl terephthalate, according to the following two-step process:

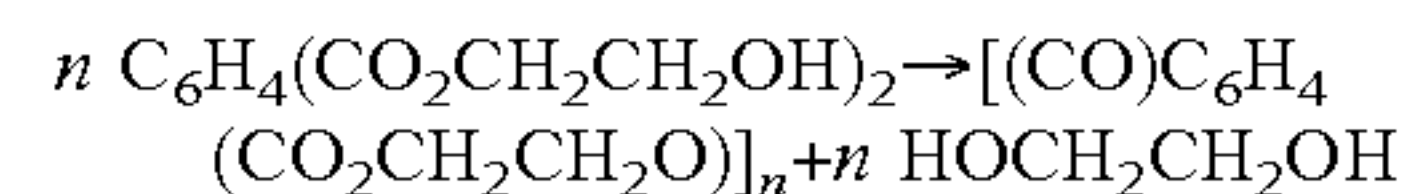
First Step

[0317]

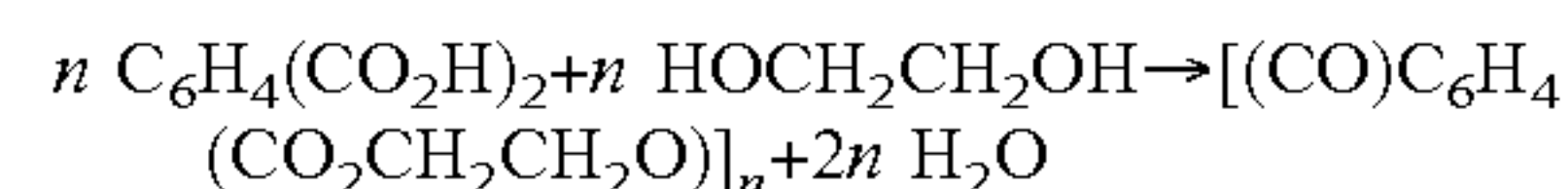


Second Step

[0318]



Alternatively, the monoethylene glycol can be the subject of an esterification reaction utilizing terephthalic acid according to the following reaction:



[0319] The polyethylene terephthalate produced according to either the transesterification or esterification of monoethylene glycol has significant applicability to numerous packaging applications such as jars and, in particular, in the production of bottles, including plastic bottles. It can also be used in the production of high-strength textile fibers such as Dacron, as part of durable-press blends with other fibers such as rayon, wool, and cotton, for fiber fillings used in insulated clothing, furniture, and pillows, in artificial silk, as carpet fiber, automobile tire yarns, conveyor belts and drive

belts, reinforcement for fire and garden hoses, seat belts, nonwoven fabrics for stabilizing drainage ditches, culverts, and railroad beds, and nonwovens for use as diaper top-sheets, and disposable medical garments.

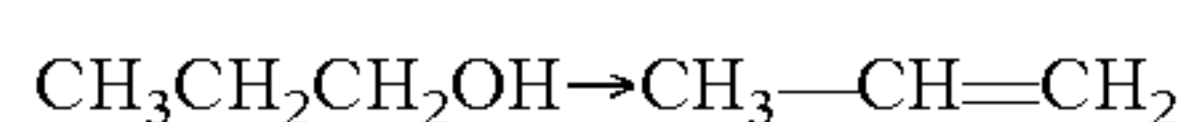
[0320] At a higher molecular weight, PET can be made into a high-strength plastic that can be shaped by all the common methods employed with other thermoplastics. Magnetic recording tape and photographic film are produced by extrusion of PET film. Molten PET can be blow-molded into transparent containers of high strength and rigidity that are also virtually impermeable to gas and liquid. In this form, PET has become widely used in bottles, especially plastic bottles, and in jars.

Isopropanol and Derivatives

[0321] In an additional embodiment, isopropanol or isopropyl alcohol (IPA) produced according to the method may be used in numerous product applications, including either in isolation or as a feedstock for the production for more complex products. Isopropanol may also be used in solvents for cosmetics and personal care products, de-icers, paints and resins, food, inks, adhesives, and pharmaceuticals, including products such as medicinal tablets as well as disinfectants, sterilisers and skin creams.

[0322] The IPA produced may be used in the extraction and purification of natural products such as vegetable and animal oil and fats. Other applications include its use as a cleaning and drying agent in the manufacture of electronic parts and metals, and as an aerosol solvent in medical and veterinary products. It can also be used as a coolant in beer manufacture, a coupling agent, a polymerisation modifier, a de-icing agent and a preservative.

[0323] Alternatively, the IPA produced according to the method of the disclosure may be used to manufacture additional useful compounds, including plastics, derivative ketones such as methyl isobutyl ketone (MIBK), isopropylamines and isopropyl esters. Still further, the IPA may be converted to propylene according to the following formula:



[0324] The propylene produced may be used as a monomer base for the production of various polypropylene oligomers by way of chain-growth polymerization via either gas-phase or bulk reactor systems. The most common catalysts consist of titanium (III) chloride, the so-called Ziegler-Natta catalysts and metallocene catalysts.

[0325] Polypropylene oligomers so produced may be classified according to tacticity and can be formed into numerous products by either extrusion or molding of polypropylene pellets, including piping products, heat-resistant articles such as kettles and food containers, disposable bottles (including plastic bottles), clear bags, flooring such as rugs and mats, ropes, adhesive stickers, as well as foam polypropylene which can be used in building materials. Polypropylene may also be used for hydrophilic clothing and medical dressings.

Commodity Chemicals and Articles

[0326] According to one embodiment, one or more first products is a commodity chemical. In one embodiment, wherein the commodity chemical is selected from ethanol, isopropanol, monoethylene glycol, sulfuric acid, propylene, sodium hydroxide, sodium carbonate, ammonia, benzene, acetic acid, ethylene oxide, formaldehyde, methanol, or any

combination thereof. In one embodiment, the commodity chemical is aluminum sulfate, ammonia, ammonium nitrate, ammonium sulfate, carbon black, chlorine, diammonium phosphate, monoammonium phosphate, hydrochloric acid, hydrogen fluoride, hydrogen peroxide, nitric acid, oxygen, phosphoric acid, sodium silicate, titanium dioxide, or any combination thereof. In another embodiment, the commodity chemical is acetic acid, acetone, acrylic acid, acrylonitrile, adipic acid, benzene, butadiene, butanol, caprolactam, cumene, cyclohexane, dioctyl phthalate, ethylene glycol, methanol, octanol, phenol, phthalic anhydride, polypropylene, polystyrene, polyvinyl chloride, polypropylene glycol, propylene oxide, styrene, terephthalic acid, toluene, toluene diisocyanate, urea, vinyl chloride, xylenes, or any combination thereof. The method according to one embodiment, wherein the commodity chemical is utilized in the sector selected from plastics, synthetic fibers, synthetic rubber, dyes, pigments, paints, coatings, fertilizers, agricultural chemicals, pesticides, cosmetics, soaps, cleaning agent, detergents, pharmaceuticals, mining, or any combination thereof.

[0327] In another embodiment, the method includes incorporating a commodity chemical into one or more articles or converting a commodity chemical into a product selected from ethanol, acetate, 1-butanol, butyrate, 2,3-butanediol, lactate, butene, butadiene, methyl ethyl ketone (2-butanone), ethylene, acetone, isopropanol, lipids, 3-hydroxypropionate, terpenes, isoprene, fatty acids, 2-butanol, 1,2-propanediol, 1 propanol, 1 hexanol, 1 octanol, chorismate-derived products, 3-hydroxybutyrate, 1,3-butanediol, 2-hydroxyisobutyrate, 2-hydroxyisobutyric acid, isobutylene, adipic acid, 1,3-hexanediol, 3-methyl-2-butanol, 2-buten-1-ol, isovalerate, isoamyl alcohol, monoethylene glycol, antiseptic hand rubs, therapeutic treatments for methylene glycol poisoning, therapeutic treatments for methanol poisoning, pharmaceutical solvent for pain medication, oral hygiene products, antimicrobial preservative, engine fuel, rocket fuel, plastics, fuel cells, home fireplace fuels, industrial chemical precursor, *cannabis* solvent, winterization extraction solvent, paint masking product, paint, tincture, purification and extraction of DNA and RNA, cooling bath for various chemical reactions, ethylene to raw material, anaesthetic, ethylene and nitrogen in fruit ripening, fertilizer, safety glass, oxy-fuel in metal cutting, welding, high velocity thermal spraying, refrigerant, raw material to polyethylene, raw material to PET, raw material to PVC, fibers, packaging, coatings, adhesives, ethylene dichloride (EDC), vinyl chloride monomer (VCM), alpha olefins, linear alpha olefins, detergent alcohols, plasticizer alcohols, vinyl acetate monomer (VAM), barrier resins, industrial ethanol, ethyl acetate, ethyl acrylate, polyethylene oligomers, Ultra-high-molecular-weight polyethylene (UHMWPE), Ultra-low-molecular-weight polyethylene (ULMWPE or PE-WAX), High-molecular-weight polyethylene (HMWPE), High-density polyethylene (HDPE), High-density cross-linked polyethylene (HDXLPE), Cross-linked polyethylene (PEX or XLPE), Medium-density polyethylene (MDPE), Low-density polyethylene (LDPE), Very-low-density polyethylene (VLDPE), Chlorinated polyethylene (CPE), films, food packaging, non-food packaging, shrink film, stretch film, containers, drums, household goods, caps, pallets, pipes, refuse sacks, carrier bags, industrial lining, ethylene oxide, ethoxylates, shampoo, kitchen cleaners, glycol ethers, ethanolamines, surfactants, personal care products, polyester

fibers, textiles, nonwovens, cover stock for diapers, road-building fabrics, filters, fiberfill, felts, transportation upholstery, paper reinforcement, tape reinforcement, tents, rope, cordage, sails, fish netting, seatbelts, laundry bags, synthetic artery replacements, carpets, rugs, apparel, sheets, pillowcases, towels, curtains, draperies, bed ticking, blankets, liquid coolant, antifreeze, preservative, dehydrating agent, drilling fluid, polyester resins, insulation materials, polyester film, de-icing fluid, heat transfer fluid, automotive antifreeze, water-based adhesives, latex paints, asphalt emulsions, electrolytic capacitors, synthetic leather, polyester resin PET, jars, bottles, plastic bottles, high-strength fibers, Dacron, durable-press blends, insulated clothing, furniture filling, pillow filling, artificial silk, carpet fiber, automobile tire yarns, conveyor belts, drive belts, reinforcement for fire and garden hoses, nonwoven fabrics for stabilizing drainage ditches, culverts, railroad beds, nonwovens for diaper top-sheets, disposable medical garments, high-strength plastics, magnetic recording tape, photographic film, as feedstock, solvents for cosmetics, inks, medicinal tablets, disinfectants, sterilizers, skin creams, purification of vegetable oil and fats, purification of animal oil and fats, cleaning agent, drying agent, aerosol solvent, derivative ketones, isopropylamines, isopropyl esters, propylene, polypropylene oligomers, polymerization modifier, coupling agent, heat resistant articles, kettles, food containers, disposable bottles, clear bags, flooring, mats, adhesive stickers, foam polypropylene, building materials, hydrophilic clothing, medical dressings, or any combination thereof.

[0328] In another embodiment, the method includes incorporating a commodity chemical into an article or converting a commodity chemical into a product selected from humectants, filters, fire extinguishing sprinkler system, fuel for warming foods, heat transfer fluids, non-reacted component in formulation, deodorizing or air purifying, softening agent, arts/craft glue/paste, toys, children products, freezer gel pack, treating wood rot and fungus, preserving biological tissues and organs, alkyd type resins, resin esters, enamels, lacquers, latex paint, asphalt emulsion, thermoplastic resin, hydrate inhibition agent, agent for removing water vapor, shoe polish, vaccines, screen cleaning solution, water-based hydraulic fluid, heat transfer for liquid cooled computers, personal lubricant, lubricant, toothpaste, anti-foaming agent in food industrial applications, flame resistant hydraulic fluids, additive for electrolytic polishing belts, industrial solvent, trash bags, shower curtains, cups, utensils, medical devices, durable goods, nondurable goods, plastic sacks, plastic lids, industrial strapping, construction materials, felt, ovenable trays, frozen food trays, microwavable tray, artificial vascular scaffolds, vascular prostheses, woven devices, polyester-based prostheses, vehicle liner material, soaps, cosmetic products, laundry detergent jugs, laundry detergent, soap microplastic, microbeads, cosmetic product microbeads, detergent pods, disinfectant with scrubbing agents, toothpaste with microbeads, face wash, conditioner, body wash, hand cleaner, exfoliating products, bath products, shower gels, powder laundry detergent, lotions, deodorants, toilet cleaners, sunscreen, shopping bags, mouthwash bottles, peanut butter containers, salad dressing and vegetable oil containers, polar fleece fiber, tote bags, paneling, milk jugs, juice bottles, bleach bottles, motor oil bottles, cereal box liners, recycling containers, floor tile, drainage pipe, benches, picnic tables, fencing, wire jacketing, sliding windows, decks, mud flaps, roadway gutters,

speed bumps, squeezable bottles, bread, dry cleaning bags, trash can liners, trash cans, compost bins, shipping envelopes, lumber, syrup bottles, ketchup bottles, straws, medicine bottles, battery cables, battery cases, disposable plates and cups, egg cartons, carry-out containers, compact disc cases, signs and displays, synthetic fibers, yarn, stable phase change material, thermal energy storage material, nylon 6,6, nylon, tires, rubber, adiponitrile, shoes, footwear, or any combination thereof.

[0329] One embodiment is directed to a method of capturing carbon by microbial fermentation, the method comprising:

[0330] 1) receiving a gas stream(s) selected from:

[0331] a) a gas stream comprising CO and/or CO₂ generated from a process comprising pyrolysis, torrefaction, reforming, partial oxidation, or gasification;

[0332] b) an industrial off gas comprising CO₂, or a direct air capture gas stream comprising CO₂;

[0333] c) a gas stream comprising methane or methane combustion products; or

[0334] d) any combination thereof;

[0335] 2) passing the gas stream(s) to a bioreactor containing a culture of one or more microorganisms;

[0336] 3) fermenting the culture in the bioreactor to produce one or more first products; and

[0337] 4) incorporating one or more first products into one or more articles or converting one or more first products into one or more second products.

[0338] The method according to one embodiment, wherein one or more first products is selected from ethanol, isopropanol, monoethylene glycol, sulfuric acid, propylene, sodium hydroxide, sodium carbonate, ammonia, benzene, acetic acid, ethylene, ethylene oxide, formaldehyde, or methanol.

[0339] The method according to one embodiment, wherein one or more first products is a commodity chemical.

[0340] The method according to one embodiment, wherein the commodity chemical is selected from ethanol, isopropanol, monoethylene glycol, sulfuric acid, propylene, sodium hydroxide, sodium carbonate, ammonia, benzene, acetic acid, ethylene, ethylene oxide, formaldehyde, or methanol.

[0341] The method according to one embodiment, wherein the commodity chemical is incorporated into one or more articles, converted into one or more second products, or any combination thereof.

[0342] The method according to one embodiment, wherein one or more first products, one or more second products, or one or more articles are selected from acetate, 1-butanol, butyrate, 2,3-butanediol, lactate, butene, butadiene, methyl ethyl ketone, acetone, lipids, 3-hydroxypropionate, terpenes, isoprene, fatty acids, 2-butanol, 1,2-propanediol, 1 propanol, 1 hexanol, 1 octanol, chorismite-derived products, 3-hydroxybutyrate, 1,3-butanediol, 2-hydroxyisobutyrate, 2-hydroxyisobutyric acid, isobutylene, adipic acid, keto-adipic acid, 1,3-hexanediol, 3-methyl butanol, 2-buten-1-ol, isovalerate, isoamyl alcohol, or any combination thereof.

[0343] The method according to one embodiment, wherein one or more first products, one or more second products, or one or more articles are selected from 1,4-butanediol, methyl isobutyl ketone, methyl isobutyl carbonyl, amyl alcohol, polyoxymethylene, polypropylene,

acrylic acid, poly acrylate, acrylate esters, polyvinyl alcohol, polyvinyl acetate, propylene oxide, polyether polyols, polyurethane, linear low density polyethylene, acrylonitrile, methyl methacrylate, or any combination thereof.

[0344] The method according to one embodiment, wherein the one or more first product comprises sulfuric acid and the one or more article is selected from fertilizer, superphosphates, detergents synthetic resins, dyestuffs, pharmaceuticals, petroleum catalysts, insecticides, anti-freeze, pigments, paints, enamels, printing inks, coated fabrics, paper, cellophane, acetate textile, viscose textiles, lubricants, non-ferrous metals, batteries, or any combination thereof.

[0345] The method according to one embodiment, wherein the one or more first product comprises propylene and the article is polypropylene, films, fibers, containers, packaging, caps, or closures.

[0346] The method according to one embodiment, wherein the one or more first product comprises ethylene.

[0347] The method according to one embodiment, further comprising incorporating the ethylene into anaesthetics, fruit ripening agents, fertilizers, safety glass, metal cutting gases, welding gases, high velocity thermal sprays, refrigerants, or any combination thereof.

[0348] The method according to one embodiment, further comprising converting the ethylene into polyethylene (PE), polyethylene terephthalate (PET), polyvinyl chloride (PVC), acetaldehyde, ethylene dichloride, ethylbenzene, or any combination thereof.

[0349] The method according to one embodiment, further comprising incorporating the polyethylene into food packaging films, non-food packaging films, shrink and stretch films, containers, drums, household goods, caps, pallets, water pipes, gas pipes, irrigation pipes, refuse sacks, carrier bags, industrial linings, or any combination thereof.

[0350] The method according to one embodiment, wherein the one or more first products comprises ethylene and the one or more second products is ethylene oxide and the method further comprises converting the ethylene oxide to monoethylene glycol, ethoxylates, glycol ethers, ethanamines, or any combination thereof.

[0351] The method according to one embodiment, further comprising utilizing the monoethylene glycol in the production of nonwoven polyester fibers, cover stock for diapers, building materials, construction materials, road-building fabrics, filters, fiberfill, felts, transportation upholstery, paper and tape reinforcement, tents, rope and cordage, sails, fish netting, seatbelts, laundry bags, synthetic artery replacements, carpets, rugs, apparel, sheets and pillowcases, towels, curtains, draperies, bed ticking, blankets, or any combination thereof.

[0352] The method according to one embodiment, further comprising converting the monoethylene glycol to polyethylene terephthalate (PET).

[0353] The method according to one embodiment, further comprising utilizing the polyethylene terephthalate (PET) in the production of a polyester resin-product selected from insulation materials, polyester film, de-icing fluids, heat transfer fluids, automotive antifreeze and coolants, water-based adhesives, latex paints and asphalt emulsions, electrolytic capacitors, paper, synthetic leather, or any combination thereof.

[0354] The method according to one embodiment, further comprising utilizing the polyethylene terephthalate (PET) in

the production of bottles, jars, high-strength textile fibers, durable-press textile blends, insulated clothing filling, furniture filling, pillow filling, artificial silk, carpet fibers, automobile tire yarns, conveyor belts, drive belts, hose reinforcement fibers, seat belts, drainage ditch stabilizing fabrics, culvert stabilizing fabrics, railroad bed stabilizing fabrics, diaper topsheets, or disposable medical garments, magnetic recording tape, photographic film, or any combination thereof.

[0355] The method according to one embodiment, wherein the one or more first products comprises isopropanol and the one or more articles is cosmetic solvents, personal care product solvents, de-icers, paints, resins, food, inks, adhesives, medicinal tablets, disinfectants, sterilisers, skin creams, vegetable oil extraction products, animal oil extraction products, cleaning agents, drying agents, medical aerosol solvents, coolants, coupling agents, polymerisation modifiers, de-icing agents, preservatives, or any combination thereof.

[0356] The method according to one embodiment, wherein the one or more first products is isopropanol the one or more second products is propylene, ketones, isopropylamines, isopropyl esters, or any combination thereof.

[0357] The method according to one embodiment, wherein the one or more first product comprises isopropanol the one or more second products is propylene.

[0358] The method according to one embodiment, further comprising polymerizing the propylene into polypropylene.

[0359] The method according to one embodiment, further comprising utilizing the polypropylene in the production of heat-resistant kettles, heat resistant food containers, disposable bottles, clear bags, rugs, mats, ropes, adhesive stickers, building materials, construction materials, hydrophilic clothing, medical dressings, or any combination thereof.

[0360] The method according to one embodiment, wherein the one or more first products comprises monoethylene glycol and the one or more articles is liquid coolant, antifreeze, preservative, dehydrating agent, drilling fluid, or any combination thereof.

[0361] The method according to one embodiment, wherein the one or more first products comprises monoethylene glycol and the one or more second products is polyethylene glycol, polyethylene terephthalate, or both.

[0362] The method according to one embodiment, further comprising utilizing the one or more second products in the production of cover stock for diapers, road-building fabrics, filters, fiberfill, felts, transportation upholstery, paper and tape reinforcement, tents, rope and cordage, sails, fish netting, seatbelts, laundry bags, synthetic artery replacements, carpets, rugs, apparel, sheets and pillowcases, towels, curtains, draperies, bed ticking, blankets, or any combination thereof.

[0363] The disclosure has been described herein with reference to certain preferred embodiments, in order to enable the reader to practice the disclosure without undue experimentation. Those skilled in the art will appreciate that the disclosure can be practiced in a large number of variations and modifications other than those specifically described. It is to be understood that the disclosure includes all such variations and modifications. Furthermore, titles, headings, or the like are provided to aid the reader's comprehension of this document, and should not be read as limiting the scope of the present disclosure. The entire

disclosures of all applications, patents and publications cited herein are herein incorporated in their entirety by reference.

[0364] More particularly, as will be appreciated by one of skill in the art, implementations of embodiments of the disclosure may include one or more additional elements. Only those elements necessary to understand the disclosure in its various aspects may have been shown in a particular example or in the description. However, the scope of the disclosure is not limited to the embodiments described and includes systems and/or methods including one or more additional steps and/or one or more substituted steps, and/or systems and/or methods omitting one or more steps.

[0365] All references, including publications, patent applications, and patents, cited herein are hereby incorporated by reference to the same extent as if each reference were individually and specifically indicated to be incorporated by reference and were set forth in its entirety herein. The reference to any prior art in this specification is not, and should not be taken as, an acknowledgement that that prior art forms part of the common general knowledge in the field of endeavor in any country.

[0366] The use of the terms “a” and “an” and “the” and similar referents in the context of describing the disclosure (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. The terms “comprising,” “having,” “including,” and “containing” are to be construed as open-ended terms (i.e., meaning “including, but not limited to”) unless otherwise noted. The term “consisting essentially of” limits the scope of a composition, process, or method to the specified materials or steps, or to those that do not materially affect the basic and novel characteristics of the composition, process, or method. The use of the alternative (e.g., “or”) should be understood to mean either one, both, or any combination thereof of the alternatives. As used herein, the term “about” means $\pm 20\%$ of the indicated range, value, or structure, unless otherwise indicated.

[0367] Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. For example, any concentration range, percentage range, ratio range, integer range, size range, or thickness range is to be understood to include the value of any integer within the recited range and, when appropriate, fractions thereof (such as one tenth and one hundredth of an integer), unless otherwise indicated.

[0368] All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the disclosure and does not pose a limitation on the scope of the disclosure unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the disclosure.

[0369] Embodiments of this disclosure are described herein. Variations of those embodiments may become apparent to those of ordinary skill in the art upon reading the foregoing description. The inventors expect skilled artisans to employ such variations as appropriate, and the inventors intend for the disclosure to be practiced otherwise than as

specifically described herein. Accordingly, this disclosure includes all modifications and equivalents of the subject matter recited in the claims appended hereto as permitted by applicable law. Moreover, any combination of the above-described elements in all possible variations thereof is encompassed by the disclosure unless otherwise indicated herein or otherwise clearly contradicted by context.

What is claimed is:

1. A method of capturing carbon by microbial fermentation, the method comprising:

- 1) receiving a gas stream(s) selected from:
 - a) a gas stream comprising CO and/or CO₂ generated from a process comprising pyrolysis, torrefaction, reforming, partial oxidation, or gasification;
 - b) an industrial off gas comprising CO₂, or a direct air capture gas stream comprising CO₂;
 - c) a gas stream comprising methane or methane combustion products; or
 - d) any combination thereof;
- 2) passing the gas stream(s) to a bioreactor containing a culture of one or more microorganisms;
- 3) fermenting the culture in the bioreactor to produce one or more first products; and
- 4) incorporating the one or more first products into one or more articles or converting the one or more first products into one or more second products.

2. The method of claim 1, wherein one or more first products is selected from ethanol, isopropanol, monoethylene glycol, sulfuric acid, propylene, sodium hydroxide, sodium carbonate, ammonia, benzene, acetic acid, ethylene, ethylene oxide, formaldehyde, or methanol.

3. The method of claim 1, wherein one or more first products is a commodity chemical.

4. The method of claim 3, wherein the commodity chemical is selected from ethanol, isopropanol, monoethylene glycol, sulfuric acid, propylene, sodium hydroxide, sodium carbonate, ammonia, benzene, acetic acid, ethylene, ethylene oxide, formaldehyde, or methanol.

5. The method of claim 3, wherein the commodity chemical is incorporated into one or more articles, converted into one or more second products, or any combination thereof.

6. The method of claim 1, wherein one or more first products, one or more second products, or one or more articles are selected from acetate, 1-butanol, butyrate, 2,3-butanediol, lactate, butene, butadiene, methyl ethyl ketone, acetone, lipids, 3-hydroxypropionate, terpenes, isoprene, fatty acids, 2-butanol, 1,2-propanediol, 1 propanol, 1 hexanol, 1 octanol, chorismite-derived products, 3-hydroxybutyrate, 1,3-butanediol, 2-hydroxyisobutyrate, 2-hydroxyisobutyric acid, isobutylene, adipic acid, keto-adipic acid, 1,3-hexanediol, 3-methyl-2-butanol, 2-buten-1-ol, isovalerate, isoamyl alcohol, or any combination thereof.

7. The method of claim 1, wherein one or more first products, one or more second products, or one or more articles are selected from 1,4-butanediol, methyl isobutyl ketone, methyl isobutyl carbinol, amyl alcohol, polyoxomethylene, polypropylene, acrylic acid, poly acrylate, acrylate esters, polyvinyl alcohol, polyvinyl acetate, propylene oxide, polyether polyols, polyurethane, linear low density polyethylene, acrylonitrile, methyl methacrylate, or any combination thereof.

8. The method of claim 1, wherein the one or more first product comprises sulfuric acid and the one or more article is selected from fertilizer, superphosphates, detergents syn-

thetic resins, dyestuffs, pharmaceuticals, petroleum catalysts, insecticides, antifreeze, pigments, paints, enamels, printing inks, coated fabrics, paper, cellophane, acetate textile, viscose textiles, lubricants, non-ferrous metals, batteries, or any combination thereof.

9. The method of claim **1**, wherein the one or more first product comprises ethylene.

10. The method of claim **9**, further comprising incorporating ethylene into anaesthetics, fruit ripening agents, fertilizers, safety glass, metal cutting gases, welding gases, high velocity thermal sprays, refrigerants, or any combination thereof.

11. The method of claim **9**, further comprising converting the ethylene into polyethylene (PE), polyethylene terephthalate (PET), polyvinyl chloride (PVC), acetaldehyde, ethylene dichloride, ethylbenzene, or any combination thereof.

12. The method of claim **11**, further comprising incorporating the polyethylene into food packaging films, non-food packaging films, shrink and stretch films, containers, drums, household goods, caps, pallets, water pipes, gas pipes, irrigation pipes, refuse sacks, carrier bags, industrial linings, or any combination thereof.

13. The method of claim **1**, wherein the one or more first product comprises ethylene and the one or more second product is ethylene oxide and the method further comprises converting the ethylene oxide to monoethylene glycol, ethoxylates, glycol ethers, ethanolamines, or any combination thereof.

14. The method of claim **13**, further comprising utilizing the monoethylene glycol in the production of nonwoven

polyester fibers, cover stock for diapers, building materials, construction materials, road-building fabrics, filters, fiberfill, felts, transportation upholstery, paper and tape reinforcement, tents, rope and cordage, sails, fish netting, seatbelts, laundry bags, synthetic artery replacements, carpets, rugs, apparel, sheets and pillowcases, towels, curtains, draperies, bed ticking, blankets, or any combination thereof.

15. The method of claim **13**, further comprising converting the monoethylene glycol to PET.

16. The method of claim **11**, further comprising utilizing the PET in the production of a polyester resin-product selected from insulation materials, polyester film, deicing fluids, heat transfer fluids, automotive antifreeze and coolants, water-based adhesives, latex paints and asphalt emulsions, electrolytic capacitors, paper, synthetic leather, or any combination thereof.

17. The method of claim **1**, wherein the one or more first product is isopropanol, and the one or more second product is selected from propylene, ketones, isopropylamines, isopropyl esters, or any combination thereof.

18. The method of claim **1**, wherein the one or more first product comprises isopropanol, and the one or more second product is propylene.

19. The method of claim **18**, further comprising polymerizing the propylene into polypropylene.

20. The method of claim **1**, wherein the one or more first product comprises monoethylene glycol, and the one or more second product is polyethylene glycol, polyethylene terephthalate, or both.

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